

Signature-Based Methods for Nonparametric Regression of Stochastic Processes

Semester Thesis

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Abstract

Regression is more prominent than ever, in theory and in practice. Developing a mathematically precise and broad view of regression lets us see large-language-modeling, time-series forecasting, and much more as fitting neatly into the (extended) framework of regression. This review focuses on two signature-based approaches to regression for stochastic processes. The first is the Nonparametric, signature-based regression scheme proposed in Schell and Alaifari [2024]. The second is a Gaussian process approach using a signature kernel, formulated within the weighted-space framework for learning on non-locally compact spaces introduced in Cuchiero et al. [2025]. In particular, we cast stochastic-process regression as the stage on which time-series forecasting can be rigorously formulated. We compare the two methods at a theoretical level and, to some extent, numerically, implementing in particular the statistical estimator proposed in Schell and Alaifari [2024]. Throughout, we briefly introduce the underlying concepts in the specific context in which they appear, ranging from general mathematical considerations on regression to the signature transform of stochastic processes and its reproducing kernel Hilbert spaces-based approximation theory for functions on paths spaces.

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1 Introduction

Learning statistical relations between two *related* sets of *data* $(x_i)_{i=1}^n, (y_i)_{i=1}^n$ is an omnipresent task in many applications, and associated theoretical insights are often directly relevant in practice. We refer to it as *regression*. In order to mathematically model regression and tackle it mathematically we need to define what we mean by *related* and the 'nature' of the *data*, or as a mathematician would say, the structure of the mathematical space where the data live.

Many of the modern technological achievements and most pressing problems such as climate modeling, natural language processing, financial modeling, genomics and many others, rely on ordered data and the task of learning statistical relations out of it. We shall refer to samples of ordered data as *paths*. As diverse as ordered data presents itself, e.g. as a sequence of letters (or more generally tokens) in a sentence, the mean temperature of the Earth in time, the price of a stock in time etc., so different can be their relation and the task at hand, e.g. the translation of a sentence in one language to another language, the 'right' answer to a posed question, the most probable continuation of our Earth's temperature given its historical values or the price of Apple's stock market tomorrow. The above touches the point that prediction into the future can be understood as a special case of the same general task of regression. In this exposition this task will be addressed mathematically in detail and numerically to some extent. Concretely, tasks that are relevant in practice, such as language modeling, already require a broader notion of *relation* between data. One has to generalize from functional relations to more general distributional relations, as an input x can have more than one sensible outputs y e.g., the entirety of all possible translations of a sentence.

In particular, we will see in Section 2 that regression shall be understood as the learning of measure-valued functions, which we will refer to as *regression kernels* or *regression operators*. Furthermore, Section 2 will introduce the concept of regression as an operator learning problem, yielding a framework general enough to encompass the regression of stochastic processes.

On the other hand, the second component of regression, the 'nature' of this sequentially ordered data is what makes regression of this type of data a hard problem. Indeed, path spaces, as Hölder spaces, continuous functions on locally compact spaces, smooth functions, absolutely continuous functions and so on, are notoriously complex and hard to characterize. More precisely, they are usually not locally-compact. Hence approximating maps that map from paths to measures, as is needed for our notion of regression, is a non-trivial task, in the sense that classical Stone-Weierstrass theorems don't apply. We will see two approaches to this. In more mathematical terms, we will introduce two different theories, yielding explicit numerics, i.e. convergence results for estimators. The first will base its approximation theory on Giles' theorem through the bounded signature and use it to extend the density property of signature components on subspace of continuous functions to density in some L^p spaces. The latter will be developed into its very own approximation theory on non-locally compact spaces, so-called weighted spaces, and uses it as a center stage where Gaussian process regression and signature kernels can meet.

Both approaches use a particular feature map from path spaces into a Hilbert space, defined by the so-called *signature transform*. The latter will be introduced in Section 3. We

will build up some density results, important in the light of approximation theory. In Section 4 we directly apply these to deduce a consistent statistical estimator of the regression operator, for which we follow the approach in Schell and Alaifari [2024]. Chapter 4 will culminate in the construction of an explicit regression estimator, together with a practical algorithm to perform the regression of stochastic processes.

In Section 5 we formalize time-series forecasting as a kernel learning problem and thus understand it as a special case of regression, as described in Section 2. We will then turn our attention to a different paradigm of regression, *Bayesian inference*, and the very closely related concept of Gaussian process regression. The related core concepts, such as reproducing kernel Hilbert spaces, some introductory remarks on Bayesian inference, together with an introduction to Gaussian Processes, are given in Section 6. Section 7 then introduces a different approximation framework on non-locally compact spaces where the previously introduced concepts surrounding Gaussian process regression combine with signature kernels to give an alternative regression scheme for stochastic processes.

In conclusion, we give some thoughts on how the two approaches compare. In particular, on closer examination, we consider how the methods handle interpolation, that is, noiseless regression of a functional ground-truth relation in the data $y_i = f(x_i)$. This comparison will not be exhaustive but hint at the broader insight that the nonparametric regression approach treated in Section 4 is more general, in that the space of learnable distribution of the data has essentially no structured restrictions, in theory. On the other hand, Gaussian process regression and the related stochastic process regression scheme of Section 7 has practical advantages as this model allows for accessible computation of quantiles of the finite dimensional distributions and hence allows for easier error estimation. We compare this with the Gaussian process approach that assumes the distribution. We complete the picture with an implementation of the algorithm, given at the end of Section 4 as a time series forecasting algorithm and compare it visually with a naïve kernel GPR implementation. A small convergence study for implementation this Algorithm, for the statistical estimator proposed in Schell and Alaifari [2024] concludes the exposition.

2 Regression

This first part will lay the conceptual foundation of the core story of this exposition. Indeed, we will see that understanding regression profoundly and embed it in a clear mathematical framework is what enables us to introduce more advanced tools of rough path theory, namely the signature transform and structurally related considerations, which eventually leads to a clean reformulation of regression for paths as a well-understood, convex optimization problem. The development culminates in the proposition of a universally consistent statistical estimator for the conditional distribution of jointly distributed stochastic processes.

We will complete the picture with an alternative approach, based on Gaussian process regression (GPR), where the introduction of the signature transform through a kernel enters naturally through the kernelized structure inherent to GPR. To embark on this journey, we will start by a popular approach to regression, with which most of the readers will be familiar. We will then argue why this is not satisfactory and give a cleaner and more general definition.

Often regression is understood as the task to learn a functional relation given a set of observed data $\{(x_i, y_i)\}_{i=1}^n \in \mathcal{X} \times \mathcal{Y}$ where we don't yet set structural assumptions on the sets $\mathcal{X}, \mathcal{Y}$. Hence the underlying assumption is the existence of a map $f : \mathcal{X} \rightarrow \mathcal{Y}$ such that

$$y_i = f(x_i), \quad \forall i = 1, \dots, n. \quad (1)$$

Notice that the assumption of a functional relation implies a purely deterministic, 'noise-free', relation between the data. In practice, however, observations y_i are typically corrupted by noise, for instance by the act of measurement which can be noisy. To model this, one introduces a noise term ϵ . The distributional assumptions on the noise turn out to be crucial from a modeling perspective. Let's first neglect this and state the regression relation for the data $\{(x_i, y_i)\}_{i=1}^n$,

$$y_i = f(x_i) + \epsilon_i \quad \forall i = 1, \dots, n. \quad (2)$$

A standard approach is to see x_i and y_i as realizations of random variables X and Y respectively¹. To formulize the regression problem in a form that allows for a general characterization of the solution, we consider the following set up.

Take some underlying probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and assume X, Y to be real valued random-variables. Furthermore, assume again that the noise terms ϵ_i are realizations of a random variable ϵ and $\epsilon|X \sim \mathcal{N}(0, \sigma^2)$ (that is to say that $\mathbb{P}(\epsilon|X) \ll \lambda$, with the Radon-Nikodym density w.r.t. to the Lebesgue measure λ being the (centered) Gaussian density $\propto \exp\left(-\frac{x^2}{2\sigma^2}\right)$). The above then becomes

$$\begin{cases} Y = f(X) + \epsilon, \\ \mathbb{E}[\epsilon|X] = 0, \end{cases}^2 \quad (3)$$

where we assume $f : (\mathbb{R}, \mathcal{B}(\mathbb{R})) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is measurable (e.g. continuous).

This simple (and, as we shall see, rather constrained) view on regression allows a straight forward, explicit expression for the solution $f(X)$. Indeed, assume the following: $X, Y \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ ³. Then, applying $\mathbb{E}[\cdot|X]$ on both sides, using that the conditional noise has 0-mean and that $f(X)$ is X measurable for measurable f , we get the general expression:

$$f(X) = \mathbb{E}[Y|X]. \quad (4)$$

Hence the regression problem for (3) boils down to learning the conditional expectation of Y given X .

¹We take them to be real-valued in this setting only. In general, consider $X : \Omega \rightarrow \mathcal{X}, Y : \Omega \rightarrow \mathcal{Y}$ where $\mathcal{X}, \mathcal{Y}$ are to be taken Polish

²In the literature, one often considers the case where the data is real valued and $\epsilon|X \sim \mathcal{N}(0, \sigma^2)$. In that case we have directly from (3) that $Y|X \stackrel{d}{=} (\epsilon + f(X))|X \sim \mathcal{N}(0, \sigma^2)$. In our case we consider X, Y to be stochastic processes, hence their law is a measure on the space of paths. These are typically infinite dimensional. But it is well known that for infinite dimensional spaces, no non-zero, translation invariant measure exists, in particular no Lebesgue measure. Whence in our case, it is typically non-sensical to give a Radon-Density w.r.t. to the Lebesgue-measure. Nevertheless, as we shall see, one can still define Gaussian measures on (infinite-dimensional) Banach-spaces and through their laws, give sense to Gaussian measure on Banach spaces.

³We will usually assume L^2 , which allows for a geometrical interpretation of condition expectation through the Hilbert projection theorem

Another way of tackling (3) is to find f that minimizes the error between y_i and $f(x_i)$ over the data $\{(x_i, y_i)\}_{i=1}^n$. It is instructive to measure the error as the least square distance, i.e. defining the solution to (131) as the least square solution, additionally requiring $f(X), Y \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ (see prior footnote) and thus setting $\mathcal{G} := \{g : \mathcal{X} \rightarrow \mathbb{R}, \text{ measurable s.t. } g(X) \in L^2(\Omega, \mathbb{P})\}$ we look for

$$\hat{f} \stackrel{!}{=} \arg \inf_{f \in \mathcal{G}} \mathbb{E}[(f(X) - Y)^2], \quad (5)$$

where we note that $f(X) \in L^p(\Omega, \mathbb{P}) \iff f \in L^p(\mathbb{R}, \mathbb{P}_X)$, for each $p \in [1, \infty)$.

By the Hilbert projection theorem⁴ the infimum exists and is unique. Indeed, the above serves as the definition of the conditional expectation for $f(X) \in L^2(\Omega, \mathcal{F}, \mathbb{P})$

$$\hat{f}(X) = \mathbb{E}[Y|X]. \quad (6)$$

Notice, that this approach acknowledges the existence of noise but doesn't make any assumption on its distribution; it simply defines the solution to minimize the error coming from the noise. Hence, we see that both approaches; with conditional noise assumptions (see (3)) and as a least square problem (see (5)) yield the same solution; $\mathbb{E}[Y|X]$.

Taking a step back, both approaches, conceptually, really represent the view that: a) the ground truth relation between the observed data $x_i \sim y_i$ is functional, and b) the fact that we don't directly observe this purely functional relation is due to noise that perturbs the "true" underlying functional relation. We claim that this view is very restricting and often not modeling the situation appropriately.

Consider for example, the task of finding the "right" translation of a sentence given in language A into a language B. A functional relation (every sentence has exactly one correct translation) is restraining. Hence it is more natural to give up a functional relation and rather look for set valued maps $f : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{Y})$ modeling the relation within the data. The final step is to understand that every such possible translation doesn't come with the same probability (or in the context of an LLM: not every token is equally probable to follow a given prompt). It is exactly the task of training such an LLM to learn these probabilities. Hence to come back to the translation task, a more appropriate modeling approach is to look for the "right" probability distribution over possible translations in the language B, given a sentence in the language A. That is, the perturbative noise sometimes needs to be understood as an inherently probabilistic aspect of the task, which should not be overlooked.

Hence we ought to be looking at measure-valued maps $f : \mathcal{X} \rightarrow \mathcal{M}^1(\mathcal{Y})$. Then we can formulate more precisely our understanding of regression as the learning of map $x \mapsto \mathbb{P}(Y|X = x)$ where we content ourselves with an intuitive understanding of the left object for the moment (a more appropriate view of the conditional probability in light of regular conditional distributions (i.e., stochastic kernels) will follow; see equation (46). For now we

⁴Indeed, due to the identity $\int_{\Omega} |g(X(\omega))|^2 \mathbb{P}(d\omega) = \int_{\mathcal{X}} |g(x)|^2 X_* \mathbb{P}(dx)$ we have $\mathcal{G} = L^2(\mathcal{X}, X_* \mathbb{P})$, where $X_* \mathbb{P}$ denotes the image measure of $\mathbb{P}$ under the measurable map $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, i.e. the *law* of X . Hence $\mathcal{G}$ is a Hilbert space and thus the Hilbert projection theorem applies.

understand $\{\mathbb{P}(Y|X = x)\}_{x \in \mathcal{X}}$ as a family of measures). Nevertheless, we point out that the latter is assuming a probability measure $\mu \in \mathcal{M}^1(\mathcal{X} \times \mathcal{Y})$ which is taken to be the joint distribution $\mathbb{P}_{X,Y}$. We can state our first approach to an abstract formulation of regression as the learning of the map $f : x \mapsto \mathbb{P}(Y|X = x)$ given a finite number of samples $\{(x_i, y_i)\}_{i=0}^n$ of the joint distribution $\mathbb{P}_{X,Y}$.

Finally, it will be advantageous to do another identification. Noticing that a version of conditional distribution is defined as $B \rightarrow \mathbb{P}(Y \in B|X) := \mathbb{E}[1_B(Y)|X]$ (this indeed defines a probability measure if $(\mathcal{Y}, \mathcal{F}_Y)$ is Borel space (see Durrett [2019], Theorem 4.1.6.)), and still assuming that one can make sense of the map $x \mapsto \mathbb{P}(Y|X = x)$, we can equivalently identify the latter as $\mathbb{P}(Y \in B|X = x) = \mathbb{E}[1_B(Y)|X = x]$. Through measure-theoretic induction we can thus make sense of the latter for each $\mathbb{P}_Y$ -measurable map $f : (\mathcal{Y}, \mathcal{F}_Y) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that $f(Y) \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ ⁵. We can now precisely formulate our newly gained interpretation of regression

Definition 2.1 (Regression). Let $(\mathcal{X}, \mathcal{F}_X), (\mathcal{Y}, \mathcal{F}_Y)$ be Borel spaces. Given finite samples $\{(x_i, y_i)\}_{i=0}^n$, of a probability measure $\mu \in \mathcal{M}^1(\mathcal{X} \times \mathcal{Y})$, we want to learn the mapping

$$f \mapsto (x \mapsto \mathbb{E}[f(Y)|X = x]) \quad (7)$$

for $f : (\mathcal{Y}, \mathcal{F}_Y) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, with $f \in L^1(\mathcal{Y}, \mathcal{F}_Y, \mathbb{P}_Y)$.

Remark. As argued before, this encompasses the learning of the conditional expectation for $f = 1_B$ where $B \in \mathcal{F}_Y$ (as usual, it is enough to take $B \in \mathcal{D}_Y$ for $\mathcal{D}_Y$ a generating π -system for $\mathcal{F}_Y$) and also notice that the purely deterministic functional learning is a special case of 7 when considering the family of measures (stochastic kernel, see equation (41)) $\{\mathbb{P}(Y|X = x)\}_{x \in \mathcal{X}} = \{\delta_{f(x)}|x \in \mathcal{X}\}$.

3 Algebraic Structures and the Signature Transform

3.1 Tensor Algebra

We will briefly introduce the signature transform and some of its important properties as used in the proceeding. The approach will be to introduce the free tensor algebra over an alphabet and identify, under identification of the alphabet with a orthonormal basis of a finite dimensional vector space, the free tensor algebra, as a graded algebra of tensor powers of the vector space, which will be used henceforth.

Hence consider some alphabet $\{1, \dots, d\}$ and denote it $\langle d \rangle$ the set of finite "words" $\omega = i_1 \cdots i_k$ (word of length $|\omega| = k$) over the alphabet. Next denote by $T(\langle d \rangle)$ the free algebra over the formal power series over the alphabet $\langle d \rangle$:

Definition 3.1 ((Extended) Tensor Algebra). Let $\langle d \rangle$ be the set of words (finite formal products) over the alphabet $\{1, \dots, d\}$. Then we call the extended tensor algebra of formal

⁵In other words, the the statement about the indicator functions is mathematically equivalent to the statement about the measurable and integrable functions i.e. knowing one of them means knowing both.

power series of $\langle d \rangle$ the following

$$T(\langle d \rangle) = \left\{ \mathbf{a} = \sum_{\omega \in \langle d \rangle} a_\omega \cdot \omega \mid a_\omega \in \mathbb{R} \right\} \quad (8)$$

and further restricting to finite formal sums, the tensor algebra

$$T(\langle d \rangle) = \left\{ \mathbf{a} = \sum_{\omega \in \langle d \rangle} a_\omega \cdot \omega \mid a_\omega \in \mathbb{R}, \text{ , } a_\omega \neq 0 \text{ for finitely many } \omega \in \langle d \rangle \right\}. \quad (9)$$

Next, let $(\mathbf{e}_1, \dots, \mathbf{e}_d)$ be an orthonormal basis of some d -dimensional real vector space V and identify the letters of the alphabet with elementary tensors $i \equiv \mathbf{e}_i \in V$ and a word by $\omega = i_1 \dots i_k \equiv \mathbf{e}_{i_1} \otimes \dots \otimes \mathbf{e}_{i_k} \in V^{\otimes k}$. Through this identification we write $T((V))$ and $T(V)$ instead of $T(\langle d \rangle)$ and $T(\langle d \rangle)$ respectively⁶.

Further we introduce the notation $\mathbf{e}_\omega := \mathbf{e}_{i_1} \otimes \dots \otimes \mathbf{e}_{i_k} \in V^{\otimes k}$. In particular, we can now identify

$$\sum_{\omega \in \langle d \rangle} a_\omega \cdot \omega \equiv \sum_{\omega \in \langle d \rangle} a_\omega \cdot \mathbf{e}_\omega = \underbrace{\sum_{k=0}^{\infty} \left(\sum_{\omega: |\omega|=k} a_\omega \cdot \mathbf{e}_\omega \right)}_{\pi_k(\mathbf{a}) \equiv \mathbf{a}_k \in V^{\otimes k}}, \quad (10)$$

where $\pi_n : T((V)) \rightarrow V^{\otimes n}$ the canonical projection on the n^{th} tensor component. The above corresponds to the identification

$$T((V)) = \left\{ \sum_{\omega \in \langle d \rangle} a_\omega \cdot \omega \mid a_\omega \in V \right\} \equiv \left\{ \sum_{k=0}^{\infty} \mathbf{a}^k \mid \mathbf{a}^k \in V^{\otimes k} \right\} = \Pi_{k=0}^{\infty} V^{\otimes k} \quad (11)$$

which, under the former identification of the alphabet and a vector space, becomes a graded algebra. Analogously we have

$$T(V) = \left\{ \sum_{k=0}^{\infty} \mathbf{a}^k \mid \mathbf{a}^k \in V^{\otimes k}, \mathbf{a}^k \neq 0 \text{ for finitely many } k \in \mathbb{N} \right\} = \bigoplus_{k=0}^{\infty} V^{\otimes k}. \quad (12)$$

Lastly, we use the inner product structure on $(V; \langle \cdot, \cdot \rangle)$ to introduce more handy notation. Due to the orthonormality of $\{\mathbf{e}_\omega\}_\omega$, where we recall the underlying identification of $\omega \equiv \mathbf{e}_{i_1} \dots \mathbf{e}_{i_k} \in V^{\otimes k}$ and by definition of the extension of the inner product on V to $V^{\otimes k}$ i.e. for pure tensors

$$\langle v_1 \otimes \dots \otimes v_k \mid w_1 \otimes \dots \otimes w_k \rangle_{V^{\otimes k}} = \langle v_1 \mid w_1 \rangle_V \dots \langle v_k \mid w_k \rangle_V \quad (13)$$

⁶As any d -dimensional real vector space is canonically isomorphic to $\mathbb{R}^d$ we usually take the later for concrete computations. This shall be implicitly understood in the following

and then extended by linearity to all elements of $V^{\otimes k}$, and further to $T((V))$ by requiring that (13) holds true on every tensor component $\pi_n(T((V))) = V^{\otimes n}$. We define

$$\langle \mathbf{a} | \mathbf{b} \rangle := \sum_{k=0}^{\infty} \langle \pi_k(\mathbf{a}), \pi_k(\mathbf{b}) \rangle_{V^{\otimes k}} = \sum_{\omega, \omega' \in \langle d \rangle} e_{\omega}^*(\mathbf{a}), e_{\omega'}^*(\mathbf{b}) \delta_{\omega\omega'} \quad (14)$$

$$= \sum_{\omega \in \langle d \rangle} \langle \omega, \mathbf{a} \rangle \cdot \langle \omega, \mathbf{b} \rangle \quad \forall \mathbf{a}, \mathbf{b} \in T((V)), \quad (15)$$

where we used that $\{e_{\omega}^* \mid \omega \in \langle d \rangle\}$ defined by $e_{\omega}^*(e_{\rho}) = \delta(\omega, \rho)$ for $\omega, \rho \in \langle d \rangle$, forms a Schauder basis for $T((V))^*$ and denote the duality pairing $e_{\omega}^*(e_{\rho}) = \langle e_{\omega}, e_{\rho} \rangle \equiv \langle \omega, \rho \rangle$ hence $e_{\omega}^*(\mathbf{a}) = \langle \omega, \mathbf{a} \rangle = a_{\omega}$ which we call the coordinates of the tensor series $\mathbf{a} \in T((V))$.

Finally, we will sometimes also refer to $\mathbf{a} = (\mathbf{a}^0, \mathbf{a}^1, \mathbf{a}^2, \dots) \in T((V))$ with $\mathbf{a}^k = \sum_{\omega: |\omega|=k} \langle \omega, \mathbf{a} \rangle \cdot \omega$ explicitly referring to the graded structure of $T((V))$. Hence the above becomes $\langle \mathbf{a} | \mathbf{b} \rangle = \sum_{k=0}^{\infty} \langle \mathbf{a}^k, \mathbf{b}^k \rangle_{V^{\otimes k}}$.
From here on forward take $V = \mathbb{R}^d$.

To conclude this introductory section about the tensor algebra we eventually define the operations (sum, product) with respect to which the tensor algebra is indeed an algebra.

$$(\text{summation}) \quad \mathbf{a} + \mathbf{b} = \mathbf{c}, \quad \mathbf{c}^k = \mathbf{a}^k + \mathbf{b}^k \quad (16)$$

$$(\text{product}) \quad \mathbf{a} \otimes \mathbf{b} = \mathbf{c}, \quad \mathbf{c}^k = \sum_{j=0}^k \mathbf{a}^j \otimes \mathbf{b}^{k-j}, \quad (17)$$

where we use the vector space isomorphism $V^{\otimes m} \otimes V^{\otimes n} = V^{\otimes(m+n)}$, hence $\mathbf{a}^j \otimes \mathbf{b}^{k-j} \in V^{\otimes k}$. For the existence of a multiplicative inverse we additionally require that $\mathbf{a}^0 = 1$ for all $\mathbf{a} \in T((V))$.

There is another way to define a product on $T((V))$ which will turn out to be the right one for Stone-Weierstrass universality type statements for signature coordinates. This second product is called *shuffle* product.

Definition 3.2 (Shuffle Product). We define $\sqcup : T(V) \times T(V) \rightarrow T(V)$

-for $f \in V, \alpha \in \mathbb{R}$: $f \sqcup \alpha = \alpha \sqcup f = \alpha f$

-for $a, b \in V$: $f \sqcup g = (f_- \sqcup g) \cdot a + (f \sqcup g_-) \cdot b$ for $f \in V^{\otimes k}, g \in V^{\otimes l}$ of the form $f = f_- \cdot a$ and $g = g_- \cdot b$

and extend the above inductively to a unique algebra product on $T((V))$.

We will have a second look at the shuffle product after having introduced the signature transform, and see that the shuffle product is more naturally defined on the algebraic dual $T(V^*)$.

3.2 Signature Transform

Having introduced the basic algebraic structure of the tensor algebra, we finally identify it as the codomain of the signature transform, whose richness lies in part in the structure of the former. Without further ado, we define the signature transform.

Definition 3.3 (Signature Transformation). Let $\Delta_{[a,b]}^k \subset \mathbb{R}^k$ be the k -dimensional simplex over the interval $[a, b] \subset \mathbb{R}$ and $X : [a, b] \rightarrow V$ be a p -finite variation path for $p < 2$. The signature of X is then an element of $T((V))$ defined by

$$\mathbf{sig}(X)_{[a,b]} = (1, \mathbf{sig}(X)_{[a,b]}^1, \mathbf{sig}(X)_{[a,b]}^2, \dots), \quad (18)$$

where

$$\mathbf{sig}(X)_{[a,b]}^k = \int_{\Delta_{[a,b]}^k} dX_{s_1} \otimes \dots \otimes dX_{s_k} \quad (19)$$

$$\equiv \left(\int \dots \int_{a \leq t_1 \dots \leq t_k \leq b} dX_{s_1}^{i_1} \dots dX_{s_k}^{i_k} \mid \omega = i_1 \dots i_k \in \langle d \rangle \right) \quad (20)$$

$$\equiv \left(\int_{\Delta_{[a,b]}^k} dX^\omega \mid \omega \in \langle d \rangle, |\omega| = k \right) \equiv \sum_{\omega \in \langle d \rangle, |\omega| = k} \left(\int_{\Delta_{[a,b]}^k} dX^\omega \right) \cdot \omega. \quad (21)$$

As promised we will prove the a crucial density theorem of the signature components in the space of continuous functions over a compact set K denoted $C(K)$ (equipped with uniform topology) for the signature transform and use it to give an equivalent definition of the shuffle product . First we observe that due to the factorial decay of the signature components (see e.g. Cass and Salvi [2024] proposition 1.2.3.) the signature transform maps into a strict subspace $\mathcal{H}$ of the extended tensor algebra $T((V))$. This is interesting because it turns out that we can equip $\mathcal{H}$ with an inner product, which makes it a separable Hilbert space, which motivates the view on the signature map as feature map on path space.

Lemma 3.4. *Let $(T((V)), \langle \cdot, \cdot \rangle)$ be the extended tensor algebra from 3.1 and let*

$$\mathcal{H}^d = \{ \mathbf{a} \in T((V)) \mid \|\mathbf{a}\|^2 = \sum_{m \geq 0} \|\pi_m(\mathbf{a})\|^2 = \sum_{\omega \in \langle d \rangle} |\langle \omega, \mathbf{a} \rangle|^2 \leq \infty \} \subset T((V)). \quad (22)$$

Then $(\mathcal{H}^d, \langle \cdot, \cdot \rangle)$ is a separable Hilbert's space with orthonormal basis $\{\omega \mid \omega \in \langle d \rangle\}$. Futher let $\mathcal{C}_d^1 = \{x \in \mathcal{C} \mid \exists \dot{x} \in L^1([0, 1]; \mathbb{R}^d) \text{ such that } x_t = x_0 + \int_0^t \dot{x}_s ds \text{ for a.e. } t \in [0, 1] \}$ the space of absolutely continuous functions. Then

$$\mathbf{sig} : \mathcal{C}_d^1 \rightarrow \mathcal{H}^d \quad (23)$$

*is **continuous** with respect to the p -variation topology for any $p \geq 1$. Further the augmented signature transform $\underline{\mathbf{sig}}(x) := \mathbf{sig}((t, x_t)_{t \in [0, 1]})$, is **injective**.*

From here on we will always work with the augmented signature transform $\underline{\mathbf{sig}}$ and by an abuse of notation denote it again as $\mathbf{sig}$.

⁷For clarity and practical interest, notice the following: In the case $V = \mathbb{R}$, we have $\pi_m(\int dX) = \int dX^{\otimes m} = \int dX \dots dX \in \mathbb{R}$ essentially due to $\mathbb{R}^{\otimes m} = \mathbb{R} \otimes \dots \otimes \mathbb{R} \cong \mathbb{R}$

⁸In the following we will mostly omit the dependence of $\mathcal{H}$ on the size of the alphabet d for clarity of presentation

In order to develop a Stone-Weierstrass like approximation theory for path space we will now proceed with the study of dense subsets of continuous maps on path spaces. For this the study of linear forms on signature transforms will turn out to be fruitful.

The following lemma is the key to the universal approximation result for signatures (see Theorem 3.6). First we introduce some notation.

Let $C_{0,p}(V) = \{x \in C_p(V, \mathbb{R}) : x(0) = 0\}$ and let $\mathcal{C}_p = C_{0,p}/\sim$, where $\sim$ denotes the tree like equivalence relation. For $x \in C_{0,p}$ let $[x]$ be its equivalence class in $\mathcal{C}_p$. It is a non-trivial statement that the equivalence relation induced by tree like equivalence is the same as the equivalence relation induced by $x \sim y \iff \mathbf{sig}(x) = \mathbf{sig}(y)$. We will use this result without proof. For clarity, observe that linear forms on $T(V^*)$ are spanned by $\{\mathbf{e}_\omega^*\}_{\omega \in \langle d \rangle}$ and every such linear form on $T(V^*)$ induces a linear form on $T((V))^*$ by $\langle \mathbf{e}_\omega^*, (A^0, A^1, \dots) \rangle := \mathbf{e}_\omega^*(A^{|\omega|})$

Lemma 3.5. *For $1 \leq p < 2$ consider, for every $f \in T((V))^*$ induced by linear form on $T(V^*)$, the functions*

$$\Phi_f : [x] \rightarrow \langle f, \mathbf{sig}(x) \rangle. \quad (24)$$

Then $\mathcal{A} = \{\Phi_f : f \in T((V))^\}$ is a subalgebra of $\mathbb{R}^{\mathcal{C}_p} = \{f : \mathcal{C}_p \rightarrow \mathbb{R}\}$ that contains constant functions and separates points in $\mathcal{C}_p$.*

Proof. First, notice that for $\pi_0 : (A^0, A^1, A^2, \dots) = A^0$ and for $\alpha \in \mathbb{R}$ we have $\Phi_{\alpha \cdot \pi_0}([x]) = \langle \alpha \cdot \pi_0, \mathbf{sig}(x) \rangle = \alpha \cdot \langle \pi_0, \mathbf{sig}(x) \rangle = \alpha \cdot \mathbf{sig}^0(x) = \alpha \cdot 1$, hence $\mathcal{A}$ contains all the constant functions in $\mathbb{R}^{\mathcal{C}_p}$. Next we show that $\mathcal{A}$ separates points. To this end, let $[x] \neq [y]$. But by the fact, stated prior to this Lemma, that $\sim$ corresponds to equivalence of signatures, we directly have that $\mathbf{sig}(x) \neq \mathbf{sig}(y)$. But, generally $\mathbf{A} = (A^0, A^1, \dots) = (B^0, B^1, \dots) = \mathbf{B} \iff \pi_k(\mathbf{A}) = A^k = B^k = \pi_k(\mathbf{B})$ for all $k \in \mathbb{N}$. Hence there exists $m \in \mathbb{N}$ such that $\pi_m(\mathbf{A}) \neq \pi_m(\mathbf{B})$ i.e. $\Phi_{\pi_m}([x]) \neq \Phi_{\pi_m}([y])$. Finally it remains to show that $\mathcal{A}$ is an algebra. Clearly $\Phi_{f+\alpha g} = \Phi_f + \alpha \Phi_g$. For the closedness under multiplication consider the following

For a word $\omega = i_1 \dots i_k$, denote

$$\mathbf{sig}^I(x) = \langle \mathbf{e}_\omega^*, \mathbf{sig}(x) \rangle = \int \dots \int_{0 \leq u_1 \leq \dots \leq u_k \leq T} \mathbf{e}_{i_1}^*(dX_{u_1}) \dots \mathbf{e}_{i_k}^*(dX_{u_k}). \quad (25)$$

Then

$$\left(\int \dots \int_{0 \leq u_1 \leq \dots \leq u_r \leq T} \mathbf{e}_{i_1}^*(dX_{u_1}) \dots \mathbf{e}_{i_r}^*(dX_{u_r}) \right) \cdot \left(\int \dots \int_{0 \leq v_1 \leq \dots \leq v_s \leq T} \mathbf{e}_{j_1}^*(dX_{v_1}) \dots \mathbf{e}_{j_s}^*(dX_{v_s}) \right) \quad (26)$$

$$= \sum_{\sigma \in \text{Shuffle}(r,s)} \int \dots \int_{0 \leq \sigma(u_1) \leq \dots \leq \sigma(v_s) \leq T} \mathbf{e}_{i_1}^*(dX_{\sigma(u_1)}) \dots \mathbf{e}_{i_r}^*(dX_{\sigma(v_s)}) = \quad (27)$$

$$\sum_{\sigma \in \text{Shuffle}(r,s)} \int \dots \int_{0 \leq w_1 \leq \dots \leq w_s \leq T} \mathbf{e}_{\sigma^{-1}(i_1)}^*(dX_{w_1}) \dots \mathbf{e}_{\sigma^{-1}(j_s)}^*(dX_{w_s}) \quad (28)$$

for $\text{Shuffle}(r,s)$ being the set of shuffle permutation of the sets $\{1, \dots, r\}, \{1, \dots, s\}$ (permutations of $\{1, \dots, r+s\}$ that preserve the order within $\{1, \dots, r\}, \{1, \dots, s\}$ respectively).

Hence we have showed that for any words $\omega = i_1 \dots i_r$ and $\rho = j_1 \dots j_s$

$$\mathbf{e}_\omega^* \mathbf{e}_\rho^* = \sum_{\sigma \in \text{Shuffle}(r,s)} \mathbf{e}_{(k_{\sigma^{-1}(1)}), \dots, k_{\sigma^{-1}(r+s)}}^* \equiv \mathbf{e}_\omega^* \sqcup \mathbf{e}_\rho^* \equiv \mathbf{e}_{\omega \sqcup \rho}^* \quad (29)$$

if restricted to the range of the signature transform of $C_0^\rho(V)$. By linearity it follows for all $f \in T(V^*)$ (and induced linear forms on $T((V))^*$) that

$$\langle f, \mathbf{sig}(x) \rangle \langle g, \mathbf{sig}(x) \rangle = \langle f \sqcup g, \mathbf{sig}(x) \rangle. \quad (30)$$

Finally we can use equation (30), to show directly that $\mathcal{A}$ is closed under multiplication, hence a subalgebra of $\mathbb{R}^{\mathcal{C}_p}$. $\square$

As a corollary of Lemma 3.5 and the classical Stone–Weierstrass theorem, we obtain the following fundamental result concerning the role of signature components as an approximation system. Although it will not be used directly in the subsequent discussion, it is conceptually important.

Theorem 3.6 (Universal Approximation with Signatures). *For $p \in [1, 2)$ and let $K \subset \mathcal{C}_p$ a compact subset (w.r.t to any topology with which $\mathcal{C}_p$ is endowed). Denote by $C(K)$ the continuous maps on K with topology of uniform convergence. Then $\mathcal{A}|_K = \{\Phi_f|_K \mid f \in T((V))^*\}$ is dense in $C(K)$ i.e. $\overline{\mathcal{A}|_K}^{\|\cdot\|_\infty} = C(K)$.*

Proof. The proof is a direct replica of the previous Lemma 3.5 to show that $\mathcal{A}|_K$ is a point separating subalgebra that contains the constant functions and the conclusion is then a direct application of Stone–Weierstrass. $\square$

Recalling that $\{\mathbf{e}_\omega^*\}_\omega$ span $T(V^*)$ and recalling that the *signature coordinates* are defined as $\mathbf{sig}(x)^\omega(x) := \langle \mathbf{e}_\omega^*, \mathbf{sig}(x) \rangle$, the above states exactly that the signature coordinates, restricted to a compact subset $K \subset \mathcal{C}_p$ are dense in $C(K)$.

In light of later use, we will shortly address the bounded signature transform. To keep it more general, which is the goal of this section, we follow Chevyrev and Oberhauser [2022]. In the latter, the main goal is essentially to generalize the universality results (common way to refer to density arguments in approximation theory) to **non**-locally compact spaces⁹ through Giles Theorem.

Recall that Giles' theorem as a generalisation of Stone–Weierstrass to non-locally compact spaces. Morally speaking, Giles achieves this by letting go of the compactness (resp. local comp.) assumption of the underlying topological space $\mathcal{X}$ and instead restricting the approximation system, that is the point separating subalgebra of $C(\mathcal{X}, \mathbb{R})$ (resp. $C_0(\mathcal{X}, \mathbb{R})$)¹⁰ to be a subalgebra of **bounded** continuous functions $C_b(\mathcal{X}, \mathbb{R})$ and switching from the uniform topology to the *strict* topology¹¹. (The strict topology on $C_b(\mathcal{X}, \mathbb{R})$ is weaker than

⁹Recall that F.Riesz's Theorem states that a TVS is finite dimensional iff it is locally compact. Hence the path space we are usually interested here, being infinite dimensional, are not locally compact

¹⁰function that vanish at infinity, i.e. for $\epsilon > 0$ there exists a compact set K_ϵ such that $\sup_{x \in \mathcal{X}/K_\epsilon} |f(x)| < \epsilon$

¹¹locally convex topology induces by family of seminorms $p_\psi(f) = \sup_{x \in \mathcal{X}} |f(x)\psi(x)|$, $\psi \in C_0(\mathcal{X}, \mathbb{R})$. The so induced topology on $C_b(\mathcal{X}, \mathbb{R})$ is weaker than the uniform topology and stronger than the topology of uniform convergence on compact sets (equals the compact open topology because the codomain $(\mathbb{R}, |\cdot|)$ is a metric space)

the uniform topology i.e. $\tau_{strict} \subset \tau_{||\cdot||_\infty}$, hence $\overline{\mathcal{A}}^{||\cdot||_\infty} \subset \overline{\mathcal{A}}^{strict}$ so the switching to a weaker topology yields a weaker statement in the sense that the dense subset is smaller). This tradeoff, of a more restrictive set of approximants and a weaker topology crucially preserves Riesz-representation theorem in the sense that $C_b(\mathcal{X}, \mathbb{R})^* \cong \mathcal{M}(\mathcal{X})$, where the left-hand side is the topological dual of $C_b(\mathcal{X}, \mathbb{R})$ under the strict topology and the right-hand side is the set of finite, regular Borel measures on $\mathcal{X}$ (see Chevyrev and Oberhauser [2022] Theorem 9 (iii)).

Chevyrev and Oberhauser then use the signature map to construct a dense subset of $C_b(\mathcal{X}, \mathbb{R})$ very much like we did it for $C(\mathcal{X}, \mathbb{R})$ in Theorem 3.6. They are then faced with the fundamental issue that the signature transform, on a non compact set, is not bounded, similarly as for polynomials. Hence in order to apply Giles Theorem, they need to find a way to make the signature transform bounded, without losing the algebraic properties, in particular the point separating property of the subalgebra. (Notice that dividing every polynomial by their respective norm makes them bounded, but doesn't preserve the point separating property). The alternative proposed in Chevyrev and Oberhauser [2022] is the use of tensor normalisation yielding the bounded signature transform, which shall be shortly introduced in the following (for completeness it shall be mentioned that Chevyrev and Oberhauser [2022] doesn't just stop there but uses duality between universality of a feature map Φ and characteristicness to eventually get the characteristicness of the bounded signature transform on the dual of $C_b(\mathcal{X}, \mathbb{R})$ w.r.t. to the strict topology which happens to be the space of probability measures $\mathcal{M}^1(\mathcal{X})$).

The following results will not be proven. For proofs; see Chevyrev and Oberhauser [2022]. First we define the key object, the tensor normalisation

Definition 3.7 (Tensor Normalisation). A tensor normalisation (tN) is a **continuous** and **injective** map of the form

$$\Lambda : T((V)) \rightarrow B_{\leq R}^{T((V))} = \{\mathbf{t} \in T((V)) \mid \|\mathbf{t}\| \leq R\} \quad (31)$$

$$\mathbf{A} \rightarrow \delta_{\lambda(\mathbf{A})} \mathbf{A} = (A^0, \lambda(\mathbf{A}) \cdot A^1, \lambda(\mathbf{A})^2 \cdot A^2, \dots), \quad (32)$$

where $\lambda : T((V)) \rightarrow (0, \infty)$ is some function.

The existence of such a map is non-trivial and is shown in Chevyrev and Oberhauser [2022]. We can now define the family of bounded signature transforms

Definition 3.8. Let $\mathcal{H}^R = \mathcal{H} \cap B_{\leq R}^{T((V))}$, then

$$\mathbf{sig}_\Lambda := \Lambda \circ \mathbf{sig} : \mathcal{C}_d^1 \rightarrow \mathcal{H}^R. \quad (33)$$

For clarity, observe that this means explicitly that for $x \in \mathcal{C}_d^1$ we have $\mathbf{sig}_\Lambda(x) = \Lambda(\mathbf{sig}(x)) = \sum_{m \geq 0} \lambda^m \mathbf{sig}^m(x)$ with $\lambda^m \mathbf{sig}^m(x) = \{\lambda^{|\omega|} \mathbf{sig}^\omega(x) \mid \omega \in \langle d \rangle; |\omega| = m\}$.

Recalling that we implicitly always take the augmented signature transform, the injectivity and continuity of the signature transform from Lemma 3.4 are preserved under composition with an injective and continuous map Λ . One key feature of this definition is that some

key algebraic properties of the signature transform are conserved. In particular the set of bounded signature components is still a point separating non-vanishing algebra of **bounded** functions. We are now able to apply Giles Theorem to the latter to get a dense family of the continuous bounded functions $C_b(\mathcal{Z}, \tau_{strict})$ (a non compact $\mathcal{Z}$). We state this formally in the next proposition

Theorem 3.9 (Density of Bounded Signature Components in $C_b(\mathcal{Z})$). *Let $\Lambda = \delta_{\lambda(\cdot)}$ be a feature normalisation, $\mathcal{Z} = (\mathcal{C}_1^d, \|\cdot\|_{1-var})$ and define*

$$\mathcal{A}_\Lambda = span_{\mathbb{R}}\{\lambda^{|\omega|}\langle\omega, \mathbf{sig}(x)\rangle \mid \omega \in \langle d \rangle\} . \quad (34)$$

Then $\mathcal{A}_\Lambda$ is dense a non-vanishing point separating algebra which is dense in $C_b(\mathcal{Z})$ with respect to the strict topology τ_{strict} on $C_b(\mathcal{Z})$.

Remark. *Notice that we have the following homogeneity property*

$$\delta_\lambda(\mathbf{sig}(x)) = \mathbf{sig}(\lambda x) . \quad (35)$$

Indeed

$$\delta_\lambda(\mathbf{sig}(x)) = (1, \lambda \cdot \mathbf{sig}^1(x), \lambda^2 \cdot \mathbf{sig}^2(x), \dots) \quad (36)$$

and explicitly

$$\lambda^m \cdot \mathbf{sig}^m(x) = \lambda^m \int \underbrace{dx \otimes \dots \otimes dx}_{m \text{ times}} = \int d\lambda x \otimes \dots \otimes d\lambda x = \mathbf{sig}^m(\lambda \cdot x) . \quad (37)$$

hence we essentially bound the signature by rescaling its domain.

The final density result about signature coordinate integrals or more precisely the components of the augmented, bounded signature is its L^p -density. The case $p = 2$ is used to get the L^2 -dense family which comes up in the identification of the regression operator in (56). Hence the following theorem is crucial for what is to come.

Theorem 3.10 ($L^p(\mu)$ Density of Bounded Signature Algebra). *Let $\mathcal{A}_\Lambda = \{\lambda^{|\omega|}\langle\omega, \mathbf{sig}(\cdot)\rangle \mid \omega \in \langle d \rangle\}$ for any $\mu \in \mathcal{M}^1(\mathcal{Z})$ and Λ any feature normalisation. Then*

$$\overline{\mathcal{A}_\Lambda}^{L^p} = L^p(\mu) \quad (38)$$

for $p \in [1, \infty)$.

The above claim is showed in Schell and Alaifari [2024] as a part of the proof of Proposition 4.5. We will extract the relevant part of the proof and be more explicit on a couple of points. What we will actually show is that for a random variable X and a space $\mathcal{H} = \{\mathbf{a} \in T((V)) \mid \|\mathbf{a}\|^2 = \sum_{m \geq 0} \|\pi_m(\mathbf{a})\|^2 = \sum_{\omega \in \langle d \rangle} |\langle \omega, \mathbf{a} \rangle|^2 \leq \infty\} \subset T((V))$ we have that the components $\langle w, Z \rangle$ of $Z \in L^p(\Omega, \sigma(X), \mathcal{H})$ can be L^p -approximated with $\mathcal{A}_\Lambda(X) = \{\rho \circ X \mid \rho \in \mathcal{A}_\Lambda\}$ functions. Noticing that $\langle w, Z \rangle \in L^p(\sigma(X))$ and arguing why w.l.o.g. $f = \langle w, Z \rangle$ for any $f \in L^p(\sigma(X))$ we conclude, that $\overline{\mathcal{A}_\Lambda(X)}^{L^p} = L^p(\mathcal{X}, \mathbb{P}_X)$. To show the claim we observe that the exact proof goes through when replacing $\mathbb{P}_X$ by a probability measure μ . The core observation is that we can L^p -approximate Z by bounded truncations which turn out to be in $\overline{\mathcal{A}_\Lambda(X)}^{L^p}$. Indeed, truncate Z to be between $-n$ and n i.e. consider $Z_n := n \wedge (Z \vee -n)$

Proof. First notice that for $Z \in L^p(\Omega, \sigma(X), \mathcal{H})$ we have that $\langle w, Z \rangle \in L^p(\sigma(X))$. Indeed

$$\mathbb{E}[|\langle w, Z \rangle|^p] = \mathbb{E}[(|\langle w, Z \rangle|^2)^{\frac{p}{2}}] \leq \mathbb{E}[(\sum_{\omega \in \langle d \rangle} |\langle w, Z \rangle|^2)^{\frac{p}{2}}] \quad (39)$$

$$= \mathbb{E}[\|Z\|^p] = (\|Z\|_{L^p})^p < \infty, \quad (40)$$

hence the components of Z are indeed in $L^p(\sigma(X))$. Let's now proof through functional monotone class Theorem (see A.2) that $\overline{\mathcal{A}_\Lambda(X)}^{L^p} \cap L^\infty$ contains all $\sigma(\mathcal{A}_\Lambda(X))$ measurable functions.

First, notice that $\mathcal{A}_\Lambda(X) \subset C_b(\mathcal{X})$ (real-valued bounded functions on $\mathcal{X}$) and hence $\mathcal{A}_\Lambda(X) \subset \overline{\mathcal{A}_\Lambda(X)}^{L^p} \cap L^\infty$.

We abbreviate $H = \overline{\mathcal{A}_\Lambda(X)}^{L^p} \cap L^\infty$. Because the pointwise multiplication of essentially bounded functions is again essentially bounded and due to the algebra structure of $\mathcal{A}_\Lambda(X)$, we have that H is closed under pointwise multiplication. What is more, it contains the constant function because its subset $\mathcal{A}_\Lambda(X)$ does so. Clearly H is real vector space (intersection of two vector spaces). It remainse to check that H is closed under monotone, bounded limits. For this, let $(f_n)_{n \geq 1}$ be an increasing sequence of functions in H that converges point-wise to f that is bounded. Then $f_n \rightarrow f$ in L^p hence $f \in H$. Whence we can apply the functional monotone class Theorem to H and conclude that H contains all $\sigma(\mathcal{A}_\Lambda(X))$ -measurable functions. (lhs are exactly the essentially bounded and $\sigma(\mathcal{A}_\Lambda(X))$ -measurable functions. Observe that we have shown that $L^\infty(\sigma(\mathcal{A}_\Lambda(X))) \subset \overline{\mathcal{A}_\Lambda(X)}^{L^p} \cap L^\infty$

It can be shown that $\sigma(\mathcal{A}_\Lambda(X)) = \sigma(X)$ (see Lemma 4.3. Schell and Alaifari [2024]) thus we have $L^\infty(\sigma(X)) \subset \overline{\mathcal{A}_\Lambda(X)}^{L^p} \cap L^\infty \subset \overline{\mathcal{A}_\Lambda(X)}^{L^p}$. Now the core observation is that we can L^p -approximate $\langle w, Z \rangle$ by bounded truncations which turn out to be in $\overline{\mathcal{A}_\Lambda(X)}^{L^p}$. Indeed truncate $\langle w, Z \rangle$ to be between $-n$ and n i.e. consider $\langle w, Z \rangle_n := (\langle w, Z \rangle \wedge n) \vee -n$ which pointwise approximates $\langle w, Z \rangle$ and also in L^p due to dominated convergence. Notice that $\langle w, Z \rangle_n \in L^p(\sigma(X)) \cap L^\infty \subset \overline{\mathcal{A}_\Lambda(X)}^{L^p}$ by the above. Hence $\langle w, Z \rangle \in \overline{\mathcal{A}_\Lambda(X)}^{L^p}$. In other words, we have shown that for any $Z \in L^p(\sigma(X), \mathcal{H})$ we that its components $\langle w, Z \rangle \in L^p(\sigma(X))$ turn out to be in $\overline{\mathcal{A}_\Lambda(X)}^{L^p}$. Why does this show that $\overline{\mathcal{A}_\Lambda(X)}^{L^p} = L^p(\sigma(X))$? Given any $f \in L^p(\sigma(X))$ and any $Z \in L^p(\sigma(X), \mathcal{H})$ (exists, e.g. take for $Z : \Omega \rightarrow \mathcal{H}$, $\mathbb{E}[Z|X]$) and define for every $\delta \in \Omega$ $\tilde{Z}(\delta) = \sum_{\omega \in \langle d \rangle, \omega \neq \tilde{\omega}} \langle \omega, Z(\delta) \rangle \cdot \omega + f \cdot \tilde{\omega}$ for any $\tilde{\omega}$ i.e. $\tilde{Z} = Z - \langle \tilde{\omega}, Z \rangle \cdot \tilde{\omega} + f \cdot \tilde{\omega}$. But all the terms on the left hand side are in $L^p(\sigma(X), \mathcal{H})$ hence $\tilde{Z}$ is as well. Hence for any $f \in L^p(\sigma(X))$ apply the above to $\tilde{Z}$ to conclude that $\overline{\mathcal{A}_\Lambda(X)}^{L^p} = L^p(\sigma(X))$.

Finally, to get the result $\overline{\mathcal{A}_\Lambda}^{L^p} = L^p(\mu)$ for some probability measure μ , we replace in the above the space $L^p_X(\mathcal{H}) = L^p(\Omega, \sigma(X), \mathbb{P}; \mathcal{H}) \cong L^p(\mathcal{X}, \mathbb{P}_X)$ with $L^2(\Omega, \mathcal{F}, \mathbb{P}; \mathcal{H})$ i.e. exchanging $\mathbb{P}_X$ with μ . Then the above shows exactly what we want. $\square$

Observe that the boundedness of the bounded signature components is crucial in the above. Indeed boundedness is essential in the functional monotone convergence theorem, which is the central tool of the proposed proof.

The stated L^p -density plays a crucial role in Schell and Alaifari [2024] in particular consistency of the proposed estimator is a specific property of the bounded signature transform.

4 Nonparametric Regression with Signatures

Having introduced the signature transform and its most important properties, we will review here a recent approach to regression of path space valued random variables developed in Schell and Alaifari [2024]. The basic framework is exactly that of regression as given in Definition 7 and the task which is treated concerns the efficient estimation of the statistics $f \rightarrow \mathbb{E}[f(X)|Y]$ for X, Y mapping to entire paths. It will turn out that a signature approach, will yield a universally consistent estimator whose approximation boils down to a convex minimisation task (ridge regression) for which explicit error bounds for a given precision threshold can be given.

The first step is to understand the estimation of the statistics 7 as an operator learning problem. As pointed out in Definition 7 we understand regression as the learning of the statistics $f \rightarrow \mathbb{E}[f(Y)|X]$ which is equivalent to the learning of the conditional probabilities. Now to put this in a more formal set up, we recall that the notion of a regular version of a conditional probability is tightly linked to the concept of stochastic kernels, i.e. to maps $K : (\mathcal{X} \times \mathcal{B}(\mathcal{Y})) \rightarrow [0, 1]$ such that for every $x \in \mathcal{X}$ the map $K(x, \cdot) : \mathcal{B} \mapsto [0, 1]$ defines a Borel measure on $(\mathcal{Y}, \mathcal{B}(\mathcal{Y}))$ and for every $B \in \mathcal{B}(\mathcal{Y})$, $x \mapsto K(x, B)$ is a $\mathcal{B}(\mathcal{X})$ -measurable map. With this in hand we can state what we mean by the *disintegration* of a measure μ on the product space $\mathcal{X} \times \mathcal{Y}$.

Omitting technical details and proofs, let $\mathcal{X}, \mathcal{Y}$ be Polish and some Borel measure $\mu \in \mathcal{M}(\mathcal{X} \times \mathcal{Y})$ there exists a kernel on $(\mathcal{Y}, \mathcal{B}(\mathcal{Y}))$ which we suggestively denote as the family of measures $\mu_{\mathcal{Y}|\mathcal{X}} \equiv \{\mu_{\mathcal{Y}|x}\}_{x \in \mathcal{X}} \subset \mathcal{M}^1(\mathcal{Y})$ such that for a given marginal $\mu_{\mathcal{X}}$

$$\mu = \mu_{\mathcal{Y}|\mathcal{X}} \otimes \mu_{\mathcal{X}} , \quad (41)$$

where $\mu_{\mathcal{Y}|\mathcal{X}}$ is unique $\mu_{\mathcal{X}}$ almost surely and the above is defined for $A \in \mathcal{B}(\mathcal{X} \times \mathcal{Y})$

$$\mu(A) = \int_{\mathcal{X}} \left(\int_{\mathcal{Y}} 1_A d\mu_{\mathcal{Y}|x} \right) \mu_{\mathcal{X}} . \quad (42)$$

Considering two random variables $X : \Omega \rightarrow \mathcal{X}$, $Y : \Omega \rightarrow \mathcal{Y}$ such that $\mu = \mathbb{P} \circ (X, Y)^{-1}$ we define for $B \in \mathcal{B}(\mathcal{Y})$ the conditional probability of $Y \in B$ given $X = x$ by

$$\mathbb{P}(Y \in B | X = x) = \int_B d\mu_{\mathcal{Y}|x} \quad (43)$$

and the conditional expectation

$$\mathbb{E}[f(Y)|X] = \int_{\mathcal{Y}} f d\mu_{\mathcal{Y}|X} , \quad (44)$$

where the above now makes formal sense of the following expression

$$\mathbb{E}[f(Y)|X = x] = \int_{\mathcal{Y}} f d\mu_{\mathcal{Y}|x} . \quad (45)$$

Due to the uniqueness of disintegration, in the sense stated above, the regression map

$$T_X : \mu \rightarrow \mu_{\mathcal{Y}|X} \equiv \{\mu_{\mathcal{Y}|x} \mid x \in \mathcal{X}\} \quad (46)$$

defines a bijection when fixing a marginal distribution $\mu_{\mathcal{X}} = \xi$. We will often think of

$$\mu_{\mathcal{Y}|X} := \mu_{\mathcal{Y}|\mathcal{X}} \circ X : \begin{cases} \Omega \times \sigma(Y) \rightarrow \mathbb{R} \\ (\omega, A) \mapsto \mu_{\mathcal{Y}|X(\omega)}(A) \end{cases} \quad (47)$$

as a random measure.

According to the discussion about regression and the identification (4), we can reformulate our regression problem in our new language as the learning of the map T_X from a finite number of samples of μ . Formally that means that for i.i.d. random variables (X_i, Y_i) with same distribution as (X, Y) we want to estimate $T_X(\mu) = \{\mu_{\mathcal{Y}|x} \mid x \in \mathcal{X}\}$ by $T_X(\frac{1}{n} \sum_{i=0}^n \delta_{(X_i, Y_i)})$. For the time being we will postpone the approximation through finite samples and focus on studying the set of conditional probabilities $T_X(\mu)$.

Recall from Riesz-Kakutani that $\mathcal{M}(\mathcal{Y}) \cong C^*(\mathcal{Y}, \mathbb{R})^{12}$ for a locally compact Hausdorff space $\mathcal{Y}$. For our purposes, the study of locally compact Hausdorff spaces is not sufficient, as path spaces are notoriously not locally compact. Nevertheless the identification (injective mapping) of probability measures on $\mathcal{Y}$ with bounded functionals still holds through

$$\mu \mapsto \left(f \mapsto \int_{\mathcal{Y}} f d\mu \right) .$$

Indeed this mapping is clearly injective as can be seen by taking $f = \mathcal{I}_A$ for measurable sets $A \in \mathcal{F}_{\mathcal{Y}}$. Furthermore we have the basic identification of objects $\{f : \mathcal{X} \rightarrow \mathcal{Y}\} \equiv \mathcal{Y}^{\mathcal{X}} = \{f(x)\}_{x \in \mathcal{X}}$ and what is known as "currying" i.e. the series of identification of the mappings $\{X \times Y \rightarrow Z\} \cong \{X \rightarrow Z^Y\} \cong \{Y \rightarrow Z^X\}^{13}$ (where the isomorphisms are understood as isomorphisms of sets). We can now apply these standard identifications to

$$T_X(\mu) = \{\mu_{\mathcal{Y}|x}\}_{x \in \mathcal{X}} \equiv (x \mapsto \mu_{\mathcal{Y}|x}) \quad (48)$$

$$\equiv \left(x \mapsto \int_{\mathcal{Y}} f d\mu_{\mathcal{Y}|x} \right) \equiv \left((x, f) \mapsto \int_{\mathcal{Y}} f d\mu_{\mathcal{Y}|x} \right) \quad (49)$$

$$\equiv \left(f \mapsto \left(x \mapsto \int_{\mathcal{Y}} f d\mu_{\mathcal{Y}|x} \right) \right) . \quad (50)$$

Furthermore, we take the domain of $T_X(\mu)$ to be $L^2(\mu_{\mathcal{Y}})$. Whence we have identification of $T_X(\mu)$ as an operator

$$\hat{T}_X(\mu) \equiv T_X(\mu) : L^2(\mu_{\mathcal{Y}}) \rightarrow L^2(\xi) , \quad (51)$$

¹² $\mathcal{M}(\mathcal{Y})$ are the finite radon measures here

¹³ $[x \rightarrow (y \rightarrow f_x(y) = f(x, y))] \equiv [(x, y) \rightarrow f(x, y)] \equiv [y \rightarrow (x \rightarrow f_y(x) = f(x, y))]$

where $\mu_{\mathcal{Y}} = (\pi_2)_*\mu$ is the marginal of $\mu \in \mathcal{M}^1(\mathcal{X} \times \mathcal{Y})$ for the projection $\pi_2 : \mathcal{X} \times \mathcal{Y} \rightarrow \mathcal{Y}$. In light of the density result 3.10, the motivation and utility for this identification becomes clear in that we can use the stated L^2 -density of the bounded signature components to approximate the regression operator in a formalism that is very well adapted to path spaces. First let us check that (51) holds. Indeed for $f \in L^2(\mu_{\mathcal{Y}})$ we have

$$\|\hat{T}_X(\mu)(f)\|_{L^2(\xi)} = \int_{\mathcal{X}} \left| \int_{\mathcal{Y}} f(y) d\mu_{\mathcal{Y}|x} \right|^2 d\xi(x) \leq \int_{\mathcal{X}} \int_{\mathcal{Y}} |f(y)|^2 d\mu_{\mathcal{Y}|x}(y) d\xi(x) \quad (52)$$

$$= \int_{\mathcal{X} \times \mathcal{Y}} |f(y)|^2 d\mu(x, y) = \int_{\mathcal{X}} \int_{\mathcal{Y}} |f(y)|^2 d\mu_{\mathcal{X}|y}(x) d\mu_{\mathcal{Y}}(y) = \int_{\mathcal{Y}} |f(y)|^2 d\mu_{\mathcal{Y}}(y) \quad (53)$$

$$= \|f\|_{L^2(\mu_{\mathcal{Y}})} \leq \infty, \quad (54)$$

which simultaneously shows that $\hat{T}_X(\mu)$ defines a bounded operator with operator norm $\|\hat{T}_X(\mu)\| \leq 1$. The following observation is clearly motivated by the results on density of signature components. Nevertheless we proceed more generally (not referring to signature components yet). Suppose there exists a separable Hilbert space $(\mathcal{H}_{\mathcal{Y}}, \langle \cdot, \cdot \rangle)$ and a measurable map $\alpha : \mathcal{Y} \rightarrow \mathcal{H}_{\mathcal{Y}}$ such that the family $H_{\alpha} \equiv \{ \langle s, \alpha(\cdot) \rangle \mid \in \mathcal{H}_{\mathcal{Y}} \}$ lies dense in $L^2(\mu_{\mathcal{Y}})$ i.e. $\overline{H_{\alpha}}^{L^2} = L^2(\mu_{\mathcal{Y}})$. Also recall that $\hat{T}_X(\mu)$ is a continuous operator, hence it is uniquely determined by its restriction to a dense family e.g. on $H_{\alpha} \subset L^2(\mu_{\mathcal{Y}})$. In other words we can identify $\hat{T}_X(\mu)$ as the unique continuous extension of $\hat{T}_X(\mu)|_{H_{\alpha}} \equiv \hat{T}_X^{\alpha}(\mu)$. Now analysing the action of $\hat{T}_X^{\alpha}(\mu)$ on $f_s(\cdot) = \langle s, \alpha(\cdot) \rangle \in H_{\alpha}$

$$\hat{T}_X^{\alpha}(\mu)(f_s) = \left(x \rightarrow \langle s, \int_{\mathcal{Y}} \alpha(y) d\mu_{\mathcal{Y}|x} \rangle \right), \quad (55)$$

where we used that the Bochner integral commutes with linear forms (see comment below Definition 1.15. of van Neerven [2008]). Indeed, the operator (55) is determined by the map $\mu_{(\cdot)}(\alpha) \equiv x \rightarrow \int_{\mathcal{Y}} \alpha(y) d\mu_{\mathcal{Y}|x}$ as for a given $f \in L^2(\mu_{\mathcal{Y}})$, choosing an approximating sequence $(s_n^f)_n \subset \mathcal{H}_{\mathcal{Y}}$ of f we recover the bounded regression operator $\hat{T}_X(\mu)$

$$\hat{T}_X(\mu)(f) = \lim_{n \rightarrow \infty} \langle s_n^f, \mu_{(\cdot)}(\alpha) \rangle, \quad (56)$$

where the equality holds in $L^2(\xi)$. The main idea is now that in order to approximate the conditional probabilities or equivalently the regression operator $\hat{T}_X(\mu)$, equation (56) tells us that we might try to approximate f by its L^2 -approximating sequence $(s_n^f)_n$ and the map $\mu_{(\cdot)}(\alpha)$ separately.

For clarity, notice that by (46) we recognise

$$\mu_{(\cdot)}(g) : x \mapsto \mu_x(g) \equiv \mathbb{E}[g(Y)|X = x] \quad (57)$$

and hence

$$\mu_X(g) := \mu_{(\cdot)}(g) \circ X \equiv \mathbb{E}[g(Y)|X] \quad (58)$$

We can rewrite (56) more explicitly as

$$\hat{T}_X(\mu)(f) = \lim_{n \rightarrow \infty} \langle s_n^f, \mathbb{E}[\alpha(Y)|X = (\cdot)] \rangle. \quad (59)$$

The remaining task is to find smart choices for $\mathcal{H}_Y$ and the map $\alpha(\cdot)$ in order to construct a universally consistent estimator of the map (65). This is where the theory of signatures in the form of the results recalled in Section 3.

Using the results from Section 4, in particular about the bounded signature transform and the respective density properties of the components inherited from the ones for the 'unbounded' signature transform we can show the following core results corresponding to Theorem 4.8. and 4.10. in Schell and Alaifari [2024] which essentially state that the general approach as laid out in (65) work for the choice

$$\mu \equiv \mathbb{P}_{(X,Y)} \quad \alpha \equiv \mathbf{sig}_\Lambda, \quad (60)$$

where Λ is any feature normalisation, $\mathbf{sig}_\Lambda \equiv \Lambda \circ \mathbf{sig}$ is the (augmented) bounded signature transform.

Hence we have $\mu_{(\cdot)}(\alpha) \equiv \mathbb{E}[\mathbf{sig}_\Lambda(Y)|X]$. Whence, according to (65), the regression task has now been reduced to find approximating systems of $f \in L^2(\mathbb{P}_Y)$ and $\mathbb{E}[\mathbf{sig}_\Lambda(Y)|X]$ respectively. Let's first address the approximation of $\mathbb{E}[\mathbf{sig}_\Lambda(Y)|X]$. Observe that $\mathbb{E}[\mathbf{sig}_\Lambda(Y)|X]$ is $\sigma(X)$ -measurable and $\mathcal{H}$ -valued. Furthermore, notice that under Bochner-integrability condition for Y we have that $\mathbb{E}[\mathbf{sig}_\Lambda(Y)|X]$ takes values in $B_{\leq R}^{T((V))}$, hence bounded, hence Bochner- L^p integrable. Hence it is thus natural to consider the space $L_X^p(\mathcal{H}_Y) := L^p(\Omega, \mathbb{P}, \sigma(X); \mathcal{H}_Y)$, where we set $\mathcal{H}_Y \equiv \mathcal{H}^{14}$.

Observe that due to Doob-Dynkin $L^p(\Omega, \mathbb{P}, \sigma(X); \mathcal{H}_Y) \cong L^p(\mathcal{X}, \mathbb{P}_X; \mathcal{H}_Y)^{1516}$. Hence to find an approximating sequence for $\mathbb{E}[\mathbf{sig}_\Lambda(Y)|X]$ we are equivalently looking for a dense subset of $L_X^p(\mathcal{H}_Y)$. More precisely, in light of the identification of the regression operator (56) and more deeply due to the fact that we want to approximate a conditional expectation which is morally a L^2 -projection (see Theorem 4.2 and its proof Schell and Alaifari [2024] thm. 4.8.), we only need to look at $L_X^p(\mathcal{H}_Y)$ i.e. the case $p = 2$. The following subspace will do:

Proposition 4.1 (Approximation System of Conditional Expected Signature). *Let $\Psi_X = \{\sum_{\omega \in \langle d \rangle} \langle \alpha_\omega, \mathbf{sig}_\Lambda(\cdot) \rangle \cdot \omega\}$ with $\alpha = (\alpha_\omega)_\omega$ fulfilling the summability condition $\sum_{\omega \in \langle d \rangle} \mathbb{E}[\langle \alpha_\omega, \mathbf{sig}_\Lambda(X) \rangle^2] < \infty$. Then $\Psi_X(X) = \{\psi(X) \mid \psi \in \Psi_X\}$ is as a dense subspace of $L_X^2(\mathcal{H}_Y)$*

According to the Theorem above we have an explicit approximating sequence in the L^2 sense that is in mean square.

¹⁴we omit the dimension d of the underlying alphabet here, for notational clarity

¹⁵Indeed the map $\phi \rightarrow \phi \circ X$ is an isomorphism from $L^2(\Omega, \mathbb{P})$ to $L^2(\mathcal{X}, \mathbb{P}_X)$ and by Doob-Dynkin it is also surjective

¹⁶We denote $\mathcal{H}_X, \mathcal{H}_Y$ the space $\mathcal{H}$ but allowing the fact that the codomains of X, Y , namely the path spaces $\mathcal{X}, \mathcal{Y}$ can be based on different dimensionality d_x, d_y i.e $\mathcal{X} = C^1(\mathbb{R}^{d_x}), \mathcal{Y} = C^1(\mathbb{R}^{d_y})$ which will affect the construction of the respective Hilbert codomain space $\mathcal{H}$ through the length of the alphabets $\langle d_x \rangle, \langle d_y \rangle$

It remains to estimate $f \in L^2(\mathbb{P}_Y)$. Recall that we have shown in Theorem 3.10 that the components of the bounded signature $\mathcal{A}_\Lambda(Y) = \text{span}_{\mathbb{R}}\{\langle \omega, \mathbf{sig}_\Lambda(Y) \rangle | \omega \in \langle d \rangle\}$ are dense in $L^p(\mathbb{P}_Y)$. In particular this is the case for $p = 2$. Again, L^2 -density means that we can approximate $f \in L^2(\mathbb{P}_Y)$ in mean square by functions of $\mathcal{A}_\Lambda(Y)$.

We have all the pieces to state the core theoretical statement of this section, establishing the approximating sequences of the regression operator. What will follow afterwards are technical adjustments to make numerical implementation feasible and an explicit expression for the universal and robust estimator of the latter. Let's state the theorems

Theorem 4.2 (Approximation of Conditional Expected Signature). *Let X be a stochastic process in the usual setting. Then*

$$\mathbb{E}[\mathbf{sig}_\Lambda(Y)|X] = \lim_{k \rightarrow \infty} \psi_{a_k}(X) \text{ in } L_X^2(\mathcal{H}_Y). \quad (61)$$

For any minimizing sequence $a_k \subset S^2(X_\Lambda)$ of

$$\inf_{\alpha \in S^2(X_\Lambda)} \mathbb{E}[\|\mathbf{sig}_\Lambda(Y) - \psi_\alpha(X)\|^2]. \quad (62)$$

Theorem 4.3 (Approximation Regression Operator). *Let X, Y be any stochastic processes and let $f \in L^2(\mathbb{P}_Y)$ we have*

$$\mathbb{E}[f(Y)|X] \stackrel{L_X^2(\mathbb{R})}{=} \lim_{j \rightarrow \infty} \langle \mathbb{E}[\mathbf{sig}_\Lambda(Y)|X], g^j \rangle \quad (63)$$

for each minimizing sequence $(g^j)_{j \geq 1}$ of

$$\inf_{g \in T(\mathbb{R}^d)} \int_Y (f(y) - \langle g, \mathbf{sig}_\Lambda(y) \rangle)^2 \mathbb{P}_Y(dy). \quad (64)$$

For such a minimising sequence $(g^j)_{j \geq 1}$ for a given f as above, we get

$$\mathbb{E}[f(Y)|X] \stackrel{L_X^2(\mathbb{R})}{=} \lim_{j \rightarrow \infty} \lim_{k \rightarrow \infty} \langle \psi_{a_k}(X), g^j \rangle. \quad (65)$$

We point out straight away one nice feature of 4.3 on which we will elaborate a little bit after the presentation of the concrete Algorithm (1). In 4.3 the approximation of $f \in L^2(\mathbb{P}_Y)$ is separated from the approximation of $\mathbb{E}[\mathbf{sig}_\Lambda(Y)|X]$. This will have practical advantages.

Still there is a fundamental shortcoming with Theorem 4.3 in light of a concrete numerical implementation. Indeed, we would need to evaluate our estimates of $\mathbb{E}[\mathbf{sig}_\Lambda(Y)|X]$ pointwise, but Ψ_X does only contain $\mathbb{P}_X$ almost everywhere defined functions. Indeed The summability condition on $(\alpha_\omega)_\omega$ in 4.1 only asserts $\|\psi_\alpha(X)\|_{L^2}^2 = \int_\Omega \|\psi_\alpha(X)\|^2 \mathbb{P}(d\omega) = \int_{\mathcal{X}} \|\psi_\alpha(x)\|^2 \mathbb{P}_X(dx) < \infty$ hence $\psi_\alpha(x) < \infty$ only for $\mathbb{P}_X$ -almost all $x \in \mathcal{X}$. In other words ψ_α is only well defined for $\mathbb{P}_X$ -almost all $x \in \mathcal{X}$ hence point-wise evaluation is problematic. We only elaborated on this part because this is where the RKHS come in and they do in a fundamentally different way then for the GPR case we will see in Section 7. Indeed, for our case at hand, we will introduce a RKHS to get a dense subspace of Ψ_X for which, pointwise evaluation of maps is well defined, making it feasible to construct estimators which rely on observed data hence pointwise evaluation of the estimators. The right RKHS structure is given as follows

Proposition 4.4. *Let $(\mathcal{H}_k, \|\cdot\|_k)$ be the RKHS associated reproducing kernel $k : (x, y) \mapsto \langle \mathbf{sig}_\Lambda(x), \mathbf{sig}_\Lambda(y) \rangle$. Then the space of function*

$$\mathcal{K}_\Lambda := \{f : \mathcal{X} \rightarrow \mathcal{H}_\mathcal{Y} \mid \{\langle f, \omega \rangle \mid \omega \in \langle d \rangle\} \subset \mathcal{H}_k \text{ and } \sum_{\omega \in \langle d \rangle} \|\langle f, \omega \rangle\|_k^2 < \infty\} \quad (66)$$

is a vector-valued RKHS with reproducing kernel $(x, y) \mapsto k(x, y)\mathcal{I}_{\mathcal{H}_\mathcal{Y}}$ and $\mathcal{K}_\Lambda$ is dense in $L^2(\mathcal{X}, \mathbb{P}_X; \mathcal{H}_\mathcal{Y}) \cong L_X^2(\mathcal{H}_\mathcal{Y})$

With this final technical step, the optimization procedure suggested in Theorem 4.3 can now be treated as a usual M-estimation scheme and enhanced to a Tikhonov regularised regression scheme. We will now give a summary of the most important result of chapter 5 in Schell and Alaifari [2024], specifically for the regression task $f \rightarrow \mathbb{E}[f(Y)|X]$ (Schell and Alaifari [2024] also treats the related regression $Z \rightarrow \mathbb{E}[Z|X]$ which will be omitted here as in we are interested in the first case in light of times series forecasting).

The basic estimation scheme we want to solve is the following Tikhonov regularization (ridge regression). For estimating $\mathbf{sig}_\Lambda(Y)$ we consider the truncated signature (at level m) and look among the RKHS h_Λ . The least squares estimator of $\mathbf{sig}_\Lambda(Y)$ is hence defined as follows

$$\hat{\Upsilon}_{\mu|\lambda}^{(m)} := \arg \min_{g \in h_\Lambda^{(m)}} \phi_{\mu|\lambda}^{(m)}(g) \quad (67)$$

for $g \in L_X^2(\mathcal{H}_\mathcal{Y}^{(m)})$ where $\mathcal{H}_\mathcal{Y}^{(m)} := \pi_m[\mathcal{H}_\mathcal{Y}]$, $\phi_\mu^{(m)}(g) := \int_{\mathcal{Y} \times \mathcal{X}} \|\pi_m(\mathbf{sig}_\Lambda(y)) - g(x)\|^2 d\mu(x, y)$ ¹⁷ and

$$\phi_{\mu|\lambda}^{(m)}(g) := \phi_\mu^{(m)}(g) + \lambda \|g\|_{h_\Lambda}^2, \quad (68)$$

where now we take $g \in h_\Lambda$ the vRKHS from Proposition 4.4.

For the estimation scheme of f we have similarly

$$l_{\mu_\mathcal{Y}, \lambda}^f := \arg \min_{l \in \mathcal{H}_\mathcal{Y}} \phi_{\mu_\mathcal{Y}|\lambda}^f(l), \quad (69)$$

where $\phi_{\mu_\mathcal{Y}}^f(l) := \int_{\mathcal{Y}} |\langle l, \mathbf{sig}_\Lambda(y) \rangle - f(y)|^2 \mu_\mathcal{Y}(dy)$ and

$$\phi_{\mu_\mathcal{Y}|\lambda}^f(l) := \phi_{\mu_\mathcal{Y}}^f(l) + \lambda \|\langle l, \mathbf{sig}_\Lambda(\cdot) \rangle\|_k^2. \quad (70)$$

Notice the subtleties in the above. The estimation scheme for $\mathbf{sig}_\Lambda(Y)$ involves minimising over the product space $\mathcal{X} \times \mathcal{Y}$ while for f we only minimise over the $\mathcal{Y}$ marginal. Furthermore the first estimation is over elements of the truncated space $L_X^2(\mathcal{H}_\mathcal{Y}^{(m)})$ as we truncate signature (clearly calculating the infinite series representing the full signature is not feasible in practice). For the second part, we anyways only look at finite dimensional signature components ($l \in \langle d \rangle$ is a finite word). Furthermore, it can be shown that if the marginal $\mu_\mathcal{Y}$ is an empirical

¹⁷Here we use that $\mathbb{E}[\|f(Y) - g(X)\|] = \int_\Omega \|f \circ \pi_\mathcal{Y}(X(\omega), Y(\omega)) - g \circ \pi_\mathcal{X}(X(\omega), Y(\omega))\| d\mathbb{P}(\omega) = \int_{\mathcal{X} \times \mathcal{Y}} \|f \circ \pi_\mathcal{Y}(x, y) - g \circ \pi_\mathcal{X}(x, y)\| d(X, Y)_* \mathbb{P}(x, y) = \int_{\mathcal{X} \times \mathcal{Y}} \|f(y) - g(x)\| d\mu(x, y)$ with $\mu = (X, Y)_* \mathbb{P}$ the push-forward measure under the map (X, Y)

measures (the ones we actually work with in practice with finite data) and the training points are pairwise distinct, the M-scheme estimation of f can also be reduced to learning on a RKHS for the kernel $k(x, y) \rightarrow \langle \mathbf{sig}(x), \mathbf{sig}(y) \rangle$. Hence both estimation schemes are of the Tikhonov form, and hence the representer theorem applies yielding explicit expressions for the minimisers $\hat{\Upsilon}_{\mu|\lambda}^{(m)}, l_{\mu_Y, \lambda}^f$ in this case (i.e. considering empirical marginal measure) given by the normal equations for ridge regression.

In our setting, we consider data $D_Y = \{Y^i \mid i = 1, \dots, n\}$ and $D_{(X,Y)} = \{(X, Y)^i \mid i = 1, \dots, N\}$ assumed to be realisations of i.i.d. copies of the random variables Y and (X, Y) , respectively. Furthermore, in particular in the case of time-series forecasting, it is interesting to generalize to a multidimensional setting where $f \equiv (f_j)_{j=1}^k$ and $\mathbb{R}^k$ -valued function¹⁸. The normal equations take the usual form

$$(A_{D_Y}^T A_{D_Y} + n\lambda_1 A_{D_Y})\alpha = A_{D_Y} C_Y \quad (71)$$

$$(A_{D_{(X,Y)}}^T A_{D_{(X,Y)}} + N\lambda_2 A_{D_{(X,Y)}})\beta = A_{D_{(X,Y)}} B_Y^f, \quad (72)$$

where then coefficients $\alpha \in \mathbb{R}^{n \times k}, \beta \in \mathbb{R}^{N \times m_d}$ ¹⁹ are the unknowns and

$$A_{D_Y} = (\langle \mathbf{sig}_\Lambda(Y^{(i)}), \mathbf{sig}_\Lambda(Y^{(j)}) \rangle)_{i,j=1}^n \quad (73)$$

$$A_{D_{(X,Y)}} = (\langle \mathbf{sig}_\Lambda(X^{(k)}), \mathbf{sig}_\Lambda(X^{(l)}) \rangle)_{k,l=1}^N \quad (74)$$

$$B_Y^f = (f_j(y_i))_{(i,j) \in (n \times k)} \quad (75)$$

$$C_Y = \langle \mathbf{sig}_\Lambda(y_i), \text{lex}^{-1}(j) \rangle_{(i,j) \in N \times m_d}. \quad (76)$$

We want to stress the well known fact that the signature kernels, seemingly terrifying to compute, can be efficiently computed as solution to the Goursat PDE (see for instance Salvi and Cass's lecture notes Remark 2.1.6. Cass and Salvi [2024]). By the representer theorem we get the explicit expression for the estimators as finite sums of the kernel function (in our case these are the signature kernels $k(x, y) = \langle \mathbf{sig}_\Lambda(x), \mathbf{sig}_\Lambda(y) \rangle$ and vector valued signature kernel $\kappa(x, y) = k(x, y)\mathcal{I}_{\mathcal{H}_Y}$ as motivated by Proposition 4.4)

$$\arg \min_{l \in \mathcal{H}_Y} \phi_{\mu_Y|\lambda}^f(l) = \sum_{j=1}^n a_j \mathbf{sig}_\Lambda(Y^j). \quad (77)$$

Here we skip the technical details and without further ado give the expression for the estimator:

$$\mathfrak{R}(n, m, N) = \sum_{i=1}^k \left(\sum_{j=1}^n \alpha_{ij}(\lambda_1) \langle \pi_m(\mathbf{sig}_\Lambda(Y^{(j)})), \hat{\Upsilon}_{N, \lambda_Y}^{\beta_{\lambda_Y}, (m)}(\cdot) \rangle_2 \right) e_i, \quad (78)$$

¹⁸Indeed consider the forecasting at the time points $(t_I = (t_{i_1}, \dots, t_{i_k}))$ we naturally consider the mapping $f = (\pi_{t_1}, \dots, \pi_{t_k})$ for $\pi_t(X(\omega)) = X_t(\omega)$ as usual and the time points in $t_{i_j} > t_0$, where t_0 is some observation time

¹⁹where we define $m_d = \sum_{i=0}^m d^i = \frac{d^{m+1}-1}{d-1}$ is the dimensionality of the truncated tensor algebra of level m over an alphabet of size d , n and N are the sizes of the data D_Y and $D_{(X,Y)}$ respectively

²⁰here lex corresponds to the lexicographic ordering of the words $l \in \langle d \rangle$ hence $\langle \mathbf{sig}_\Lambda(y_i), \text{lex}^{-1}(j) \rangle$ is the lexicographic linearisation (in the sense of `numpy.flatten()` of the signature components)

where $f = (f_1, \dots, f_k) \in \mathcal{L}^2(\mathbb{P}_Y, \mathbb{R}^k)$, n the dimensionality of the data, $\langle \cdot, \cdot \rangle_2$ the Euclidean norm on $\mathbb{R}^{m_d}$.

$$\hat{\Upsilon}_{N, \lambda_2}^{\beta_{\lambda_2}, (m)}(\cdot) = \sum_{k=1}^{m_d} \left(\sum_{l=1}^N \beta_{kl}(\lambda_2) \langle \mathbf{sig}_{\Lambda}(X^l), \mathbf{sig}_{\Lambda}(\cdot) \rangle \right) e_k, \quad (79)$$

where $(e_i)_{i=1}^k$ is the standard basis for $\mathbb{R}^k$ and the coefficients $\alpha(\lambda_1)$ and $\beta(\lambda_2)$ solve the normal equations (71) depending on the regularisation parameters λ_1, λ_2 . Eventually we display a concrete algorithm for the estimation of the regression operator $f \mapsto T_X(\mu)(f)$.

Algorithm 1 Computing the regression Operator $f \mapsto \mu.(f)$ via Signatures (Algorithm I)

Goal: For processes $X \in \mathbb{R}^{d_X}$ and $Y \in \mathbb{R}^{d_Y}$ compute the mapping

$$L^2(P_Y; \mathbb{R}^k) \ni f \longmapsto \psi_f \in L^2(P_X; \mathbb{R}^k), \quad \psi_f := \mu.(f) \text{ s.t. } \psi_f(X) = \mathbb{E}[f(Y) \mid X].$$

1. Input: Samples $\{(x_j, y_j)\}_{j \in [N]}$ and $\{(y_\ell, f(y_\ell))\}_{\ell \in [n]}$ of the associations (X, Y) and $(Y, f(Y))$, respectively, for $f = (f_1, \dots, f_k) \in L^2(P_Y; \mathbb{R}^k)$.

1: *Hyperparameters:* tensor normalisations (Ξ, Λ) , resolution parameters $\lambda_f, \lambda_y, m > 0$.

2. Compute the weights $\beta_{\lambda_y} = (\beta_{j\nu}^{(\lambda_y)}) \in \mathbb{R}^{N \times m_{d_Y}}$ and $\alpha_{\lambda_f}^f = (\alpha_{li}^{(\lambda_f)}) \in \mathbb{R}^{n \times k}$ by solving

$$\begin{aligned} (A_X^\top A_X + N\lambda_y A_X) \beta_{\lambda_y} &= A_X C_Y, \\ (A_Y^\top A_Y + n\lambda_f A_Y) \alpha_{\lambda_f}^f &= A_Y B_Y^f, \end{aligned} \tag{110}$$

with $m_{d_Y} := (d_Y^{m+1} - 1)/(d_Y - 1)$ and the data-dependent matrices

$$\begin{aligned} A_X &= (\langle \text{sig}_\Xi(x_i), \text{sig}_\Xi(x_j) \rangle)_{i,j \in [N]}, & C_Y &= (\langle \text{sig}_\Lambda(y_i), \text{lex}^{-1}(j) \rangle)_{\substack{i \in [N] \\ j \in [m_{d_Y}]}} \\ A_Y &= (\langle \text{sig}_\Lambda(y_i), \text{sig}_\Lambda(y_j) \rangle)_{i,j \in [n]}, & B_Y^f &= (f_j(y_i))_{\substack{i \in [n] \\ j \in [k]}}. \end{aligned}$$

Here $\text{lex}: [d_Y]^* \rightarrow \mathbb{N}_0$ is the short-lex ordering of words in $[d_Y]^*$.

3. Using the weights β_{λ_y} and the standard basis $(\tilde{e}_\nu)$ of $\mathbb{R}^{m_{d_Y}}$, define

$$\hat{\Upsilon}_{N, \lambda_y}^{\beta_{\lambda_y}, (m)} := \sum_{\nu=1}^{m_{d_Y}} \left[\sum_{j=1}^N \beta_{j\nu}^{(\lambda_y)} \langle \text{sig}_\Xi(x_j), \text{sig}_\Xi(\cdot) \rangle \right] \tilde{e}_\nu. \tag{111}$$

4. With the weights $\alpha_{\lambda_f}^f$ and the quantity $\hat{\Upsilon}_{N, \lambda_y}^{\beta_{\lambda_y}, (m)}$, set

$$\hat{\psi}_f := \sum_{i=1}^k \left[\sum_{\ell=1}^n \alpha_{\ell i}^{(\lambda_f)} \langle \pi_{[m]}^{\text{lex}}(\text{sig}_\Lambda(y_\ell)), \hat{\Upsilon}_{N, \lambda_y}^{\beta_{\lambda_y}, (m)}(\cdot) \rangle \right] e_i, \tag{112}$$

for the standard basis (e_i) of $\mathbb{R}^k$ and with $\pi_{[m]}^{\text{lex}}: H_{d_Y} \rightarrow \mathbb{R}^{m_{d_Y}}$, $(q_w) \mapsto \sum_{\nu=1}^{m_{d_Y}} q_{\text{lex}^{-1}(\nu)} \tilde{e}_\nu$.

5. Output: $\hat{\psi}_f$ (estimator of the regression function ψ_f).

This is a good point to come back to the comment made below 4.3. For this, recall that we differentiate between two types of data: the samples from the random variable (X, Y) and the samples from the random variable $(Y, f(Y))$. The former are generally more costly in practice as they correspond to running the whole experiment for an input $X = x$ and getting an output $Y = y$. The latter only needs the samples from Y for which the experiment doesn't have to be done again. As pointed out in the comment below 4.3, the regression operator (65) splits the regression problem in the independent approximation of f and $\mathbb{E}[\text{sig}_\Lambda(Y)|X]$. Hence the algorithm allows to use the costly data to solve the normal equations to calculate β_{λ_y} and then $\hat{Y}_{N, \lambda_y, m}^{\beta_{\lambda_y}}$ by itself. This can be done once and for all. The estimation of $f \in \mathcal{L}^2(\mathbb{P}_Y)$ can then be done separately using only the "cheap" data of the form $(y, f(y))$, which essentially reuse the costly data. Hence the estimation for different f can be done separately, hence not requiring the generation of new samples (x, y) for every new f .

For completeness we state the culminating result of Schell and Alaifari [2024], the consistency result for the estimator $\hat{\psi}_f$ of $\mathbb{E}[f(Y)|X]$ for a well chosen parameter regime.

Proposition 4.5 (Consistency of Estimator). *For all $f \in \mathcal{L}^2(\mathbb{P}_Y)$, accuracy $\epsilon > 0$ and any $\delta > 0$ we have*

$$\sup_{\theta \in \mathcal{A}(\epsilon, \delta)} \mathbb{P}(\|\mu_{(\cdot)}(f) - \hat{\psi}_f^\theta\|_{L^2(\mathbb{P}_X)} \geq \epsilon) \leq \delta, \quad (80)$$

where $\mathcal{A}(\epsilon, \delta) = \{(\lambda_1, \lambda_2), n, N, m \mid \lambda_1 \leq \underbrace{\lambda_f^I(\epsilon)}_{(a)}, n \geq \underbrace{n_{f, \lambda_1}(\epsilon, \delta)}_{(b)}, m \geq \underbrace{m_{f, \lambda_1}(\epsilon, D_Y)}_{(c)},$
 $\lambda_2 \leq \underbrace{\lambda_{\lambda_1, m}^{II}(\epsilon, D_Y)}_{(d)}, N \geq \underbrace{N_{\lambda_1, \lambda_2, m}(\epsilon, \delta, D_Y)}_{(e)}\}$

We refer again to Schell and Alaifari [2024] for the proof and definition of the auxiliary parameters and their respective threshold functions (a)-(e).

Proposition 4.5 then asserts universal consistency of the estimator (i.e. convergence in probability to the regression operator, for every $f \in \mathcal{L}^2(\mathbb{P}_Y)$ as defined in Proposition 2.6. of Schell and Alaifari [2024])

5 Forecasting

This section tries to make the connection between time series prediction and the theory developed in Schell and Alaifari [2024] as laid out in Section 4. In order to do that, we first need to be explicit about how time-series forecasting (TSF) is understood in this context and elaborate on its mathematical formulation.

In words, the goal of time-series forecasting is, having knowledge about the past of a process up to a finite observation time, we want give an *optimal* prediction of the process at a time in the future, where the future is understood as everything after the observation time. What will give rise to different approaches is in particular what one defines to be optimal or the right measure for optimality. We consider one natural choice for optimality namely

in the mean squared sense, that is optimality is measured in L^2 -norm. We will adopt this classic choice to formulate a first version of TSF.

Definition 5.1 (Pointwise MSE Time-Series Forecasting). Let $X \equiv (X_t)_{t \geq 0}$ be a square integrable stochastic process on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ i.e. $X_t \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ for all $t \geq 0$. Let $t_0 > 0$ be an observation horizon that is the time up to which the agent observes the stochastic process. Then

$$\sigma(X_s | s \leq t_0) \equiv \mathcal{F}_{t_0}^X \quad (81)$$

models the information the agent has about the stochastic process up to that observation horizon. For $h > 0$ the agent wants to find the *optimal* predictor $\hat{X}_{t_0+h}$. We define

$$\hat{X}_{t_0+h} = \arg \min_{Y \in L^2(\mathcal{F}_{t_0}^X)} \|X_{t_0+h} - Y\|_{L^2},^{21} \quad (82)$$

where one looks for the optimal forecast among quantities that are measurable to us (observer) up to the observation horizon t_0 . Hence the above, models optimality in the mean square sense (MSE).

The following theorem gives a link between the regression and TSF. Informally it states that the argmin appearing in equation (82) can be given explicitly if one restricts to square integrable forecasts. More precisely, in order to get an explicit expression for $\hat{X}_{t_0+h}$ we restrict the *argmin* in (82) to the subspace $H_{t_0} := L^2(\Omega, \mathcal{F}_{t_0}^X, \mathbb{P})$. Notice that H_{t_0} is closed linear subspace of $L^2(\Omega, \mathcal{F}, \mathbb{P})$, hence a Hilbert space in its own right. In particular it is complete, hence the infimum i.e. the predictor is in H_{t_0} . A short proof of this can be found in the appendix A.4. The restriction to H_{t_0} simplifies the task a lot, as we have now effectively reduced to a Hilbert space projection problem as is shown in Theorem 5.2.

Theorem 5.2 (Mean Square Optimal Predictor). *Let X be a square integrable stochastic process and $t, h \geq 0$, then we have that*

$$\|\hat{X}_{t+h} - X_{t+h}\|_{L^2} = \inf_{Y \in \mathcal{F}_t^X} \|\hat{X}_{t+h} - Y\|_{L^2} \quad (83)$$

$$\iff \quad (84)$$

$$\hat{X}_{t+h} = \mathbb{E}[X_{t+h} | \mathcal{F}_t^X]. \quad (85)$$

Proof. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be the underlying probability space. Recall from fundamental Hilbert space theory that for a closed linear subspace H_t of a Hilbert space L^2 and for any $X_{t+h} \in L^2(\Omega)$, there exists a unique random variable $\hat{X}_{t+h}$ such that $\|\hat{X}_{t+h} - X_{t+h}\|_{L^2(\Omega)} = \inf_{Y \in H_t} \|Y - X_{t+h}\|$ and $\hat{X}_{t+h}$ is characterised by the condition $\hat{X}_{t+h} - X_{t+h} \in H_t^\perp$. But $\forall Z \in H_t$: $\mathbb{E}[Z\mathbb{E}[X_{t+h} | \mathcal{F}_t^X]] = \mathbb{E}[ZX_{t+h}]$ hence $X_{t+h} - \mathbb{E}[X_{t+h} | \mathcal{F}_t^X] \in H_t^\perp$. It thus follows from uniqueness of the Hilbert space projection that $\hat{X}_{t+h} = \mathbb{E}[X_{t+h} | \mathcal{F}_t^X]$. $\square$

²¹Notice that we can rightly write a minimum instead of an infimum as the existence of the latter is guaranteed by the Hilbert projection theorem for the Hilbert subspace $L^2(\mathcal{F}_{t_0}^X) \in L^2(\mathcal{F})$

Notice that in the proof above, we were very explicit. One could have simply noticed that due to measurability of X we have that $\mathcal{F}_t^X \subset \mathcal{F}$ is a sub- σ algebra hence $L^2(\Omega, \mathcal{F}_t^X, \mathbb{P}) \subset L^2(\Omega, \mathcal{F}, \mathbb{P})$ is a closed subspace. Then recall that $\mathbb{E}[X_{t+h}|\mathcal{F}_t]$ is by definition the optimal projection, where optimality is understood in the sense that it minimises the L^2 -norm. Hence from that point of view, Theorem 5.2 follows from the defining property of conditional expectation as the L^2 projection on the closed subspace $L^2(\Omega, \mathcal{F}_t^X, \mathbb{P})$.

Observe that estimating the conditional expectation in Theorem 5.2 is a special case of the regression problem, as we defined it in definition 7 as the learning problem of the statistics $f \mapsto \mathbb{E}[f(Y)|X]$.

Indeed, consider a fixed observation horizon t_0 , take as the regressand the whole stochastic process X (process whose future is to predict) and as regressor the process stopped at the observation time, $X^{t_0} = (X_{t \wedge t_0})_t$. Choose $f \equiv \pi_{t_0+h}$ for a pointwise prediction at time $t_0 + h$ for some $h > 0$.

First, notice that the map $\pi_{t_0+h} : \mathcal{Y} \rightarrow \mathbb{R}^d$ is a valid choice for f in the light of Theorem 4.3 if we additionally assume $\pi_{t_0+h}(Y) = Y_{t_0+h} \in L^2(\Omega, \mathcal{F}, \mathbb{P})$. Indeed, from Theorem 4.3 we require $f \in L^2(\mathbb{P}_Y)$, hence

$$\|\pi_{t_0+h}\|_{L^2(\mathcal{Y}, \sigma(Y), \mathbb{P}_Y)}^2 = \int_{\mathcal{Y}} |\pi_{t_0+h}(y)|^2 \mathbb{P}_Y(dy) = \int_{\Omega} |\pi_{t_0+h}(Y(\omega))|^2 \mathbb{P}(d\omega) \quad (86)$$

$$= \int_{\Omega} |Y_{t_0+h}(\omega)|^2 \mathbb{P}(d\omega) = \|Y_{t_0+h}\|_{L^2(\Omega, \mathcal{F}, \mathbb{P})}^2 < \infty. \quad (87)$$

Recall that morally speaking, we understood TSF as the task of learning the σ -algebra, $\sigma(X) = \sigma(X^{-1}(B)|B \in \mathcal{B}(C(I; \mathbb{R}^d)))$ from $\sigma_{t_0}(X) := \sigma(X_t | 0 \leq t \leq t_0)$ for an observation horizon t_0 . If we want to understand TSF as a regression problem for regressand X and regressor X^{t_0} we need to show that the σ algebras $\sigma(X^{t_0}) = \sigma((X^{t_0})^{-1}(B)|B \in \mathcal{B}(C(I; \mathbb{R}^d)))$ and $\sigma_{t_0}(X)$ coincide. Indeed if this wasn't the case, the translation of the stated regression problem to a TSF problem would refer to different prior information modelled by the different σ -algebras. Fortunately this is not the case. This is exactly the content of Lemma 3.5. in Schell and Alaifari [2024]. We restate the Lemma below and due to its conceptual importance in the link between regression and TSF, we give the prove in some detail. For that, first recall that we define a stochastic process $S = (S_t)_t$ to be a Borel measurable map $S : (\Omega, \mathcal{F}) \rightarrow (C(I, \mathbb{R}^d), \mathcal{B}(C(I, \mathbb{R}^d)))$, where the Borel σ -algebra is taken w.r.t. to the uniform topology on $C(I, \mathbb{R}^d)$, for $d \in \mathbb{N}$ and $I \subset \mathbb{R}$ some real interval.

Lemma 5.3. (*Prediction*) *Let $(S_t)_{t \in I}$ be a stochastic process and $t_0 \in I$ and let $S^{t_0} = (S_{t \wedge t_0})_{t \in I}$. Then $\sigma(S^{t_0}) = \sigma_{t_0}(S) := \sigma(S_t | t \leq t_0)$.*

Proof. We show the claim as a corollary of problem 4.2. of Karatzas and Shreve [2019] (page 60). It states that

1. $\mathcal{B}(C[0, \infty)) = \sigma(\mathcal{C}) =: \mathcal{G}$ for $\mathcal{C}$ being the cylinder sets i.e. sets of the form $\{\omega \in C[0, \infty) : (\omega(t_1), \dots, \omega(t_n)) \in A, n \geq 1, A \in \mathcal{B}(\mathbb{R}^n)\}$ for $n \in \mathbb{N}$, $A \in \mathcal{B}(\mathbb{R}^n)$ and $C[0, \infty)$ equipped with the topology induced by the metric $d(\omega_1, \omega_2) = \sum_{i=1}^{\infty} \frac{1}{2^i} \max_{0 \leq t \leq i} (|\omega_1(t) - \omega_2(t)| \wedge 1)$ (uniform on compact sets)

2. $\mathcal{G}_t = \phi_t^{-1}(\mathcal{B}(C[0, \infty)))$ for $(\phi_t \omega)(s) = \omega(t \wedge s)$ where $\mathcal{G}_t$ the σ -field generated by the cylinder sets with $(t_1, \dots, t_n) \in [0, t]$. We denote the set of these cylinder sets (restricted to $[0, t]$) by $\mathcal{C}_t$.

The first claim states that the Borel σ -algebra for the open sets of topology of uniform convergence on compact sets coincides with the cylinder σ -algebra (the one relevant for the construction of the Wiener measure). We show the proof of (2.) by assuming (1.) to be shown, as only (2.) is directly relevant for the proof of the lemma.

Indeed, we show (2.) by the double inclusion $\mathcal{C}_t = \phi_t^{-1}(\mathcal{C})$. First, notice that for cylinder set $C = \{\omega \in C[0, \infty) : (\omega(t_1), \dots, \omega(t_n)) \in A, n \geq 1, A \in \mathcal{B}(\mathbb{R}^n)\}$ we have

$$\phi_t^{-1}(C) = \{\omega \in C[0, \infty) : (\omega(t_1 \wedge t), \dots, \omega(t_n \wedge t)) \in A, n \geq 1, A \in \mathcal{B}(\mathbb{R}^n)\} \in \mathcal{G}_t. \quad (88)$$

Hence $\phi_t^{-1}(\mathcal{C}) \subset \mathcal{C}_t$ as preimages behave well under countable unions and intersection, $\mathcal{G}_t$ is stable under these operations as a σ -field and by (1.) the cylinder sets $\mathcal{C}$ generate $\mathcal{B}(C[0, \infty))$. For the reverse inclusion notice that for $C \in \mathcal{C}_t$ there exist $t_1, \dots, t_n \in [0, t]$ and $A \in \mathcal{B}(\mathbb{R}^n)$ such that

$$C_t = \{\omega \in C[0, \infty) : (\omega(t_1) \dots \omega(t_n)) \in A\} = \quad (89)$$

$$\{\omega \in C[0, \infty) : (\omega(t_1 \wedge t) \dots \omega(t_n \wedge t)) \in A\} = \phi_t^{-1}(C) \in \phi_t^{-1}(\mathcal{C}) \quad (90)$$

as seen before. Hence $\mathcal{C}_t \subset \phi_t^{-1}(C)$ and therefore $\mathcal{C}_t = \phi_t^{-1}(C)$. The claim follows by claim (1.). Indeed, using that preimages commute with countable unions and intersections we get from $\mathcal{C}_t \subset \phi_t^{-1}(C)$ that $\mathcal{G}_t = \sigma(\mathcal{C}_t) = \sigma(\phi_t^{-1}(\mathcal{C})) = \phi_t^{-1}(\sigma(\mathcal{C})) = \phi_t^{-1}(\mathcal{B}(C[0, \infty)))$ where the last inequality follows from part (1.).

Now back to the proof of 5.3. It follows from (2.), upon observing that $S^{t_0} = \phi_{t_0} \circ S$

$$\begin{aligned} \sigma(S^{t_0}) &= \sigma((S^{t_0})^{-1}(B) | B \in \mathcal{B}(C[0, \infty))) \\ &= \sigma((S^{-1} \circ \phi_{t_0}^{-1})(B) | B \in \mathcal{B}(C[0, \infty))) \\ &= \sigma(S^{-1}(A) | A \in \phi_{t_0}^{-1}(B), B \in \mathcal{B}(C[0, \infty))) \\ &\stackrel{(2.)}{=} \sigma(S^{-1}(A) | A \in \mathcal{G}_t) = \sigma(S_s | s \leq t) = \sigma_{t_0}(S). \end{aligned}$$

□

Next, we generalize our TSF framework from learning a point estimator to learning the whole conditional distribution of the predicted value given an observation.

Definition 5.4 (Probabilistic Time-Series Forecasting). Let $X \equiv (X_t)_{t \geq 0}$ be a stochastic process on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and $t_0 > 0$ be an observation horizon that is the time up to which the agent observes the stochastic process. Then

$$\sigma(X_s | s \leq t_0) \equiv \mathcal{F}_{t_0}^X \quad (91)$$

models the information the agent has about the stochastic process up to that observation horizon. For $h > 0$ the agent wants to learn the whole conditional distribution of the random

variable of interest, i.e. the distribution of X_{t_0+h} given the information he observed up to t_0 . That is, he wants to learn

$$\mathbb{P}(X_{t_0+h}|X^{t_0}) : \mathcal{F}_{t_0}^X \ni A \rightarrow \mathbb{P}(X_{t_0+h} \in A|X^{t_0}) = \mathbb{E}[\mathcal{I}_A(X_{t_0+h})|X^{t_0}] := \mu_{X^{t_0}}(A),^{22} \quad (92)$$

where $\mu_{X^{t_0}}$ denotes a regular version of disintegration of $\mu = \mathbb{P}_{(X_{t_0+h}, X^{t_0})}$ w.r.t. to the marginal law of X^{t_0} , i.e. $\mu_{X^{t_0}}(A) = (x \rightarrow \mu_x(A)) \circ X^{t_0} : \omega \rightarrow \mu_{X^{t_0}(\omega)}(A)$ for every $A \in \mathcal{F}_{t_0}^X$. In other words, we understand it as a random measure/stochastic kernel

$$\mathbb{P}(X_{t_0+h}|X^{t_0}) : \begin{cases} \Omega \times \mathcal{F}_{t_0}^X \rightarrow [0, 1] \\ (\omega, A) \rightarrow \mu_{X^{t_0}(\omega)}(A) \end{cases} \quad (93)$$

exactly as we did in (47).

Definition 5.4 thus states, that TSF is the learning of the random map $f \mapsto (\mathbb{E}[f(X_{t+h})|X^t = x])_{x \in \mathbb{R}}$ for $f \in \{1_A | A \in \mathcal{B}(\mathbb{R})\}$. To make the connection with the framework of Section 4, in particular with Theorem 4.3 notice $f(Y_{T+h}) = (f \circ \pi_{T+h})(Y)$, but for $f = 1_A$ for $A \in \mathcal{B}(\mathbb{R})$ we clearly have $\int_{\mathcal{Y}} |(1_A \pi_{T+h})(y)|^2 \mathbb{P}_Y(dy) = \int_{\Omega} |(1_A Y_{T+h}(\omega))|^2 \mathbb{P}(d\omega) \leq \int_{\Omega} \mathbb{P}(d\omega) = 1$ hence we notice that learning the map $A \mapsto \mathbb{P}(X_{t+h} \in A|X^t)$ is equivalent to learning the map $f \mapsto \mathbb{E}[f(Y)|X^t]$ for $f = 1_A \circ \pi_{t+h} \in L^2(\mathbb{P}_Y)$ hence exactly in the setting of Theorem 4.3. Also notice that the argument above trivially generalizes to tuples of prediction times $t_I = (t_{i_1}, \dots, t_{i_k})$ and the corresponding projections π_{t_I} .

We finally state the general definition of TSF which shall be used in the following when TSF is referenced.

Definition 5.5 (Time-Series Forecasting (TSF)). Let $(X_t)_{t \geq 0} : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (\mathcal{X}, \mathcal{F}_{\mathcal{X}})$ be a stochastic process indexed by time $t \in [0, T]$ and $\mathcal{X}$ a Polish space. Let $I = (i_1, \dots, i_k) \in [0, T]^k$, for some $k \in \mathbb{N}$ be a finite tuple and $X_{t_I} = (X_{t_{i_1}}, \dots, X_{t_{i_k}}) \in \mathcal{X}^k$. Then, for a time $t_0 \in [0, T]$ assume knowledge of the past of X up to time t_0 , that is $\mathcal{F}_{t_0}^X := (X_t | t \leq t_0)$. TSF is the estimate of the stochastic kernel

$$\mathbb{P}(X_{t_I}|X^{t_0}) : \begin{cases} \Omega \times \mathcal{F}_{t_0}^X \rightarrow [0, 1] \\ (\omega, A) \rightarrow \mu_{X^{t_0}(\omega)}(A) \end{cases} \quad (94)$$

for $A \in \sigma(X_{t_I}) = \sigma(X_{t_{i_1}}, \dots, X_{t_{i_k}})$, where $\mu_{X^{t_0}}(\cdot) = (x \rightarrow \mu_{\mathcal{X}=\mathbb{R}|x}) \circ X^{t_0}$ for $(x \rightarrow \mu_{\mathcal{X}=\mathbb{R}|x}) = T_X(\mu)$ the disintegration of the joint law $\mu = \mathbb{P}_{(X^{t_0}, X_{t_I})}$.²³

²²We want to avoid the notation $\mathbb{E}[\mathcal{I}_A(X_{t_0+h})|X^{t_0}(\omega)]$, eventhough we could make sense of it as $\mu_{X^{t_0}(\omega)}(A) = \mathbb{P}(X_{t_0+h} \in A|X^{t_0}(\omega)) = \mathbb{E}[\mathcal{I}_A(X_{t_0+h})|X^{t_0}(\omega)]$ but the latter has to be understood as $\int_A d\mu_{X^{t_0}}$ and should no be interpreted with taking the conditional expectation w.r.t. to $X^{t_0}(\omega) \in \mathbb{R}$ which doesn't make sense (what is the σ -field w.r.t. to which the conditional expectation is taken with? The trivial one? Then $\mathbb{E}[Y|X(\omega)] = \mathbb{E}[Y|X(\omega')] \forall \omega, \omega' \in \Omega$

²³Assuming $\mathcal{X}$ to be Polish, in particular separable, $\mathbb{P}_{(X^{t_0}, X_{t_I})}$ is a Borel measure on $(\mathcal{X} \times C([0, \infty); \mathcal{X}), \mathcal{B}(\mathcal{X} \otimes C([0, \infty); \mathcal{X}))$ as $C([0, \infty); \mathcal{X})$ equipped with the compact-open topology (induced by the metric given in the proof of Lemma 5.3 is a complete and separable space)

Up to now, we have treated signature based regression, by studying deeply identifications of the regression map (46) as an operator, such that signature based learning could be leveraged to estimate the regression operator from finite data. Next, we will dedicate a little bit of time to introducing another approach on regression, namely Gaussian process regression. For this, we briefly introduce two central notions: Kernel reproducing Hilbert spaces as the key structural concept and Bayesian inference as the key operational concept.

6 Gaussian Process Regression

In this subsection we will give an alternative approach on regression through the lense of the Bayesian paradigm, with Gaussian priors eventually corresponding to Gaussian process regression (GPR). GPR will turn out to be a functional viewpoint on the Bayesian inference which nevertheless is structurally advantageous due to its kernel representation. Recalling our first general discussion about regression in Section 2, a standard initial approach in the literature is to frame regression as the learning of the weights of a *quasi functional* relation between data $\{x_i, y_i\}_{i=1}^n$ from (X, Y) with $x_i \in \mathcal{X}$, $y_i \in \mathcal{Y}$, i.e.

$$y_i = f(x_i) + \epsilon_i \quad (95)$$

for some function $f : \mathcal{X} \rightarrow \mathcal{Y}$, where we consider noisy observation y_i perturbed by Gaussian noise $\epsilon_i | X \sim \mathcal{N}(0, \sigma^2)$. The standard approach is to reduce the general problem of (95) by parametrizing the function f so to restrict the class of possible functions and effectively reduce to a linear regression problem²⁴. Alternatively, a mathematically fruitful approach in learning theory is to restrict to a so-called reproducing kernel Hilbert space (RKHS). The usefulness of this restriction is then justified by the powerful representer theorem. In order to give a self-contained view of what is to follow, we give a short introduction to RKHS. The following is far away from being an exhaustive exposition and doesn't aspire to be one. We proceed with the definition of an RKHS.

Definition 6.1 (RKHS). Let $\mathcal{H}_k$ be a Hilbert space of real-valued functions on $\mathcal{X}$ with inner-product $\langle \cdot, \cdot \rangle$. Then $\mathcal{H}_k$ is a reproducing kernel Hilbert space (RKHS) if the evaluation functionals

$$J_x : \begin{cases} \mathcal{H}_k \rightarrow \mathbb{R} \\ f \rightarrow f(x) \end{cases} \quad (96)$$

are each continuous, w.r.t. to the norm-topology induced by $\langle \cdot, \cdot \rangle$.

Each map J_x thus defines a continuous linear functional on the Hilbert space $\mathcal{H}_k$ so by the Riesz-Fischer Theorem, there exists a (unique) support vector $k_x \in \mathcal{H}_k$ such that $J_x(f) = \langle f, k_x \rangle \forall f \in \mathcal{H}_k$. In particular (recall that $\mathcal{H}_k$ contains real-valued functions on $\mathcal{X}$) $k_x(y) = J_y(k_x) = \langle k_x, k_y \rangle$. We set $k(x, y) := k_x(y)$ which defines a positive definite form (kernel) by definition. The above equality now reads $k(x, y) = \langle k(x, \cdot), k(y, \cdot) \rangle$ and $f(x) = \langle f, k(x, \cdot) \rangle$. This allows us to characterize an RKHS via the positive definite kernel k :

Proposition 6.2. Let $\mathcal{X}$ be a set with $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ a positive definite (pd) kernel and $\mathcal{H}_k$ a Hilbert space of real-valued functions on $\mathcal{X}$ with inner-product $\langle \cdot, \cdot \rangle$. Then $\mathcal{H}_k$ is a reproducing kernel Hilbert space (RKHS) iff

- $k(\cdot, x) \in \mathcal{H}_k$, $\forall x \in \mathcal{X}$ and,
- $f(x) = \langle f, k(\cdot, x) \rangle$, $\forall f \in \mathcal{H}_k$ (Reproducing Property).

²⁴or non-linear for that matter, as a linear regression over non-linearly featurized input data

Notice the latter definition implies in particular again that $k(x, y) = \langle k(\cdot, x), k(\cdot, y) \rangle_k$ and more importantly $\|J_x\|_{op} = \sup_{f \in B_{\mathcal{H}_k}(0)} |f(x)| = \sup_{f \in B_{\mathcal{H}_k}(0)} |\langle f, k(\cdot, x) \rangle| = \|k(\cdot, x)\| < \infty, \forall x \in \mathcal{X}$, due to Cauchy-Schwarz, which is applicable because $k(\cdot, x) \in \mathcal{H}_k, \forall x \in \mathcal{X}$ by assumption. Combined with the argument before 6.2 this shows the proposition. Next, given a kernel, we give an explicit construction of the corresponding RKHS (uniquely determined by k due according to Moore-Aronszajn Theorem. For this consider

Lemma 6.3 (Reproducing Kernel Hilbert Space). *Given a Banach space B and positive definite, symmetric kernel $k : B \times B \rightarrow \mathbb{R}$, we define*

$$\mathcal{H}_k := \text{span}_{\mathbb{R}}\{k(\cdot, x) \mid x \in B\} \quad (97)$$

equipped with the bilinear form $\langle x, y \rangle_0 := \sum_i^n \sum_j^m a_n b_m k(x_n, y_m)$ ²⁵. Then the completion of the preHilbert space $(\mathcal{H}_0, \langle \cdot, \cdot \rangle_0)$, with respect to the norm induced by $\langle \cdot, \cdot \rangle_0$ yields the Hilbert space $\mathcal{H}_k$, which is a RKHS.

Proof. Indeed, by the above characterization we need to check that (i) $k(\cdot, x) \in \mathcal{H}_k, \forall x \in \mathcal{X}$ and (ii) $f(x) = \langle f, k(\cdot, x) \rangle, \forall f \in \mathcal{H}_k$. The first condition is clear by construction. For the second notice for $f(\cdot) = \sum_{i=1}^n \alpha_i k(\cdot, x_i)$ we have $\langle f, k(\cdot, x) \rangle = f(x)$ and for $f = \lim_{n \rightarrow \infty} \sum_{i=1}^n \alpha_i k(\cdot, x_i)$ we have the same conclusion by continuity of the scalar product: $\langle \lim_{n \rightarrow \infty} \sum_{i=1}^n \alpha_i k(\cdot, x_i), k(\cdot, x) \rangle = \lim_{n \rightarrow \infty} \langle \sum_{i=1}^n \alpha_i k(\cdot, x_i), k(\cdot, x) \rangle = \lim_{n \rightarrow \infty} \sum_{i=1}^n \alpha_i k(x, x_i) = f(x)$, hence (i), (ii) hold for all $f \in \text{span}_{\mathbb{R}}\{k(\cdot, x) \mid x \in B\} = \mathcal{H}_k$, thus $\mathcal{H}_k$ is an RKHS for the kernel k . In particular

$$\mathcal{H}_k = \{f = \sum_{i=1}^{\infty} c_i k(\cdot, x_i) : (c_1, \dots) \in \mathbb{R}, (x_1, \dots) \in \mathcal{X} \text{ such that} \quad (98)$$

$$\|f\|_k^2 = \sum_{i,j=1}^{\infty} c_i c_j k(x_i, x_j) < \infty\} \quad (99)$$

□

A natural way how kernels arise, is by means of a feature map. Indeed if $\phi : \mathcal{X} \rightarrow E$, where $(E, \langle \cdot, \cdot \rangle)$ is some Hilbert space, then $k(x, y) := \langle \phi(x), \phi(y) \rangle$ defines a positive definite kernel. (The predominant example for us is $\phi \equiv \mathbf{sig}$ with $\mathcal{X}$ equal to an appropriate path space e.g. $\mathcal{C}_d^1$ and $(E, \langle \cdot, \cdot \rangle) \equiv (T(\mathbb{R}^d), \langle \cdot, \cdot \rangle)$ the tensoralgebra with the hilbertspace structure)

We finish with a comment on RKHS in the setting of weighted spaces (see Section 7 which will clarify the corresponding RKHS definition given in Cuchiero et al. [2025]).

To finish this intermezzo on RKHS we state the celebrated representer theorem, which shows, that a solution to the Tikhonov minimisation problem takes a very handy form, namely the finite superposition of kernel functions.

²⁵where $x = \sum_{i=1}^n a_n k(\cdot, x_n), y = \sum_{i=1}^n b_n k(\cdot, y_n)$

Theorem 6.4 (Representer Theorem). *Given a kernel map $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ and the corresponding RKHS $\mathcal{H}_k$ and let $\mathbb{X} = (x_1, \dots, x_n) \in \mathcal{X}^n$, $\mathbb{Y} = (y_1, \dots, y_n) \in \mathcal{Y}^n$ and consider the following extremal problem*

$$\hat{f} \in \arg \min_{f \in \mathcal{H}_k} \mathcal{E}[(x_1, y_1, f(x_1)), \dots, (x_n, y_n, f(x_n))] + g(\|f\|_{\mathcal{H}_k}) \quad (100)$$

with $\mathcal{E} : (\mathcal{X} \times \mathcal{Y} \times \mathbb{R})^n \rightarrow \mathbb{R}$ some error function and $g : \mathbb{R}_+ \rightarrow \mathbb{R}$ an increasing function.²⁶ Then any solution of (100) takes the form

$$\hat{f}(x) = \sum_{i=1}^n c_i k(x, x_i) = c^T K_{x\mathbb{X}} \quad (101)$$

for some real values c_i , $i = 1, \dots, n$ and using the notation $c = (c_1, \dots, c_n) \in \mathbb{R}^n$ and $(K_{x\mathbb{X}})_i = k(x, x_i)$. Also, notice that $\|f\|_k^2 = \sum_{i=1}^n \sum_{j=1}^n c_i c_j k(x_i, x_j) = c^T K_{\mathbb{X}\mathbb{X}} c$. For a proof see Schölkopf et al. [2001].

Let's come back to the regression problem (95) and consider the case $\mathcal{Y} = \mathbb{R}$. Furthermore we look at the extremal problem in Theorem 6.4 with $\mathcal{E} = |\cdot|^2$ and $g(\cdot) = \lambda \cdot (\cdot)^2$ for some $\lambda > 0$. We obtain the following extremal problem, which is known as ridge regression:

$$\hat{f} \stackrel{!}{=} \arg \min_{f \in \mathcal{H}_k} \sum_{i=1}^n |y_i - f(x_i)|^2 + \lambda \|f\|_{\mathcal{H}_k}^2. \quad (102)$$

For this regularized least-square problem, the optimal coefficients $(c_i)_{i=1}^n$ of the representation (101) are given by normal equations. Indeed, insert the Ansatz (101) into (102) to get

$$\hat{c} \stackrel{!}{=} \arg \min_{c \in \mathbb{R}^n} \|y - K_{\mathbb{X}\mathbb{X}} c\|_2^2 + \lambda c^T K_{\mathbb{X}\mathbb{X}} c, \quad (103)$$

where we use vector notation: $c = (c_1, \dots, c_n)^T \in \mathbb{R}^n$, $y = (y_1, \dots, y_n)^T \in \mathbb{R}^n$ and $(K_{\mathbb{X}\mathbb{X}})_{ij} = k(x_i, x_j)$. Notice that due to the representer Theorem, we have recovered an optimization problem over $\mathbb{R}^n$ which is a lot simpler than optimizing over the possibly infinite-dimensional RKHS. By setting the first derivate of (103) with respect to c to 0, we get

$$c = (K_{\mathbb{X}\mathbb{X}} + \lambda \mathcal{I}_n)^{-1} y, \quad (104)$$

where $(K_{\mathbb{X}\mathbb{X}} + \lambda \mathcal{I}_n)^{-1}$ is invertible due to positive definiteness, enforced by the regularization term. We see from the above that the coefficients in the representer form of the solution are a function of the data $\{(x_i, y_i)\}_{i=1}^n$, as expected.

Then, the solution to (102) is

$$\hat{f}(x) = K_{x\mathbb{X}}^T (K_{\mathbb{X}\mathbb{X}} + \lambda \mathcal{I}_n)^{-1} y. \quad (105)$$

Furthermore, we introduce the "weight space" view which will be a good starting point to deduce GPR. For this, we further specialize to the case where $\mathcal{X} = \mathbb{R}^d$ and we take any feature map $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^l$ whose corresponding RKHS is associated with the induced kernel

$$k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R} \quad (106)$$

$$k(x, y) = \langle \phi(x), \phi(y) \rangle. \quad (107)$$

²⁶For our purposes we take $g(\cdot) = \lambda \cdot (\cdot)^2$, where $\lambda > 0$. Hence the above specializes to ridge regression.

For this regularized least square problem, the coefficients $(c_i)_{i=1}^n$ from Theorem 6.4 can be given explicitly as solution to normal equation. Indeed, For the data samples $\mathbb{X} = ((x_i)^j)_{ij} \in \mathbb{R}^{n \times d}$, where x_i^j is the j^{th} component of the i^{th} data point $x_i \in \mathbb{R}^d$ we use the notation $\Phi(\mathbb{X}) = (\phi(x_1), \dots, \phi(x_n)) \in \mathbb{R}^{l \times n}$.

Featurized regression is then linear regression for the featurized inputs, i.e. taking the Ansatz $f(x) = \beta^T \phi(x)$ for weights $\beta \in \mathbb{R}^l$. By 'Ansatz' we really mean the representer theorem. Hence a better way of seeing this transition is that, for linear regression, we consider the linear kernel $k(x, y) = x^T y$ which together with the representer theorem, yields

$$f(x) = \sum_{i=1}^n c_i k(x, x_i) = x^T \underbrace{\left(\sum_{i=1}^n c_i x_i \right)}_{\beta} = x^T \beta = \beta^T x. \quad \text{Finally, we compose with the feature}$$

map ϕ to get the featurized version.

In this case, ridge regression can be equivalently stated for the weights,

$$\hat{\beta} = \arg \min_{\omega \in \mathbb{R}^l} \sum_{i=1}^n |y_i - \beta^T \phi(x_i)|^2 + \lambda \|\beta\|_l = \|y - \beta^T \Phi(X)\|^2 + \lambda \|\beta\|_l, \quad (108)$$

where $\|\cdot\|_l$ denotes the Euclidean norm in $\mathbb{R}^l$. Equation (108) corresponds to the so called "weight space" view on featurized (linear) regression. From this point of view, Gaussian process regression will arise naturally in light of Bayesian inference as we will see shortly.

We have already motivated that Tikhonov regression (ridge regression) admits a closed form as solution (105). Its featurized generalization is straightforward. Using the notation $(\tilde{K}_{\mathbb{X}\mathbb{X}})_{ij} = (k(\phi(x_i), \phi(x_j)))_{ij}$ we get

$$\hat{f}(x) = \tilde{K}_{x\mathbb{X}} (\tilde{K}_{\mathbb{X}\mathbb{X}} + \lambda \mathcal{I}_n)^{-1} y \quad (109)$$

with $(\tilde{K}_{x\mathbb{X}})_i = \phi(x_i)^T \phi(x)$ and $(\tilde{K}_{\mathbb{X}\mathbb{X}})_{ij} = \phi(x_i)^T \phi(x_j)$.

We now turn to a different conceptual approach to the regression problem of (95), resuming the two general approaches, discussed in Section 2.

Instead of trying to estimate f directly from the data, we try to learn the joint distribution $\mathbb{P}_{XY}^\theta$ which is assumed to be parametrized by $\theta \in \Theta$ i.e. we are considering a family distributions $(\mathbb{P}_{XY}^\theta)_{\theta \in \Theta}$. Hence, the task becomes to find the best guess $\hat{\theta}$. Once we have learned our estimate for $\hat{\theta}$ we take, as our predictive estimate $f(X) = \mathbb{E}_{\hat{\theta}}[Y|X]$, as motivated in Section 2.

Here the two paradigms of statistical inference come in.

The *frequentist*-approach is to choose θ that maximizes the probability of seeing the observed data $\mathcal{D}_{1:n} = \{(x_i, y_i)\}_{i=1}^n$:

$$\hat{\theta}_{MLE} = \arg \max_{\theta \in \Theta} \mathbb{P}_{XY}^\theta(\mathcal{D}_{1:n}), \quad (110)$$

where, for the frequentist, θ is a parameter, not a random variable²⁷ and $\mathbb{P}^\theta(\mathcal{D}_{1:n})$ is the probability of observing the data $\mathcal{D}_{1:n}$ under the parameter θ . Contrast this to the *Bayesian* view point where θ is considered a random variable $\theta : \Omega \rightarrow \Theta$. We then seek to maximize the *a posteriori* probability after having observed the data $\mathcal{D}_{1:n} = \{(x_i, y_i)\}_{i=1}^n$:

$$\hat{\theta}_{MAP} = \arg \max_{\theta \in \Theta} \mathbb{P}_{XY}(\theta | \mathcal{D}_{1:n}) . \quad (111)$$

Here $\mathbb{P}(\theta | \mathcal{D}_{1:n})$ is the posterior distribution of the random variable θ . Bayesian inference now assumes prior knowledge or a prior guess in form of a *prior* distribution $\mathbb{P}(\theta)$. One then ‘updates’ this prior distribution to a *posterior*-distribution $\mathbb{P}(\theta | \mathcal{D}_{1:n})$ via Bayes formula:

$$p(\theta | \mathcal{D}_{1:n}) = \frac{p(\mathcal{D}_{1:n} | \theta) p(\theta)}{\int_{\Theta} p(\mathcal{D}_{1:n} | \theta) p(\theta) d\lambda(\theta)} , \quad (112)$$

where $p(\theta)$ is the Radon-Nykodim density of $\mathbb{P}(\theta)$ w.r.t. to some reference density ν on Θ and $p(\mathcal{D}_{1:n} | \theta)$ the corresponding Nykodim density of the stochastic kernel $\kappa(\theta, A)$ w.r.t. the same reference measure ν (for every $\theta \in \Theta$).

Already it is clear that (112) is nicely defined for Θ finite dimensional vector space, where we can take the reference measure ν to be the Lebesgue measure. For the case where Θ is a (possibly infinite-dimensional) Banach spaces, the above doesn’t hold trivially, as it is not clear with respect to what measure to take the Nikodym derivative. For a generalization to Banachspace valued random variables, see (Dashti and Stuart [2015], Theorem 3.4.) and (Kanagawa et al. [2022], page 16, formula (15)) for the special case of centered Gaussian noise.

Finally, observe that for the featurized regression setting with Gaussian noise $\epsilon | X \sim \mathcal{N}(0, \sigma^2)$, we immediately get that $Y | X, \beta \sim \mathcal{N}(\beta^T \phi(X), \sigma^2)$ (see footnote on page 3 in Section 2), hence for Gaussian noise, the conditional distribution, and hence the joint distribution is parametrized by the weights β . We can thus take $\theta \equiv \beta$.²⁸ Thus the prior on the parameter space Θ becomes a prior on the weights $\beta \in \mathbb{R}^l$.

In the following, we will focus on a Gaussian priors on weights

$$\beta \sim \mathcal{N}(0, \Sigma_\beta) \quad ^{29} \quad (113)$$

with a prior variance matrix Σ_β . This choice needs no justification a priori, but one can still give a couple of reasons. The, is a structural benefit as will be discussed in the following. The second is that the normal distribution has maximum entropy among all distributions of $\mathbb{R}^d$ with known mean and variance (see Rasmussen and Williams [2006]). The following discussion will use basic results of Gaussian distributions without proof, as the goal is to

²⁷In the frequentist perspective there exists a true *value* of θ in nature. This value is a fixed quantity, not a random one

²⁸What we mean to point out here is a recurring theme in what follows. It is inherent to Bayesian inference to set a prior distribution on some object. This prior distribution induces a prior distribution on deduced objects further down the road e.g. prior on weights yields a prior on Gaussian process. Here we observe that, fixing a prior distribution on the weights β determines a prior distribution of $Y | X$

²⁹Taking the distribution to be centered is not crucial but eases the notation

understand Gaussian Process regression structurally and its deduction is not the main goal here. For more details; see Rasmussen and Williams [2006].

Let $f(x) = \phi(x)^T \beta$ with prior $\beta \sim \mathcal{N}(0, \Sigma_\beta)$. Then

$$\mathbb{E}[f(x)] = \phi(x)^T \mathbb{E}[\beta] = 0 \quad (114)$$

$$\mathbb{E}[f(x)f(x')] = \phi(x)^T \mathbb{E}[\beta\beta^T] \phi(x') = \phi(x)^T \Sigma_p \phi(x') , \quad (115)$$

and because $f(x) = \sum_{i=1}^l \phi(x)_i \beta_i$ is again Gaussian, as a linear combination of Gaussian random variables, the above shows that any tuple $(f(x_1), \dots, f(x_n))$ is jointly Gaussian (Gaussian vector). In other words, the stochastic process $\{f_x\}_x = \{f(x)\}_x$ is a Gaussian process with mean 0 and covariance function $K(x, x') = \phi(x)^T \Sigma_p \phi(x') = \phi(x)^T \Sigma_p^{1/2} \Sigma_p^{1/2} \phi(x')$ (where the index set $x \in \mathcal{X}$ is not in general as simple as we usually assume, i.e. $t \in [0, T]$, and we used that Σ_β is p.d. and hence admits a Cholesky decomposition $\Sigma_\beta = \Sigma_\beta^{1/2} \Sigma_\beta^{1/2}$; often $\mathcal{X} = \mathbb{R}^d$).

In this special case, we recognize one practically important feature of using a Gaussian prior: the feature map ϕ only appears in the composite form $k(x, x')$. Hence, from a modeling perspective we can directly choose an adapted kernel function k rather than bothering with feature maps. This is often called the *kernel trick*. We have previously used the terminology of Gaussian processes. We recall here the corresponding notion through its definition

Definition 6.5 (Gaussian Processes). Let $\mathcal{X}$ be some (non-empty) set, $(\Omega, \mathcal{F}, \mathbb{P})$ a probability space and $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ a positive semidefinite kernel and $m : \mathcal{X} \rightarrow \mathbb{R}$ some real valued function. Then a stochastic process

$$Z : (\Omega, \mathbb{P}) \rightarrow (\mathcal{X} \rightarrow \mathbb{R}) \quad (116)$$

$$\omega \mapsto (x \mapsto Z(\omega, x)) , \quad (117)$$

i.e. a $\mathcal{F}/\mathcal{B}(\mathbb{R})^{\otimes \mathcal{X}}$ -measurable mapping³⁰, is called a Gaussian Process with mean $m : \mathcal{X} \rightarrow \mathbb{R}$ and covariance function $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ if all its finite dimensional marginals are Gaussians i.e. if for any tuple $\mathbb{X} = (x_1 \dots x_n) \in \mathcal{X}^n$ we have that the law $(Z(x_1) \dots Z(x_n))_* \mathbb{P} = \mathcal{N}(m_{\mathbb{X}}, k_{\mathbb{X}\mathbb{X}})$ is a multivariate Gaussian distribution, where $m_{\mathbb{X}} = m(\mathbb{X}) = (m(x_1), \dots, m(x_n))$ and $(k_{\mathbb{X}\mathbb{X}})_{ij} = k(\mathbb{X}, \mathbb{X}) = k(x_i, x_j)$. We write $Z \in \mathcal{GP}(m, k)$

In particular any Gaussian process is uniquely defined by its mean and covariance function.

The latter yields a dual view on Bayesian featured regression with Gaussian priors. Indeed we can equivalently consider a prior on the level of functions³¹, which means the prior becomes an initial choice of the covariance function and the mean of the Gaussian process $\mathbf{f}$. We point out, the observation previously made, that this new point of view most importantly introduces the "kernel trick" in that the feature map is now implicit in the choice of the covariance function.

³⁰we take $\mathcal{B}(\mathbb{R})^{\otimes \mathcal{X}}$ to be the cylinder σ -algebra which is generally a lot smaller than with the Borel σ -algebra $\mathcal{B}(\mathbb{R}^{\mathcal{X}})$ being taken w.r.t. to the product topology on $\mathbb{R}^{\mathcal{X}}$

³¹By this we mean the following: The law of a Stochastic process, as defined, in the particular case of a Gaussian Process, is a probability measure on path space, i.e. on functions. Hence setting a prior distribution on a Gaussian process is setting a prior distribution on sample paths i.e. on functions

The GPR approach is now to the point of view that f in 7 is a realisation of the random function resp. the stochastic process $\mathbf{f} \sim \mathcal{GP}(m, k)$, where the prior process $\mathbf{f}$ defines a prior distribution on the space of functions $\mathcal{X}^{\mathbb{R}}$ through its law. The Bayesian regression algorithm is now as usual, through updating the prior to a posterior after observing data. In particular GP regression has the structural property that it remains in the class of Gaussian measures as the posterior of a Gaussian distribution, defined as prior distribution, conditioned on the observed data $f|\mathcal{D}_{1:n}$, is again Gaussian. So on a functional level (more precisely random functions, i.e. stochastic processes) we stay in the class of Gaussian processes. Let us be more explicit to get the key results for GPR: the posterior distributions / prediction equations. For this we follow the review paper (Kanagawa et al. [2022]). We start with a functional Gaussian prior or in other words, a Gaussian process

$$\mathbf{f} \sim \mathcal{GP}(m(x), k(x, x')) \quad (118)$$

for some mean and covariance function that should reflect our prior knowledge on the underlying distribution of the data. The following posterior distribution represents the predictive quality of GPR.

Theorem 6.6 (Posterior GPR). *Let $\mathbb{X} = (x_1, \dots, x_n) \in \mathcal{X}^n$ and $\mathbb{Y} = (y_1, \dots, y_n) \in \mathbb{R}^n$ be i.i.d. observations of the random variables X and Y on which we want to condition our prior Gaussian process $\mathbf{f} \sim \mathcal{GP}(m, k)$. Then*

$$\mathbf{f}|Y \sim \mathcal{GP}(\bar{m}, \bar{k}) \quad (119)$$

that is, the regular conditional distributions

$$\mathbb{P}(f(x_1), \dots, f(x_n) | y_1, \dots, y_n) \equiv \mathbb{P}(\mathbf{f}(\mathbb{X}) | \mathbb{Y}), \quad (120)$$

$$\mathbf{f}(\mathbb{X}) = (f(x_1), \dots, f(x_n)) \quad (121)$$

with Lebesgue density

$$d\mathbb{P}(\mathbf{f}(\mathbb{X}) = \mathbb{Z} | \mathbb{Y}) = \frac{1}{(2\pi)^{n/2} |\bar{K}_{\mathbb{X}\mathbb{X}}|^{1/2}} \exp\left(-\frac{1}{2}((\mathbb{Z} - \bar{m}_{\mathbb{X}})^\top \bar{K}_{\mathbb{X}\mathbb{X}}^{-1} (\mathbb{Z} - \bar{m}_{\mathbb{X}}))\right) \quad (122)$$

$$d\mathbb{P}(\mathbf{f}(\mathbb{X}) = \mathbb{Z} | \mathbb{Y}) = \frac{1}{(2\pi)^{n/2} |\bar{K}_{\mathbb{X}\mathbb{X}}|^{1/2}} \exp\left(-\frac{1}{2}((\mathbb{Z} - \bar{m}_{\mathbb{X}})^\top \bar{K}_{\mathbb{X}\mathbb{X}}^{-1} (\mathbb{Z} - \bar{m}_{\mathbb{X}}))\right) \quad (123)$$

³² for $\mathbb{Z} = (z_1, \dots, z_n) \in \mathbb{R}^n$ and where

$$\bar{m}(x) = m(x) + K_{x\mathbb{X}}(K_{\mathbb{X}\mathbb{X}} + \sigma^2 \mathcal{I}_n)^{-1}(\mathbb{Y} - m_{\mathbb{X}}) \quad (124)$$

$$\bar{k}(x, x') = k(x, x') - K_{x\mathbb{X}}(K_{\mathbb{X}\mathbb{X}} + \sigma^2 \mathcal{I}_n)^{-1} K_{\mathbb{X}x'} \quad (125)$$

$$K_{\mathbb{X}x} = K_{x\mathbb{X}}^T = (k(x_1, x), \dots, k(x_n, x))^T \quad (126)$$

This result is structurally important because the closedness of Gaussian distribution under conditionning and marginalisation makes GPR nice, in the sense that we have explicit expressions for the predictive distribution which are easy to sample from and furthermore,

³²By Kolmogorov's Extension Theorem and properties of Gaussian processes, this defines a unique Gaussian process, which we denote by $\mathbf{f}|Y$

the marginals are well known for Gaussian distributions. What is more, the above result is more subtle than one might think. On first glance one would expect that it is a straight forward application of the classical Bayes rule, but one should be aware that Bayes' rule doesn't trivially hold for infinite-dimensional spaces. But the distribution of a Gaussian process is a distribution of a functional space which is infinite dimensional in general. Indeed the proof of 6.6 does not involve Bayes rule! For the proof we will use the following classical result for Gaussian measures, without proof.

Lemma 6.7 (Conditioned marginal of Gaussian law). *Let $a \in \mathbb{R}^n, b \in \mathbb{R}^m$ be multivariate Gaussian vectors in $\mathbb{R}^n$*

$$\begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_a \\ \mu_b \end{bmatrix}, \begin{bmatrix} A & C \\ C^T & B \end{bmatrix}\right), \quad (127)$$

where $A \in \mathbb{R}^{n \times n}, B \in \mathbb{R}^{m \times m}$ are the covariance matrices of $\mathbf{a}, \mathbf{b}$ respectively and $\mu_a \in \mathbb{R}^n, \mu_b \in \mathbb{R}^m$ the means, and $C \in \mathbb{R}^{n \times m}$ the covariance matrix. Then

$$b|a \sim \mathcal{N}(\mu_b + C^T A^{-1}(a - \mu_a), B - C^T A^{-1}C). \quad (128)$$

With this in hand we give the proof of Theorem 6.6

Proof. Consider the data $\mathbb{Z} = (z_1, \dots, z_m) \in \mathbb{R}^m$, the corresponding Gaussian vector $\mathbf{f}_{\mathbb{Z}} = (f(z_1), \dots, f(z_m)) \in \mathbb{R}^m$ and the noisy observations $\mathbb{Y} = (y_1, \dots, y_n) \in \mathbb{R}^n$. By (95) and by Definition 6.5 of a Gaussian process, $\mathbb{Y}$ and $\mathbf{f}$ are multivariate Gaussian hence

$$\begin{bmatrix} \mathbb{Y} \\ \mathbf{f}_{\mathbb{Z}} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} m_{\mathbb{X}} \\ m_{\mathbb{Z}} \end{bmatrix}, \begin{bmatrix} k_{\mathbb{X}\mathbb{X}} + \sigma^2 \mathcal{I}_n & k_{\mathbb{X}\mathbb{Z}} \\ k_{\mathbb{Z}\mathbb{X}} & k_{\mathbb{Z}\mathbb{Z}} \end{bmatrix}\right). \quad (129)$$

Setting $a = \mathbb{Y}|\mathbb{X}, b = \mathbf{f}_{\mathbb{Z}}, \mu_a = m_{\mathbb{X}}, \mu_b = m_{\mathbb{Z}}, A = k_{\mathbb{X}\mathbb{X}} + \sigma^2 \mathcal{I}_n, B = k_{\mathbb{Z}\mathbb{Z}}, C = k_{\mathbb{X}\mathbb{Z}}$ yields the result by Lemma 6.7. $\square$

The distribution of $\mathbf{f}|\mathbb{Y}$ is called the *posterior distribution* and the equations for the posterior mean and covariance in Theorem 6.6 are known as the predictive equations of GPR. In particular GPR itself yields a posterior distribution over functions, from which one can sample. If one asks for a canonically predicted function $f : \mathcal{X} \rightarrow \mathbb{R}$ for the regression problem 6.6, the maximum a posteriori estimate (MAP) of the posterior distribution is the *posterior mean function* $\bar{m}(x)$. In this sense, GPR predicts a concrete function as a particular sample of the a posteriori distribution it learned. We state the following key result which links GPR and kernel ridge regression taken from Kanagawa et al. [2022].

Theorem 6.8. *Let $\{(x_i, y_i)\}_{i=1}^n \in (\mathcal{X} \times \mathbb{R})^n$ and let $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ a positive definite kernel. Then*

$$\bar{m} = \hat{f} \quad (130)$$

if $\sigma^2 = \lambda$, where

1. $\bar{m}$ is the posterior mean function from the posterior distribution of GPR for the GP prior $\mathbf{f} \sim \mathcal{GP}(0, k)$ and with $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$, $i \in \{1, \dots, n\}$
i.i.d. under the modelling assumption $y_i = \mathbf{f}(x_i) + \epsilon_i$, $i \in \{1, \dots, n\}$
2. $\hat{f}$ is the solution of the Tikhonov regression 102 for the data $\{(x_i, y_i)\}_{i=1}^n$, regularization parameter λ and the RKHS $\mathcal{H}_k$.

Compare this result with the well known fact that sample paths of $\mathbf{f} \sim \mathcal{GP}(m, k)$ are almost surely not in the corresponding RKHS $\mathcal{H}_k$.

We now address the question on whether the regression framework defined in 7 encompasses the GPR framework as well, or if the latter is something else all together. This question essentially boils down to the relation between Bayesian regression and the frequentist counterpart and in particular the well known fact that Bayesian regression with a Gaussian prior is equivalent to LS regression with an L^2 -regularization, that is ridge regression. The latter is a special instance of the general insight that encoding prior knowledge in the Bayesian view, as a prior, is equivalent to the choice of a regularization term in LS regression.

More precisely Theorem 6.8 states the equivalence of GPR and ridge regression with respect to their prediction on an algorithmic level. We formulate our observations formally

Proposition 6.9 (Equivalence of Regression Frameworks). *Let $X : \Omega \rightarrow \mathcal{X}$ and $Y : \Omega \rightarrow \mathbb{R}$ where $\mathcal{X}$ is Polish. Consider the regression problem of estimating $f : \mathcal{X} \rightarrow \mathbb{R}$ in*

$$\begin{cases} Y = f(X) + \epsilon \\ \epsilon \sim \mathcal{N}(0, \sigma^2) \end{cases} \quad (131)$$

and consider the signature kernel $k(x, y) = \langle \underline{\mathbf{sig}}^\Lambda(x), \underline{\mathbf{sig}}^\Lambda(y) \rangle$ for some feature normalization Λ and let $\mathcal{H}_k$ be the corresponding RKHS.

Furthermore, consider n observed, i.i.d. realization of X and Y , $D_n^{X,Y} = \{(x_i, y_i)\}_{i=1}^n$.³³

Then the following estimators are asymptotically equivalent i.e. consistent estimators of f .

1. $\hat{f} = \arg \min_{\tilde{f} \in \mathcal{H}_k} \sum_{i=1}^n \|y_i - \tilde{f}(x_i)\|^2 + \sigma^2 \|\tilde{f}\|_k^2$
2. $\bar{m}$ posterior mean function of GPR for the prior $\mathbf{f} \sim \mathcal{GP}(0, k)$
3. Consistent estimator $\hat{T}_n(\mathcal{I}_{\mathbb{R}}) = \langle l_n^{\mathcal{I}_{\mathbb{R}}}, \hat{\Theta}_n^{\underline{\mathbf{sig}}^\Lambda} \rangle$ of 4.3, where $(l_n^{\mathcal{I}_{\mathbb{R}}})_{n \geq 0}$ such that $\lim_{n \rightarrow \infty} \|\langle l_n^{\mathcal{I}_{\mathbb{R}}}, \underline{\mathbf{sig}}^\Lambda(\cdot) \rangle - g\|_{L^2(\mu_Y)} = 0$ is an approximation series of $\mathcal{I}_{\mathbb{R}} \in L^2(\mathbb{P}_Y)$ and $\hat{\Theta}_n^{\underline{\mathbf{sig}}^\Lambda}$ is an estimator of $\mu.(\underline{\mathbf{sig}}^\Lambda) = \mathbb{E}[\underline{\mathbf{sig}}^\Lambda(Y)|X = \cdot]$.

³³and also i.i.d. observations of $\{\epsilon_i\}_{i=1}^n$

Proof. The equivalence of $\hat{f}$ and $\bar{m}$ is the content of Theorem 6.8 and holds exactly, not only asymptotically. Recall that under the centered Gaussian noise assumption, $f(X) = \mathbb{E}[Y|X]$ is the unique global minimizer of (131) under the mean square loss. That $\hat{T}_n(\mathcal{I}_{\mathbb{R}})$ is a consistent estimator of $\mathbb{E}[Y|X]$, is the content of Theorem 5.9. in Schell and Alaifari [2024] applied for the special case of $Z = g(Y)$ with $g \equiv \mathcal{I}_{\mathbb{R}} \in L^2(\mathbb{P}_Y)$. For this notice, that $\hat{T}_n(\mathcal{I}_{\mathbb{R}})$ is a consistent estimator of $\mathbb{E}[Y|X]$ according to Theorem 5.9. of Schell and Alaifari [2024]. $\square$

More than that, considering the regression estimator $\mathfrak{R}_{II}$ of Algo II as introduced in Schell and Alaifari [2024] (equation (116)) we derive an exact equality of the estimators.

Corollary 6.10. *In the setting of (6.9) and under the same assumptions, the following are algorithmically equivalent, i.e. for the same i.i.d. data samples $\{(x_i, y_i)\}_{i=1}^n$, the following estimators are identical.*

1. $\hat{f} = \arg \min_{\tilde{f} \in \mathcal{H}_k} \sum_{i=1}^n \|y_i - \tilde{f}(x_i)\|^2 + \sigma^2 \|\tilde{f}\|_k^2$
2. $\bar{m}$ posterior mean function of GPR for the prior $\mathbf{f} \sim \mathcal{GP}(0, k)$
3. $\mathfrak{R}_{II}$ of Algo II as introduced in Schell and Alaifari [2024] equation (116) for $\mathcal{I}_{\mathbb{R}} \in L^2(\mathbb{P}_Y)$ fixed.

Proof. Choosing the prior $\mathbf{f} \sim \mathcal{GP}(0, k)$ (i.e. $m(x) \equiv 0$) we get from the posterior mean equation for GPR (see Theorem (6.6)) that $\bar{m}(x) = K_{x\mathbb{X}}(K_{\mathbb{X}\mathbb{X}} + \sigma^2 \mathcal{I}_n)^{-1} \mathbb{Y}$. But we have seen that this is exactly the representer form of the kernel ridge solution derived in (105). For the chosen bounded signature kernel, this corresponds exactly to the estimator $\mathfrak{R}_{II}$ of Algo II as introduced in Schell and Alaifari [2024] equation (116) with hyperparameter λ chosen to be the same λ as used in the Ridge regression and GPR (i.e. $\lambda = \sigma^2$). Hence $\mathfrak{R}_{II}(D_n^{X,Y}) = \hat{f}(D_n^{X,Y}) = \bar{m}(D_n^{X,Y})$. $\square$

Remark. *The signature regression operator $\hat{T}_n(f)$ is a priori a more general estimator than the ridge regression estimate and the GPR prediction. However, for fixed $f \in L^2(\mathbb{P}_Y)$ Algo II is exactly the kernelized ridge regression for the signature kernel, as stated in the previous lemma. Hence, under the Gaussian noise assumption, the equivalence becomes exact and not only asymptotically as consistent estimators. The freedom in the choice of $f \in L^2(\mathbb{P}_Y)$, as is the case in Algo I, is also the conceptual difference in signature regression and GPR as it precisely reflects the fact, that for fixed f , equation (131) corresponds to Gaussian marginal $Y|f(X)$, by the Gaussian noise assumption. The above shows that in this case, signature regression as understood in Algo II and GPR with signature kernel as covariance function, are algorithmically identical. The a priori freedom in the choice of f in Algo I corresponds to the generality of the learnable relation $X \sim Y$ i.e. no a priori assumption on the marginal distribution $\mu_{x \in \mathcal{X}}$.*

Finally, notice that in particular, Proposition 4.5 shows consistency of ridge regression and GPR for the proposed kernel.

After having introduced and studied GPR, ridge regression and their relation, the next section brings the pieces together and, following Cuchiero et al. [2025], motivates a mathematical framework that allows for a rigorous treatment of signature kernel GPR.

7 GPR Approach on Signaturised Regression in Weighted Spaces

After having introduced a particular take on signature regression in Section 4, crucially using L^2 -density of the bounded signature components to get a universally consistent estimator for the regression operator, which is calculated through solving a ridge regression, we will address another signature regression scheme. More precisely, a Gaussian process regression scheme using a signature kernel. For this, we follow Cuchiero et al. [2025], introducing weighted spaces as an independent concept. We present the most important notions needed to follow the latter. The basic objects on which the approximation theory presented in Cuchiero et al. [2025] is built, are *weighted spaces*, a concept based on so-called *admissible weight functions* itself.

Definition 7.1 (Weighted Space). Let $(\mathcal{X}, \tau_{\mathcal{X}})$ be regular Hausdorff space. Then a function

$$\psi : \mathcal{X} \rightarrow (0, \infty) \quad (132)$$

is called an *admissible weight function* if all pre-images $\psi^{-1}((0, R]) \subset \mathcal{X}$ are compact for all $R > 0$ with respect to $(\mathcal{X}, \tau_{\mathcal{X}})$. Furthermore, let $(\mathcal{Y}, \|\cdot\|_{\mathcal{Y}})$ be a Banach space and ψ an admissible weight function. Then we define

$$\mathcal{B}_{\psi}(\mathcal{X}, \mathcal{Y}) = \{f : \mathcal{X} \rightarrow \mathcal{Y} \text{ such that } \|f\|_{\psi} := \sup_{x \in \mathcal{X}} \frac{\|f(x)\|_{\mathcal{Y}}}{\psi(x)} < \infty\}, \quad (133)$$

as the closure of bounded continuous function $C_b^0(\mathcal{X}, \mathcal{Y}) \subset \mathcal{B}_{\psi}(\mathcal{X}, \mathcal{Y})$ w.r.t. to the norm $\|\cdot\|_{\psi}$ and call this a *weighted space* with respect to ψ .

Morally speaking, these are the sets of functions whose growth is bounded by a weight function ψ .

The main motivation for this notion, introduced in Cuchiero et al. [2025], is to obtain a generalized version of Stone-Weierstrass for spaces that are not locally compact. The notion of a weighted spaces replaces the one of a locally compact Hausdorff space. In particular, with the new Stone-Weierstrass result, one can consider weighted versions of path spaces, which are notoriously not locally compact, hence enabling the formulation of density theorems for path spaces using the signature transform. Notice that we already have encountered an Ansatz to this problem during the quick study of feature normalisations, respectively, the bounded signature transform, and Giles' theorem (for a detailed treatment see Chevyrev and Oberhauser [2022]).

Both approaches also allow for results on characteristicness due to properties of the dual. In the current approach this is due to a weighted Riesz-representation theorem, stating that $B_{\psi}(\mathcal{X}, \mathbb{R})^* = \mathcal{M}_{\psi}(\mathcal{X})$, where $\mathcal{M}_{\psi}(\mathcal{X})$ denotes the space of finite signed Radon measures μ ³⁴(for a proof see Dörsek and Teichmann [2010]). Let's finally state the weighted Stone-Weierstrass version.

³⁴finite in the sense $\int_{\mathcal{X}} \psi |\mu|(dx) < \infty$ with the total variation measure $|\mu|$

Theorem 7.2 (Weighted Stone Weierstrass). *Let $\mathcal{A} \subset \mathcal{B}_\psi(\mathcal{X})$ be a subalgebra that is point separating and nowhere vanishing of ψ -moderate growth³⁵. Then, $\mathcal{A}$ is dense in $\mathcal{B}_\psi(\mathcal{X})$*

Let us turn to GPR. Recall from Section 6 that an RKHS on $\mathcal{X}$ is defined as a Hilbert space of real-valued functions H with continuous evaluation functionals. We reformulate and adapt this notion to the setting of weighted spaces.

For a RKHS $\mathcal{H}_k$ of a weighted space $(\mathcal{X}, \psi)$ we require that it is a subspace of $\mathcal{B}_\psi(\mathcal{X})$. Then, Definition 6.1 is equivalent to the requirement that the inclusion $\mathcal{H}_k \hookrightarrow \mathcal{B}_\psi(X)$ is continuous. Indeed this implies that the evaluation functionals on $\mathcal{H}_k$ are continuous. This follows from

$$|f(x)| \leq \psi(x) \|f\|_\psi \leq C \psi(x) \|f\|_k, \quad (134)$$

where we use that $\|f\|_\psi \leq C \|f\|_k$ for some $C > 0$, due to continuity of the embedding $i : \mathcal{H}_k \rightarrow \mathcal{B}_\psi(X)$.

In the following we will introduce what we will understand to be the Bayesian prior on the level of Gaussian processes $Z : \Omega \rightarrow \mathcal{B}_\psi(\mathcal{X})$ as a centered Gaussian process³⁶. Before, we give an introduction to Cameron Martin spaces, in order to build the arena for GPR and in particular priors of Gaussian processes. In particular, the following will shed light on the consistency of the notions as defined in Cuchiero et al. [2025] and the classical literature i.e. Bogachev [1998], Hairer [2009]

7.1 Cameron Martin Space

In this interlude, we do some clarifying considerations to make clear that the definition of the Cameron Martin space in Cuchiero et al. [2025] is the same as in Hairer [2009]³⁷, just specialised to the case of weighted spaces. Along the way we shall introduce the fundamental definitions required. For the identification we will in particular use the weighted Riesz-representation theorem stating $B_\psi(\mathcal{X}, \mathbb{R})^* \cong \mathcal{M}_\psi(\mathcal{X})$ (see Theorem 2.4. Dörsek and Teichmann [2010])³⁸. More precisely it states that every bounded functional in $B_\psi(\mathcal{X}, \mathbb{R})^*$ acts as an integral operator w.r.t. to some measure $\mu \in \mathcal{M}_\psi(\mathcal{X})$. That is $\forall F \in B_\psi(\mathcal{X}, \mathbb{R})^*, \exists! \mu \in \mathcal{M}_\psi(\mathcal{X})$ such that $F(f) = \int_{\mathcal{X}} f(x) \mu(dx)$. Let's recall the standard definition of the covariance operator as given in Hairer [2009].

Definition 7.3 (Covariance Operator). Let B be a Banach space and μ a Gaussian measure on B . Then we define the covariance operator

$$C_\mu : B^* \times B^* \rightarrow \mathbb{R} \quad (135)$$

$$(f, g) \mapsto C_\mu(f, g) = \int_B f(x)g(x) \mu(dx), \quad (136)$$

³⁵we say that $\mathcal{A}$ is of ψ -moderate growth if there is a subalgebra $\tilde{\mathcal{A}} \subset \mathcal{A}$ s.t. for all $f \in \tilde{\mathcal{A}}$ there exists $\lambda > 0$ such that the map $(x \rightarrow \exp(\lambda|f(x)|)) \in \mathcal{B}_\psi(\mathcal{X})$

³⁶i.e. $l \circ Z$ is a centered Gaussian on $\mathbb{R}$ because by definition the one dimensional projections of a Gaussian law are real Gaussians, hence $l_*(\mathbb{P} \circ Z^{-1}) = \mathbb{P} \circ Z^{-1} \circ l^{-1} = \mathbb{P}(l \circ Z^{-1})$

³⁷for a more general treatment covering locally convex topological spaces X , see Bogachev [1998]

³⁸ $\mathcal{M}_\psi(\mathcal{X})$ is the Banach space of signed measures $|x| : \mathcal{F} \rightarrow \mathbb{R}$ on $(\Omega, \mathcal{F})$ such that $\int_\Omega \psi(\omega) |x|(\mathrm{d}\omega) < \infty$

where we call μ a Gaussian measure if it is a Borel measure on the Banach space B such that the push-forward $f_*\mu$ is a Gaussian measure on $\mathbb{R}$ for all $f \in B^*$ (i.e. its fourier transform is given by $\widehat{f_*\mu}(x) = \exp(-\sigma^2/2x^2 + imx)$). Note that the operator C_μ can be identified through $\hat{C}_\mu(f)(g) := C_\mu(f, g)$ with the operator $\hat{C}_\mu : B^* \rightarrow B^{**}$. Furthermore, recall that in general we have the canonical injection of B in B^{**} given by the evaluation functional $J : f \rightarrow (J(f) : F \rightarrow F(f))$ allowing us to identify B as a subspace of B^{**} , hence in general B^{**} is larger than B . If there is equality we call B reflexiv. In this sense we observe that the range of the operator $\hat{C}_\mu$ actually lies in the subset B of B^{**} . We state this formally in a small lemma.

Lemma 7.4. *Let $\hat{C}_\mu$ be the operator defined by $\hat{C}_\mu : B^* \rightarrow B^{**}$, $f \mapsto (\hat{C}_\mu(f) : g \mapsto C_\mu(f, g))$. Then we can identify*

$$\hat{C}_\mu f = \int_B x f(x) \mu(dx) , \quad (137)$$

where the integral is understood in the Bochner sense. Whence we have $\text{Im}(\hat{C}_\mu) \subset B \subset B^{**}$

Proof. The expression (137) needs to be understood under the previously explained identification of B as a subset of B^{**} or more explicitly, identifying $x \equiv J(x)$, where J is the canonical embedding of B in B^{**} . Let's denote the expression on the right hand side $F(f) := \int_B x f(x) \mu(dx) \in \mathcal{B}$. As an element of $\mathcal{B}$ we identify it with $J(F(f))$. Considering its action on $\mathcal{B}$ the claim follows readily. Indeed for the above we understand (137) as the identification $J(\hat{C}_\mu) \equiv \hat{C}_\mu$. Under this identification, the claim follows immediately, using that the Bochner-integral commutes with bounded linear forms. Indeed

$$J(F(f))(g) = g\left(\int_B x f(x) \mu(dx)\right) \quad (138)$$

$$= \int_B g(x) f(x) \mu(dx) = C_\mu(f, g) . \quad (139)$$

But this shows that $F(f)(g) = \hat{C}_\mu f(g) \forall g \in \mathcal{B}$ and hence $F(f) = \hat{C}_\mu f \forall f \in \mathcal{B}$ and thus $F = \hat{C}_\mu$. $\square$

Furthermore, let us consider a Gaussian process $Z : \Omega \rightarrow \mathcal{B}_\psi(X)$ with law $\mu \equiv \mathbb{P} \circ Z^{-1}$ (which is a measure on the Banachspace $\mathcal{B}_\psi(\mathcal{X})$). Using the identification of Lemma 7.4 we get

$$\hat{C}_\mu(F) = \int_{\mathcal{B}_\psi(\mathcal{X})} f \cdot F(f) \mu(df) = \int_\Omega Z(\omega) F(Z(\omega)) \mathbb{P}(d\omega) \quad (140)$$

$$= \mathbb{E}[ZF(Z)] . \quad (141)$$

Lastly, due to the Riesz-representation for weighted spaces (i.e. using that any bounded functional on $\mathcal{B}_\psi(\mathcal{X})$ is of the form $F(f) = \int_{\mathcal{X}} f(x) \lambda(dx)$ for some finite signed Radon measure $\lambda \in \mathcal{M}_\psi(\mathcal{X})$, which gives a one-to-one correspondance between signed measures on $\mathcal{X}$ and bounded operators on $\mathcal{B}_\psi(\mathcal{X})$) we get

$$\hat{C}_\mu(\lambda) = \mathbb{E}\left[Z \int_{\mathcal{X}} Z(x) \lambda(dx)\right] , \quad (142)$$

where the last expression coincides with the ad hoc definition of the covariance operator given in Cuchiero et al. [2025]

$$\mathcal{M}_\psi(\mathcal{X}) \ni \mu \rightarrow C_Z(\mu) = \mathbb{E}[Z \int_{\mathcal{X}} Z(x) \mu(dx)]. \quad (143)$$

Lastly, the respective Cameron-Martin (CM) space is defined as the closure of the range of the covariance operator. To see this, we proof a small exercise of Hairer [2009][Exercise 4.27.] along with the corresponding definition of the CM space of a Gaussian measure μ .

Definition 7.5 (Cameron-Martin Space of μ). Given a Gaussian measure μ on a Banach space B we define its Cameron-Martin space $\mathcal{H}_\mu$ as the completion of the linear subspacce

$$\mathring{\mathcal{H}}_\mu = \{h \in B : \exists h^* \in B^* \text{ with } C_\mu(h^*, l) = l(h) \ \forall l \in B^*\} \quad (144)$$

with respect to the norm $\|h\|_\mu^2 = \langle h, h \rangle_\mu = C_\mu(h^*, h^*)$. Equipped with the scalar product $\langle h, g \rangle_\mu = C_\mu(h^*, g^*)$, it becomes a Hilbert space.

Lemma 7.6. $\mathring{\mathcal{H}}_\mu$ is the range of the operator $\widehat{C}_\mu$ defined in Lemma 7.4.

Proof. We use the identification of Lemma 7.4 and the underlying identification of B as a subspace of B^{**} (by identifying $B \ni f = J(f) \in B^{**}$). If $\widehat{C}_\mu(h^*) = h$ for some $h \in B$, then $l(\widehat{C}_\mu) = l(h) \ \forall h \in B^*$ i.e. $H \in \mathring{\mathcal{H}}_\mu$ (hence $Im(\widehat{C}_\mu) \subset \mathring{\mathcal{H}}_\mu$). Conversely if $h \in \mathring{\mathcal{H}}_\mu$ there exists h^* such that $l(C_\mu(h^*, l) = l(h) \ \forall l \in B^*$ or rewritten $\widehat{C}_\mu h^* = l(h) \ \forall l \in B^*$. But because B^* is point separating (Hahn-Banach), the latter implies $\widehat{C}_\mu h^* = h$ i.e. $h \in Im(\widehat{C}_\mu)$ so we get $\mathring{\mathcal{H}}_\mu \subset Im(\widehat{C}_\mu)$, hence the claim. $\square$

Thus we recover exactly the definition of the Cameron Martin space H_Z of the Gaussian prior Z used in chapter 6 of Cuchiero et al. [2025], defined as the closure of

$$\mathring{\mathcal{H}}_Z := \{C_Z(\mu) : \mu \in \mathcal{M}_\psi(\mathcal{X})\} \quad (145)$$

with respect to the scalar product $\langle C_Z(\mu_1), C_Z(\mu_2) \rangle = \mathbb{E}[\int_{\mathcal{X}} Z(x) \mu_1(dx), \int_{\mathcal{X}} Z(x) \mu_2(dx)]$. Indeed it is the range of the covariance operator, whose equivalence has just been argued. -

For the sake of completeness, we state without proof a characterization of Cameron Martin spaces that gives operational meaning as well as intuition for the latter.

Theorem 7.7 (Cameron Martin Spaces). *Let B be a Banachspace, and μ be a Gaussian measure on B . Then for $h \in B$, define the translation map $T_h : B \rightarrow B$ by $T_h(x) = x + h$. Then the image measure $T_h^* \mu$ is absolutely continuous with respect to μ if and only if $h \in \mathcal{H}_\mu$.*

That is, the Cameron Martin space for a given Gaussian measure contains exactly the directions in the Banach space B , in which we can translate the measure, without changing its null sets. For a proof; see Hairer [2009], Theorem 4.44.

7.2 Signature-kernel GPR and Gaussian Priors

We are back to the discussion of Cuchiero et al. [2025] and its most important theorem for our purposes, Theorem (7.9). For this we further investigate the relation between the RKHS induced by some kernel k and the Cameron Martin space H_Z induced by the Gaussian process Z . Intuitively, this is done by making a connection between the covariance function, which uniquely determines its C.M. space and the kernel which uniquely determines its RKHS.

The case where Z takes values in $C(\mathcal{X})$ for $\mathcal{X}$ a compact metric space (such that $C(\mathcal{X}) \subset l^\infty(\mathcal{X})$) is nicely worked out in Theorem 2.1. of van der Vaart and van Zanten [2008]. In the current setting, Z takes values in $\mathcal{B}_\psi(\mathcal{X})$ which in general (i.e. $\mathcal{X}$ being a completely regular Hausdorff topological space) is only σ -compact and not locally compact. Hence we conceptually reproduce the proof for the equivalence, given in Cuchiero et al. [2025].

Proposition 7.8. *Let $\mathcal{H}_Z = \overline{\mathring{\mathcal{H}}_Z}$ where the closure is taken w.r.t. to $\langle C_Z(\mu_1), C_Z(\mu_2) \rangle = \mathbb{E}[\int_{\mathcal{X}} Z(x)\mu_1(dx), \int_{\mathcal{X}} Z(x)\mu_2(dx)]$ and for $\mathring{\mathcal{H}}_Z := \{C_Z(\mu) : \mu \in \mathcal{M}_\psi(\mathcal{X})\}$. Then $\mathcal{H}_Z$ is a RKHS.*

Proof. We prove the claim by showing that both conditions of Proposition (6.2) hold for $\mathring{\mathcal{H}}_Z$. This equivalently implies that the evaluation functionals on $\mathring{\mathcal{H}}_Z$ are bounded w.r.t. to the induced norm. The evaluation functionals then uniquely extend to bounded operators on the closure, hence showing that $\overline{\mathring{\mathcal{H}}_Z}$ is a Hilbert space satisfying the characterising conditions of Proposition (6.2), hence a RKHS.

Indeed, we can give an explicit expression for the kernel. Recall that the kernel must have the reproducing property, i.e.

$$\langle k(x, \cdot), C_Z(\mu) \rangle = C_Z(\mu)(x) = \mathbb{E}[Z(x) \int_{\mathcal{X}} Z(x)\mu(dx)]. \quad (146)$$

It is not hard to come up with the guess for $k(x, \cdot)$;

$$\langle C_Z(\delta_x), C_Z(\mu) \rangle = \mathbb{E}[\int_{\mathcal{X}} Z(x')\delta_x(dx') \int_{\mathcal{X}} Z(y)\mu(dy)] = \mathbb{E}[Z(x) \int_{\mathcal{X}} Z(y)\mu(dy)] = C_Z(\mu)(x). \quad (147)$$

Hence, we set $k(x, \cdot) = C_Z(\delta_x)$ and deduce $k(x, y) = C_Z(\delta_x)(y) = \mathbb{E}[Z(y) \int_{\mathcal{X}} Z(x')\mu(dx')] = \mathbb{E}[Z(y)Z(x)] = \langle C_Z(\delta_x), C_Z(\delta_y) \rangle$. In particular $k(x, \cdot) = C_Z(\delta_x) \in H_Z$ by definition, thus by Proposition (6.2), $\mathcal{H}_Z$ is an RKHS with kernel $k(x, y) = \langle C_Z(\delta_x), C_Z(\delta_y) \rangle$. $\square$

We now come to the main theorem of this section (Theorem 6.4. Cuchiero et al. [2025]) giving, in particular, an explicit construction for the Gaussian prior $Z : \Omega \rightarrow \mathcal{B}_\psi(\mathcal{X})$ for a given kernel k as part of the modeling choice.

Theorem 7.9. *Given a kernel k on the weighted space $(\mathcal{X}, \psi)$, and a countable family $(h_i)_i$ in $\mathcal{B}_\psi(\mathcal{X})$ such that we can express the kernel as $h = \sum_i h_i \otimes h_i$ on $\mathcal{X} \times \mathcal{X}$ and the constraint $\sum_i \|h_i\|_{\mathcal{B}_\psi(\mathcal{X})} < \infty$ holds. Then we can construct the respective Gaussian prior Z with covariance k as the $\mathcal{B}_\psi(\mathcal{X})$ -limit of the random $\sum_i Z_i h_i$ where Z_i are standard i.i.d. Gaussian random variables. What is more, the Cameron-Martin space of Z coincides with the RKHS for the kernel k .*

Proof. The purpose of the summability condition is only to apply the Ito-Nisio theorem (M. Ledoux and M. Talagrand [1991], Theorem 2.4.) asserting that the Banach-valued random series $\sum_i h_i Z_i$ converges iff $\mu_n \circ f \rightarrow \mu \circ f$ weakly for all $f \in B_\psi(\mathcal{X})$, where μ_n is the law of the partial sum $S_n := \sum_{i=1}^n Z_i$ and for some limit distribution $\mu \in \mathcal{M}_1(B_\psi(\mathcal{X}))$. In particular, we have that $f(S_n) \sim \mathcal{N}(\sum_{i=1}^n f(h_i), 0)$, hence $\widehat{\mu_n \circ f^{-1}}(t) = \exp(im_n t)$ for $m_n = \sum_{i=1}^n f(h_i)$. For the convergence in the Banach space, notice

$$\mathbb{E}[\sum_i |Z_i| \|h_i\|_{\mathcal{B}_\psi(X)}] = \sum_i \mathbb{E}[|Z_i| \|h_i\|_{\mathcal{B}_\psi(X)}] \leq C \sum_i \|h_i\|_{\mathcal{B}_\psi(X)}, \quad (148)$$

which is finite by assumption.

Thus by Itô-Nisio we know that the family of probability measures $(\mu_n \circ f^{-1})_{n \geq 1}$ converges weakly for every $f \in B_\psi(\mathcal{X})$, hence so do their characteristic functions (in points of continuity). By continuity of the exponential function it follows that $\widehat{\mu \circ f^{-1}}(t) = \exp(imt)$ for $m = \lim_{n \rightarrow \infty} m_n$. Hence for all $f \in B_\psi(\mathcal{X})$ we have shown that $\mu \circ f^{-1}$ is Gaussian, hence the limit random variable Z with law μ is Gaussian and $B_\psi(\mathcal{X})$ -valued. Furthermore

$$\mathbb{E}[Z(x)Z(y)] = \mathbb{E}[\sum_i h_i(x)Z_i h_j(y)Z_j] = \quad (149)$$

$$\sum_{ij} h_i(x)h_j(y) \underbrace{\mathbb{E}[Z_i Z_j]}_{\delta_{ij}} = \sum_i \mathbb{E}[h_i(x)h_i(y)] = k(x, y).^{39} \quad (150)$$

Hence Z has the sought after covariance structure.

Furthermore, the pointwise covariance determines the covariance operator introduced previously $C_Z(\mu) = \mathbb{E}[Z \int_X Z(x)\mu(dx)]$ because for $C_Z(\mu) = \int_X k(x, \cdot)\mu(dx)$ holds (and $k(x, y)$ is equal to the pointwise covariance). Hence $C_Z(\mu)$ determines the RKHS and also, by definition, the Cameron-Martin space, whence they coincide. $\square$

To give further context to Theorem (7.9), recall that practically GPR begins with the choice of a Gaussian prior or equivalently a kernel map k . Theorem 7.9 then shows that if one finds a countable system of functions $(h_i)_i \subset B_\psi(\mathcal{X})$ such that we can rewrite the kernel as $k = \sum_i h_i \otimes h_i$ (i.e. $k(x, y) = \sum_i h_i(x)h_i(y)$), where the main condition to be checked is that the series $\sum_i \|h_i\|_\psi$ is finite, then there exists a Gaussian process Z (i.e. a prior) whose kernel is exactly the k , fixed by the practitioner. More precisely, the contribution of the theorem should be understood as a recipe for how to construct such a prior, rather than a mere existence proof. Indeed existence is a consequence of Kolmogorov extension theorem, as the finite dimensional marginals are multivariate Gaussians $\mu_{x_1, \dots, x_n}$ with a given covariance function $K_{ij} = (k(x_i, x_j))_{ij}$ which forms a consistent family of measures.

Furthermore, we can give a little more intuition about the meaning of Theorem (7.9). Using the interpretation derived from Theorem (7.7) and the interpretation of H_k through

³⁹The interchange of summation and expectation value is justified due to the absolute convergence of the double sum

the representer theorem (6.4), the statement $H_Z = H_k$, states that the directions that leave the prior measure invariant, are exactly the function, that can be learned to arbitrary precision through ridge regression, given enough data. More than that, due to the fact that the Cameron Martin space of the prior is stable under the Bayesian updating of GPR (see Proposition (7.10)), the Cameron Martin space of the prior, characterizes the distributions that can be learned through GPR completely. The invariance of the Cameron Martin spaces in GPR is stated and proven in the next proposition.

Proposition 7.10 (Invariance of H_Z und GPR). *Given a prior $Z \sim \mathcal{GP}(0, k(x, x'))$ for a given kernel $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$. Let Y be observed data and $Z|Y$ the posterior according to GPR. Then*

$$H_Z = H_{Z|Y} . \quad (151)$$

Whence we never leave H_Z during the GPR updates from priors to posteriors, nor do the successive Cameron-Martin spaces shrink during Bayesian updating.

Proof. The proof is subtle in the fact that one uses that the posterior is absolutely continuous w.r.t. to the prior. One is tempted to conclude this from Baye's rule, however the later doesn't hold for infinite dimensional Banachspaces in its original form. Nevertheless the posterior distribution is still absolutely continuous w.r.t. to the prior (see Dashti and Stuart [2015], Theorem 3.4.).

We can conclude by using a little bit of heavy machinery, namely the Feldman-Hajek Theorem A.3, one of the central theorems in the theory of Gaussian measures. The latter formalizes the insight that two Gaussian measures on the same locally convex space are either singular or equivalent. What is more, we use a classic corollary that if $\mu \approx \nu$ then their corresponding Cameron-Martin spaces are equal (see Appendix A.2). Hence, it suffices to show that the measure of the posterior is absolutely continuous w.r.t. to the prior measure.

As remarked, for the finite dimensional case this follows directly from Bayes' formula. For the general Banach spaced case, notice two things. First, the derivation of the posterior distribution given in Theorem 6.6 doesn't involve Baye's rule hence also holds for the general Banach space case. From this we conclude that the posterior is also Gaussian in the general case. Second using the generalized treatment in (Dashti and Stuart [2015] Theorem 3.4: infinite dimensional Bayes' rule) the posterior is indeed absolutely continuous w.r.t. to prior density. Hence we conclude by Feldman-Hajek and the stated corollary that the Cameron-Martin spaces are equal. $\square$

In other words, Proposition 7.10 states that under GPR, for a given Gaussian prior distribution μ_{prior} , we can only learn equivalent Gaussian measures as posterior distributions.

Eventually, Theorem 7.9 allows for a clean treatment of GPR using a signature kernel. From here on we will omit a lot of details and technical intricacies introduced in the rigorous development in Cuchiero et al. [2025], especially concerning geometric rough paths and the corresponding Carnot-Carathéodory norm on the *free step- N nilpotent Lie group* $G^N(\mathbb{R}^d)$ obtained by exponentiating the corresponding Lie-algebra $G^N(\mathbb{R}^d) = \exp^{(N)}(\mathfrak{g}^N(\mathbb{R}^d))$. For the details, we refer to sections 5.1., 5.2. and 6 of Cuchiero et al. [2025].

The goal is to introduce the signature kernel for two paths $X, Y \in \mathcal{C}_d^1$ defined as $k(X, Y) = \langle \mathbf{sig}(X), \mathbf{sig}(Y) \rangle = \sum_{\omega \in \langle d \rangle} \langle e_\omega, \mathbf{sig}(X) \rangle \langle e_\omega, \mathbf{sig}(Y) \rangle$.

More precisely, we introduce the following weighted kernel, which is still positive definite

$$k(X, Y) = \sum_{\omega \in \langle d \rangle} a_\omega^2 \langle e_\omega, \mathbf{sig}(X) \rangle \langle e_\omega, \mathbf{sig}(Y) \rangle \quad (152)$$

Adding the weights a_ω allows for an additional degree of freedom enabling the summability condition of Theorem 7.9. Indeed, in Cuchiero et al. [2025] it is shown that for the following choice of the weight function

$$\psi(X) = \exp(\beta \|X\|_{cc,p-var,\alpha}^\gamma) \quad (153)$$

and the basis functions

$$h_\omega(X) = a_\omega \langle e_\omega, \mathbf{sig}(X) \rangle \quad (154)$$

Hence, Theorem 7.9 gives an explicit approximating construction of the sought after Gaussian process Z , with covariance structure as in Equation 152. In addition, the signature kernel is universal for $\mathcal{B}_\psi(\mathcal{X})$, i.e., its RKHS is dense in $\mathcal{B}_\psi(\mathcal{X})$. In other words, signature kernel regression is compatible with GPR and all functions in $\mathcal{B}_\psi(\mathcal{X})$ are learnable with the GPR approach.

Hence, one might be interested in asking how large the space $\mathcal{B}_\psi(\mathcal{X})$ is. A general answer does not seem possible, and in the case of signature kernel regression, we just notice that due to Fernique's Theorem, for $\gamma = 2$ in (153), $\exists \beta$ such that $\exp(\beta \|X\|_{cc,p-var,\alpha}^2)$ is integrable. Hence, for $f \in \mathcal{B}_\psi(\mathcal{X})$ we have $\int_{\mathcal{X}} |f(x)| d\mu(x) \leq \|f\|_\psi \int_{\mathcal{X}} \exp(\beta \|X\|_{cc,p-var,\alpha}^2) d\mu(x) < \infty$. Hence for this choice of the parameters in ψ we find that $\mathcal{B}_\psi(\mathcal{X}) \subset L^1(\mathcal{X}, \mu)$ where μ is the law of Z i.e. a Gaussian measure on $\mathcal{B}_\psi(\mathcal{X})$.

8 Comparison of Signature-Based and Gaussian Process Regression

In this section we will try to bring across the point that the signature regression approach, based on bounded signature transform introduced in Section 4, and the Signature based GPR approach introduced in the penultimate section both tackle regression as the learning of a measure valued functional. Indeed, regression, as discussed here, is the learning of the conditional distribution of Y given X where the data $\{(x, y)\} \subset \mathcal{X} \times \mathcal{Y}$ are assumed to be realizations of the random variables X, Y . In more mathematical terms, we have motivated that we actually want to learn is the stochastic kernel

$$\begin{aligned} \mathcal{X} &\rightarrow \mathcal{M}^1(\mathcal{Y}) \\ x &\rightarrow \mu_{\mathcal{Y}|x}, \end{aligned}$$

¹ $\|\cdot\|_{cc,p-var,\alpha}^\gamma$ is the sum of the Carnot-Carathéodory distance on the free N-step Lie-algebra $G^N(\mathbb{R}^d)$ (domain of the geometric rough path) and the p-variation norm (indeed Cuchiero et al. [2025] considers α -geometric rough paths that have finite p-variation).

²such that $k(X, Y) = \sum_{\omega \in \langle d \rangle} a_\omega \mathbf{sig}^\omega(X) \mathbf{sig}^\omega(Y) = \sum_{\omega \in \langle d \rangle} (h_\omega \otimes h_\omega)(X, Y)$ fulfilling the summability condition $\sum_{\omega \in \langle d \rangle} \|h_\omega\|_\psi < \infty$

where $\mu_{\mathcal{Y}|\mathcal{X}} \equiv (\mu_{\mathcal{Y}|x})_{x \in \mathcal{X}}$ is the disintegration of the joint distribution $\mu \in \mathcal{M}^1(\mathcal{X} \times \mathcal{Y})$ (i.e. $\mu = \mathbb{P}_{X,Y}$) with respect to some marginal $\xi \in \mathcal{M}^1(\mathcal{X})$. What followed in the discussion in Section 2 and Section 4 were merely identifications with operators that enabled us to use density of signature components in order to find appropriate approximation systems, allowing us to use signatures efficiently for regression of path-valued random variables. The learnable distributions are a priori any disintegration of a Borel measure $\mu \in \mathcal{M}^1(\mathcal{X} \times \mathcal{Y})$ w.r.t. to some fixed marginal $\xi \in \mathcal{M}^1(\mathcal{X})$. Contrast this with the Gaussian Process regression case. As discussed, the learnable distributions here are Gaussian distributions over the path space $\mathcal{Y}^{\mathcal{X}}$ that are absolutely continuous w.r.t to the prior distribution⁴⁰. Looking a little closer, learning the distribution over the function space directly, or equivalently learning the Gaussian process, boils down to learning the right Gaussian marginals at every point $x \in \mathcal{X}$. Indeed, by definition, a Gaussian process $\mathbf{f} \sim \mathcal{GP}(m(x), k(x, x'))$ is a particular collection of random variables $\mathbf{f} = \{f(x) : x \in \mathcal{X}\}$ where $f(x) : \Omega \rightarrow \mathcal{Y}$ for every $x \in \mathcal{X}$. Hence, for $\mathcal{Y} = \mathbb{R}$ at every $x \in \mathcal{X}$, we model the marginal as a Gaussian random variable $f(x) \sim \mathcal{N}(m(x), k(x, x))$

Specializing in interpolation, i.e. learning of a purely (deterministic) functional relation notice that in the approach taken by Schell and Alaifari [2024] no further assumptions or restrictions are made on the space of learnable functions, other than measurability and integrability. Hence, recalling that the learning of the kernel (46) encompasses the learning of purely functional relations (using Dirac measures $\delta_{f(x)}$), we see that the only requirement for $f : \mathcal{X} \rightarrow \mathcal{Y}$ is that $(\delta_{f(x)})_{x \in \mathcal{X}}$ is a well-defined stochastic kernel, i.e., in particular, for every $B \in \mathcal{B}(\mathcal{Y})$ we need that the map $x \mapsto \delta_{f(x)}(B)$ is $\mathcal{B}(\mathcal{X})/\mathcal{B}(\mathcal{Y})$ -measurable. This entails that we need $f : (\mathcal{X}, \mathcal{B}(\mathcal{X})) \rightarrow (\mathcal{Y}, \mathcal{B}(\mathcal{Y}))$ to be measurable.⁴¹

In the case of GP regression, in particular as presented in Section 7, we can learn functional relations in $\mathcal{B}_{\psi}(\mathcal{X})$, which is merely a subset of $L^1(\mathcal{X}, Z_*\mathbb{P})$ for a Gaussian process prior $Z : \Omega \rightarrow \mathcal{B}_{\psi}$, as remarked before (by defining the GPR predicted function to be the posterior mean function).

Hence, the GPR approach is generally more restrictive, but we gain something in return. Indeed, in the GPR case, the marginals at each point $x \in \mathcal{X}$ are Gaussians, whose quantiles are easily computable and very well known. This, to us, is the biggest difference between the two approaches to nonparametric signature-based regression which can be stated as a tradeoff between generality of the learnable distributions and accessibility of uncertainty quantification.

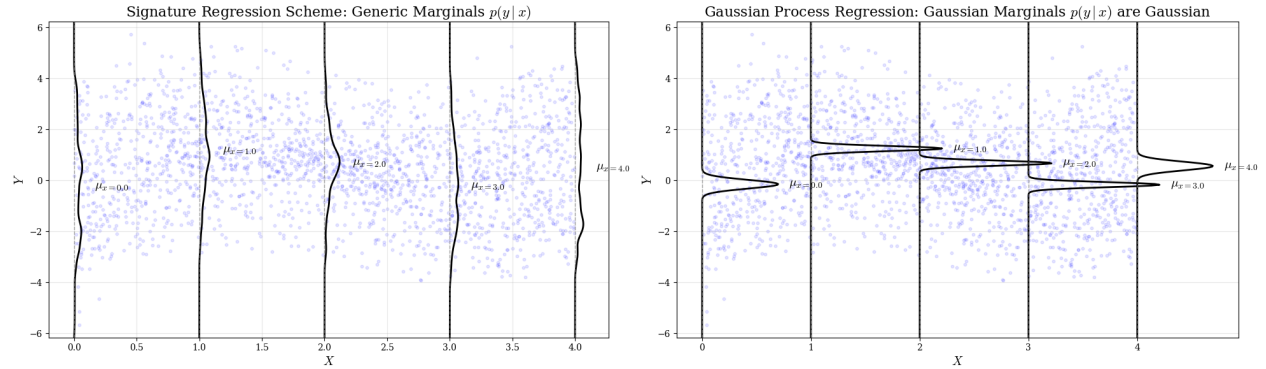
Indeed, to summarize the above, the bounded signature approach has essentially no a priori restriction on the learnable distributions. But due to the generality, for an uncertainty estimation, we need to calculate quantiles of generally Nonparametric distribution, which can be done empirically but is then much more cumbersome and costly than for the GPR

⁴⁰A priori no further structure is defined on $\mathcal{Y}^{\mathcal{X}}$, but if we want to talk about a Gaussian measure on $\mathcal{Y}^{\mathcal{X}}$ we usually put at least a locally convex topology on the latter, for which we can develop a rich theory of Gaussian measures as in Bogachev [1998]. In the case of Cuchiero et al. [2025], one shows that the weighted space (X, ψ) , for some appropriate weight function, neatly ties in the framework of GPR

⁴¹Indeed, denoting the map $\phi : x \mapsto \delta_{f(x)}(B)$ we need $\phi^{-1}(A) \in \mathcal{B}(\mathcal{X})$, $\forall A \in \mathcal{B}(\mathcal{Y})$ i.e. $\{x \in \mathcal{X} \mid \delta_{f(x)}(B) \in A\}$ which is in $\mathcal{B}(\mathcal{Y})$ iff f is $\mathcal{B}(\mathcal{X})/\mathcal{B}(\mathcal{Y})$ -measurable

approach. For the GPR case, we learn Gaussian marginals, hence uncertainty estimation for point predictions or whole paths are easily computable.

The previous discussion is easily motivated and visually supported by the following plots



(a) Empirical Distribution for samples from Nonparametric distribution

(b) Empirical marginals for Gaussian samples

8.1 Numerical Experiments

In this subsection, we implement nonparametric signature regression Algorithm (1), to perform time series forecasting. In the following, we will refer to Algorithm (1) as Algo I. For comparison, we implement GPR with a custom signature kernel, using e. path values for discretized times, after the observation horizon $S > 0$. Predicting the whole path does not directly correspond to TSF as treated in Section (5), but is natural from a GPR point of view, since the GPR prediction corresponds to the posterior mean function, which can clearly be reduced to times after S to get what we mean by forecasting of future values. Furthermore, prediction of the whole path is just a special case of the TSF framework of Section (5) as explained next.

Observe that numerically, we use a time discretisation of the interval $[0, T]$ with grid size $dt = T/N$ for some discretisation level N . Hence, we are looking at the projection operator π_{t_I} for $t_I = (0, T/N, 2T/N, \dots, (N-1)T/N, T)$. As previously motivated, we want to consider the prediction of the whole path as TSF too, and not just the future values. This is compatible with Definition 5.5 by taking $f = \mathcal{I}_A \circ \pi_{t_I} \in L^2(\mathbb{P}_Y)$. Indeed, notice that in both cases; NS and GBM, and in particular the interpolation of their numerical discretisation $\hat{X}$ are continuous functions, hence they are bounded on the real compact interval $[0, T]$. Whence, we can take $A = [a, b]$ with $a = \inf\{\bar{X}_t(\omega) \mid t \in [0, T], \omega \in \Omega\}$ and $b = \sup\{\bar{X}_t(\omega) \mid t \in [0, T], \omega \in \Omega\}$, which are both finite by Heine-Borel, and consider $f = \mathcal{I}_A \circ \pi_{t_I} \in L^2(\mathbb{P}_Y)$.

As explained in Section 5, we use (X, X^{t_0}) as pairs (regressant, regressor) for training for nonparametric signature regression and for the implementation of signature kernel GPR. For NSPR, we use the following for the efficient computation of signatures and signature kernels: Anh Tong [2023] and M. Tamayo-Rios [2025]. The numerical experiments are conducted on two different data sets: noisy sinus data (NS), which are of the form $y_i = \sin(x_i) + \epsilon$, where

$\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ and geometric Brownian motion paths (GBM) $X_t = X_0 \exp\left((\mu - \frac{\sigma^2}{2})t + \sigma W_t\right)$ solving the stochastic differential equation

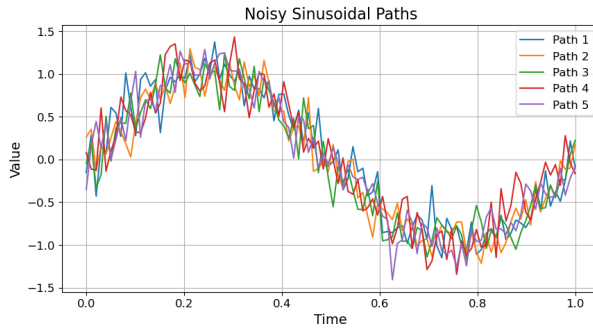
$$dX_t = \mu X_t dt + \sigma X_t dW_t. \quad (155)$$

The table below lists all parameters of the model. For parameter values that change for the different numerical experiments, no value is indicated in the following table. Rather, these values will be listed when needed correspondingly. Thus if nothing else is stated, the values of the parameters are as follows.

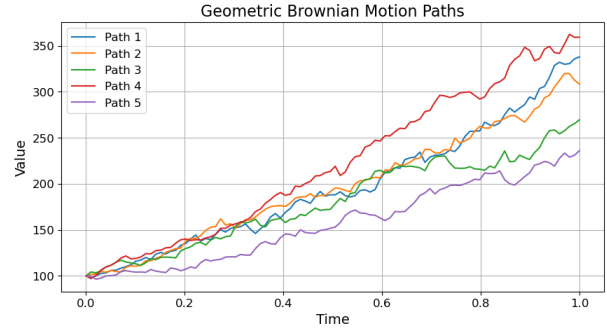
Parameter	Value	Description
T	1	Final time horizon
N	100	Number of time steps for discretization
M	-	Number of Sample paths for training
depth	8	Signature truncation depth
S	0.9	Observation horizon
μ	1.0	Drift coefficient for GBM
σ	0.2	volatility coefficient for GBM
σ_ϵ	0.2	variance of Gaussian noise on NS, i.e. $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon)$
λ_1	1.0	ridge regression regularization parameter for X^S stopped paths
λ_2	1.0	ridge regression regularization parameter for X unstopped paths
α	-	noise parameter for GP regression for GBM/NS prediction

Table 1: Model parameters common for all models and all experiments

To illustrate the nature of the respective data, the following plots show 5 samples paths for each type of data.

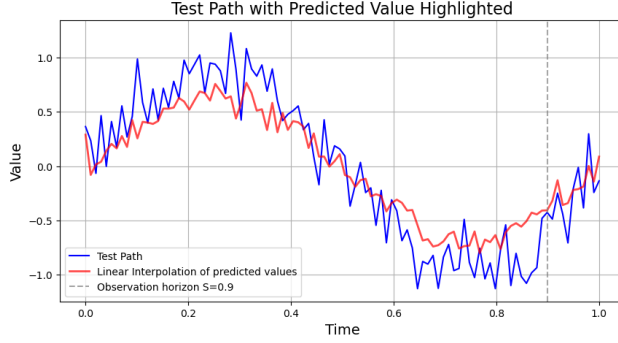


(a) $M=5$ noisy sinus paths.

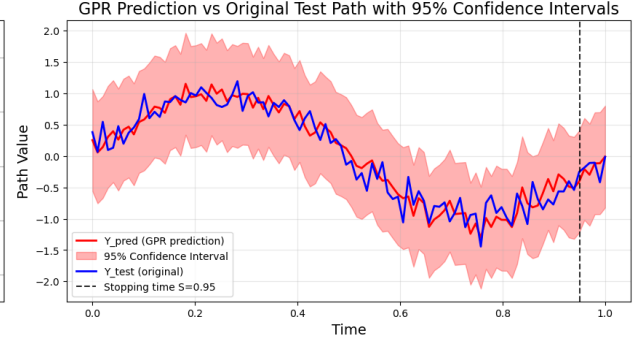


(b) $M=5$ geometric Brownian motion paths with parameters $\mu = 0.2$, $\sigma = 0.2$.

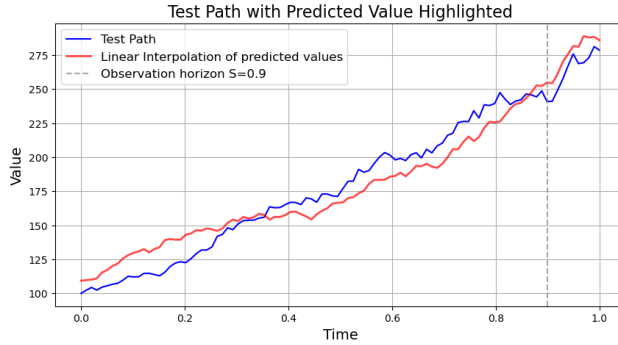
For testing, we generate a test path, with the same parameters as for training. We compare the whole predicted path with the test path, with particular interest in the performance at $t > S$. In the case of GPR we also visually indicate the 95% confidence interval at each point, which is a byproduct of `sklearn.gaussian_process`. This interval is depicted in the right figure below as the shaded red area.



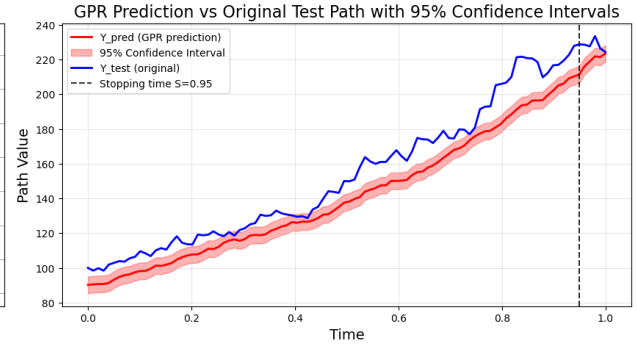
(a) NS predictions with Algo I for $M = 10$ training paths. The prediction of Algo I is taken to be all points of the discretised interval $[0, T]$, hence the path as a whole is predicted.



(b) NS predictions with GPR I: $M=10$, $\alpha = 0.1$. The predicted path is the posterior mean function $\bar{m}(x)$. For every point, a 95% confidence interval is calculated and visualised as the red shaded area.



(a) GBM predictions with Algo I for $M = 10$ training paths. The prediction of Algo I is taken to be all points of the discretised interval $[0, T]$, hence the path as a whole is predicted.



(b) GBM predictions with GPR I: $M=10$, $\alpha = 0.1$. The predicted path is the posterior mean function $\bar{m}(x)$. For every point, a 95% confidence interval is calculated and visualised as the red shaded area.

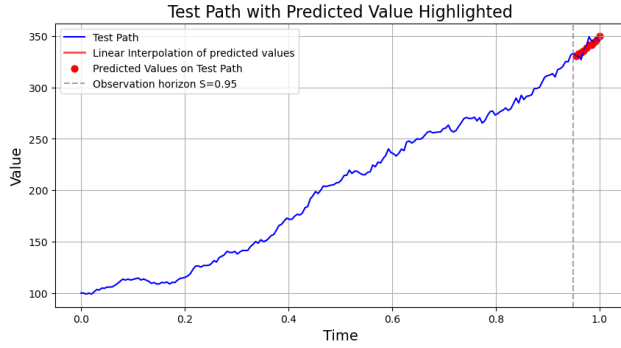
For better computability and performance we perform a Z-score normalisation on the data, prior to the calculation of the signature and rescale after prediction. What is more, the implementation of the bounded signature follows Chevyrev and Oberhauser [2022] and solve the polynomial suggested in Proposition [14] with ψ as suggested in Example [4], satisfying

Corollary 15, hence inducing a tensor normalization.

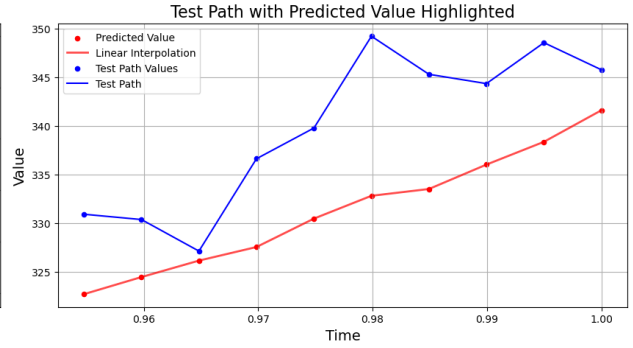
$$\psi(\sqrt{x}) = \begin{cases} x & \text{if } x \leq C, \\ C + C^{1+a}(C^{-a} - x^{-a})/a & \text{if } x > C, \end{cases} \quad (156)$$

for $a > 0$ and $C \geq 1$. For our experiments, we chose $a = 0.5$ and $C = 2$.

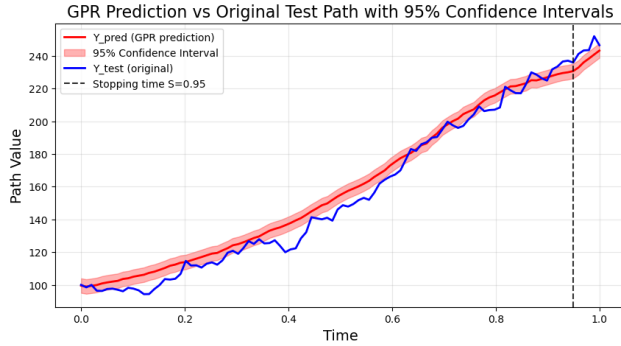
Following more closely what we would naturally understand as prediction and what is underlying the developments of 5.5, we predict only for time points $t_i > S$, for the Nonparametric signature regression and the signature kernel GPR implementation.



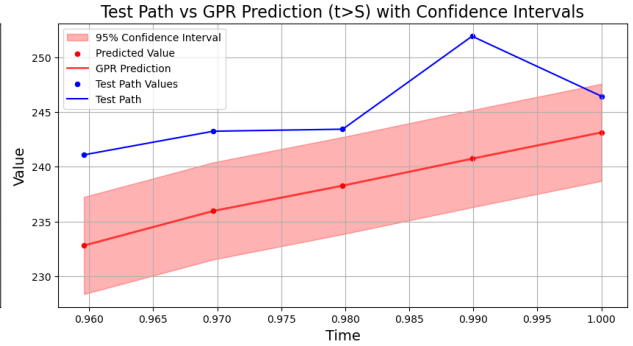
(a) GBM prediction with Algo I using $M = 100$ training paths and $N = 200$ time points.



(b) closeup on predicted values for GBM prediction with Algo I for $t_i > S = 0.95 \cdot T$



(a) GBM prediction with GPR: $M=100$, $N=200$, $S = 0.95$. If we want to the prediction of GPR on future values $t_i > S$, we simply ignore $t_i \leq S$, yielding the closeup on the right plot.



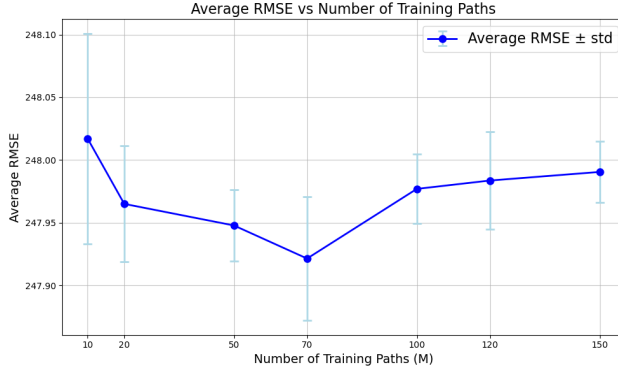
(b) closeup on predicted values for GBM prediction with GPR, interpreted as GPR TSF for $t_i > S$.

These plots clearly indicate, at least visually, that both methods learn from the training data and yield sensible predictions for the test path and its future values. In particular, in the case of the GBM data, both algorithms seem to learn the drift μ quite well. To get a good estimate of the volatility, we suspect that one would need much more training data. The

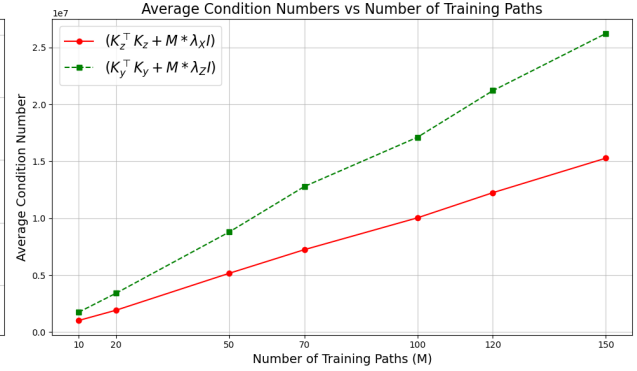
currently learned volatility depends more on the inherent noise parameters λ_1, λ_2 and α of the regression models, rather than actually learning the volatility of the underlying stochastic model (155). This leads us to the limitations of the current experiments.

The main limiting factor for the precision of the prediction is the computational limitations of the machine on which the experiments were conducted (MacBook Air 2023, Chip: Apple M2). This limited the number of paths M , the discretization level/resolution of the paths N and the truncation level (depth) of the signature and their covariance kernel, which were feasible for the estimation of the estimators considering runtime. The Python implementations underlying all the experiments presented are provided as Jupyter notebooks; see Joel Waespi [2025].

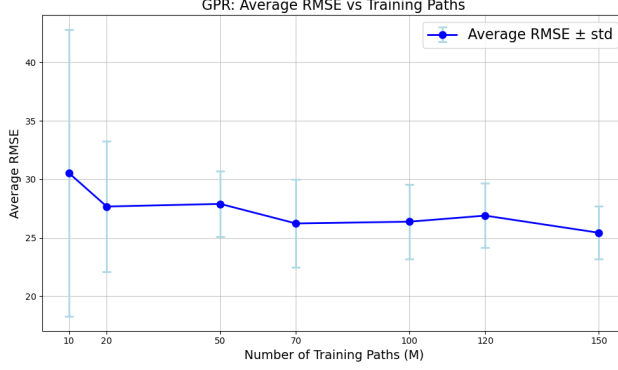
In order to study the convergence for the implementation of the statistical estimators of Algo I and GPR, we plot the root mean squared error (RMSE) for increasing number of training paths $M \in \{10, 20, 50, 70, 100, 120, 150\}$ evaluated on the same test path for a common grid with $N = 200$. More precisely, we repeat every run for every value of M , 10 times and report the mean RMSE. Furthermore, we also report the condition number of the matrices that are inverted during the ridge regression in Algo I and GPR, that is, matrices of the form $K_Z^T K_Z + \lambda \mathcal{I}$, where Z denotes stopped paths X or full paths Y .



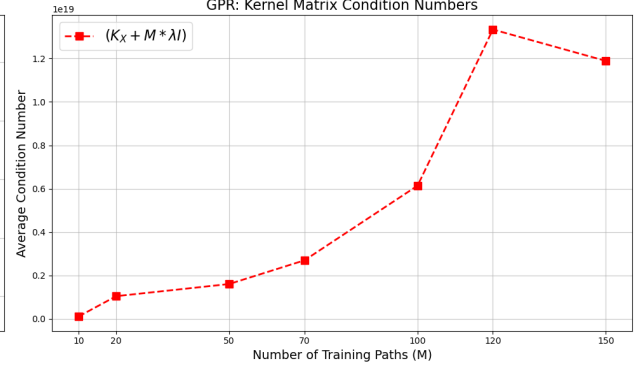
(a) Average RMSE for Algo I and GBM data: $M=\{10, 20, 50, 70, 100, 120, 150\}$, $N=200$. Additionally, the standard deviation is marked in light blue



(b) Average condition number of inverted matrices¹ in Algo I for GBM data: $M=\{10, 20, 50, 70, 100, 120, 150\}$, $N=200$.



(a) Average RMSE for GPR and GBM data:
 $M=\{10, 20, 50, 70, 100, 120, 150\}$, $N=200$.
 Additionally, the standard deviation is marked
 in light blue



(b) Average condition number of inverted
 matrices for GPR and GBM data:
 $M=\{10, 20, 50, 70, 100, 120, 150\}$, $N=200$.

The reported average RMSE seemingly fails to continuously converge as the number of training paths increases in the case of Algo I and slowly converges for GPR. First, we remark that the number of paths that were used (maximum of 150 paths) is clearly not enough to draw any definitive conclusion concerning numerical convergence. Nevertheless, we can observe and state hypotheses that are to be tested using more capable computing methods and devices. In both cases, Algo I and GPR, we observe the contrary action of the increased precision of the statistical estimator and the decreasing numerical stability, caused by the higher condition number, corrupting the gain in precision. Indeed, we can visually confirm that an increasing condition number corresponds to an increasing RMSE and vice versa. For Algo I it seems that at about 70 paths the destabilization of the high condition number starts to dominate the performance.

For the GPR, the condition number increases as well, but on a smaller scale, with a dip for the maximal number of paths. The convergence seems to be more steady, in particular due to a drop in condition number for $M = 150$. From these simple experiments, we can put forward the hypothesis with regards to RMSE performance and in the realm of $M \leq 150$ and discretization $N = 200$, the GRP approach shows better results. Again, this hardly allows for any conclusion regarding a comparison of both methods due to low number of paths, discretization, and signature depth being hardly enough to study the limiting behavior of the statistical estimator, in particular in the Nonparametric regression case.

9 Conclusion

This project reviewed two paradigms for regression of stochastic processes viewed as the learning of stochastic kernels and studied two complementary signature-based schemes: non-parametric regression via signatures and Gaussian process regression using signature kernels.

¹The computation of the coefficients $\alpha_{\lambda_f}^f$ and β_{λ_y} require inversion of $A_Z^T A_Z + M \lambda A_Z$, for $Z \in \{X, Y\}$.

²The computation of the posterior mean $\bar{m}(x) = m(x) + K_{x\mathbb{X}}(K_{\mathbb{X}\mathbb{X}} + \sigma^2 \mathcal{I}_n)^{-1}(\mathbb{Y} - m_{\mathbb{X}})$ require inversion of $K_{\mathbb{X}\mathbb{X}} + \sigma^2 \mathcal{I}_n$, where we can decompose $K_{\mathbb{X}\mathbb{X}} = L^T L$, for $Z \in \{X, Y\}$.

In particular, the thesis presented regression in a general framework as a kernel learning problem, which allowed time-series forecasting to be rigorously formulated as a special case of regression. We introduced the fundamental concepts underlying both approaches.

For the Nonparametric regression approach, this were most notably the signature transform and bounded signatures. The L^2 -density of bounded signature coordinates enabled the estimation of regression kernels through ridge regression and tensor normalization.

For the GPR approach, we briefly introduced elements of Gaussian measure theory to understand Gaussian priors in the context of Gaussian process regression. We then focused on an alternative approximation framework for non-locally compact spaces based on weighted spaces. Combined with a primer on reproducing kernel Hilbert spaces, this provided a rigorous foundation for signature-kernel GPR. We compared the two methods from three perspectives: (i) the range of learnable distributions and corresponding finite-dimensional marginals, (ii) their interpretation as deterministic functional learning problems, and (iii) their empirical performance in time-series forecasting. We found that the nonparametric signature approach imposes essentially no structural restrictions on the learnable conditional laws. In contrast, signature-kernel GPR on weighted spaces restricts learnable laws to Gaussian marginals but provides immediate uncertainty quantification via closed-form finite-dimensional distribution of the posterior, highlighting a trade-off between generality and practicality.

Empirically, both methods proved to be feasible for time-series forecasting. Experiments on noisy sine waves and geometric Brownian motion demonstrated the practical applicability of both estimators. The RMSE trends reflected the opposing effects of gained statistical accuracy and numerical ill-conditioning when increasing the number of training samples, underscoring the importance of regularization and preconditioning at larger sample sizes. Overall, nonparametric signature regression maximizes modeling flexibility, while signature-kernel GPR provides accessible uncertainty estimates. The nature of the task itself appears to determine which of the two methods is best suited.

References

- Alexander Schell and Rima Alaifari. *On Nonparametric Regression of Stochastic Processes via Signatures*. arXiv preprint, 31 May 2024.
- Rick Durrett. *Probability: Theory and Examples*. Lecture notes, January 11, 2019.
- Terry J. Lyons, Michael J. Caruana, and Thierry Lévy. *Differential Equations Driven by Rough Paths*. École d’Été de Probabilités de Saint-Flour XXXIV–2004, Springer, 2007.
- Olav Kallenberg. *Foundations of Modern Probability*, 3rd edition. Springer, 2020.
- Ilya Chevyrev and Harald Oberhauser. *Signature Moments to Characterize Laws of Stochastic Processes*. arXiv preprint, 30 July 2022.
- Jan van Neerven. *Stochastic Evolution Equations*. Lecture Notes ISEM 2007/08.
- Thomas Cass and Christopher Salvi. *Lecture Notes on Rough Paths and Applications to Machine Learning*. arXiv preprint, 9 April 2024.
- Ioannis Karatzas and Steven E. Shreve. *Brownian Motion and Stochastic Calculus*, 2nd edition. Springer, 2019.
- Andreas Krause and Jonas Hübotter. *Probabilistic Artificial Intelligence*. Lecture notes, January 28, 2025.
- Carl E. Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.
- Bernhard Schölkopf, Ralf Herbrich, and Alex J. Smola. *A Generalized Representer Theorem*. Springer, 2001.
- Motonobu Kanagawa, Philipp Hennig, Dino Sejdinovic, and Bharath K. Sriperumbudur. *Gaussian Processes and Kernel Methods: A Review on Connections and Equivalences*. arXiv preprint, 30 July 2022.
- Christa Cuchiero, Philipp Schmock, and Josef Teichmann. *Global Universal Approximation of Functional Input Maps on Weighted Spaces*. arXiv preprint, 3 February 2025.
- Philipp Dörsek and Josef Teichmann. *A Semigroup Point of View on Splitting Schemes for Stochastic (Partial) Differential Equations*. arXiv preprint, 11 November 2010.
- A. W. van der Vaart and J. H. van Zanten. *Reproducing Kernel Hilbert Spaces of Gaussian Priors*. arXiv preprint, 21 May 2008.
- Masoumeh Dashti and Andrew M. Stuart. *The Bayesian Approach to Inverse Problems*. arXiv preprint, 2 July 2015.
- Vladimir I. Bogachev. *Gaussian Measures*. American Mathematical Society, 1998.

- Martin Hairer. *An Introduction to Stochastic Partial Differential Equations*. Lecture Notes, 2009.
- M. Ledoux and M. Talagrand. *Probability in Banach spaces: Isoperimetry and Processes*. Springer, Berlin, 1991.
- Anh Tong. <https://github.com/anh-tong/signax>. Github repository, 2023.
- M. Tamayo-Rios. <https://github.com/geekbeast/powersig>. Github repository, 2025.
- Joel Waespi. <https://github.com/Joelwaa/SemesterProject-Numerical-Experiments>. Github repository, 2025.

A Appendix

A.1 Pettis Measurability Theorem

Let $(A, \mathcal{A})$ be a measurable space and let F be a norming subspace of E^* . For a function $f : A \rightarrow E$ the following assertions are equivalent.

1. f is strongly $\mathcal{A}$ -measurable⁴²
2. f is separably valued and $\langle f, x^* \rangle$ is $\mathcal{A}$ -measurable for all $x^* \in E^*$ ⁴³⁴⁴
3. f is separably valued and $\langle f, x^* \rangle$ is $\mathcal{A}$ -measurable for all $x^* \in F$

In particular, for a separable Banach space E $f : A \rightarrow E$ is strongly $\mathcal{A}$ -measurable iff it is weakly $\mathcal{A}$ -measurable.

A.2 Functional Monotone Class

Let $\mathcal{M}$ be a set of bounded real valued functions on some non-empty set S . Suppose that $\mathcal{M}$ is closed under multiplication and let $\mathcal{A} = \sigma(\mathcal{M})$. Let H be a real vector space of bounded real-valued functions on S and assume that H has the following properties

- $\mathcal{M} \subset H$
- H contains the constant function 1
- H is closed under monotone bounded convergence i.e., if $0 \leq f_1 \leq f_2 \leq \dots$ is a sequence in H and $f = \lim_{n \rightarrow \infty} f_n$ is bounded, then also $f \in H$

Then H contains all bounded $\mathcal{A}$ -measurable functions.

A.3 Feldman-Hajek

We state the celebrated Feldman-Hajek Theorem. For this we follow Bogachev [1998], in particular Theorem 2.7.2. and the Corollary 2.7.3.

Theorem A.1 (Feldman-Hajek). *Two Gaussian measures on the locally convex space are either equivalent or mutually singular*

Corollary A.2. *Let μ and ν be Gaussian measures on a locally convex space X . If $\mu \sim \nu$ then $H(\mu) = H(\nu)$ as sets*

Lastly, a version specializing to locally convex spaces X , that are in addition Hilbert spaces $\mathcal{H}$. This is taken from Dashti and Stuart [2015] Theorem 7.35.

⁴² $f : A \rightarrow E$ is strongly $\mathcal{A}$ -measurable if there exists a sequence of $\mathcal{A}$ -simple functions $f_n : A \rightarrow E$ such that $\lim_{n \rightarrow \infty} f_n \rightarrow f$ pointwise on A .

⁴³ f is *separably valued* if there exists a separable closed subspace $E_0 \subset E$ such that $f(A) \subset E_0$. In particular, f is separably valued if E is a separable Banach space.

⁴⁴here we define $\langle f, x^* \rangle(\xi) = \langle f(\xi), x^* \rangle$. If these maps are $\mathcal{A}$ -measurable, we say that f is *weakly $\mathcal{A}$ -measurable*

Corollary A.3. *Let μ, ν be Gaussian measures on a Hilbert space $\mathcal{H}$ characterised by their respective mean and covariance function $m_\mu, K_\mu, m_\nu, K_\nu$. Then $\mu \sim \nu$ iff*

- *There Cameron Martin space are the same, where in the Hilbert space setting they are given by the image of $K^{1/2}$.⁴⁵ We call that space H*
- $m_\mu - m_\nu \in H$
- $(K_\mu^{-1/2} K_\nu^{1/2})(K_\mu^{-1/2} K_\nu^{1/2})^* - \mathcal{I}$ *is Hilbert-Schmidt on $\bar{H}$*

A.4 H_t is a Closed Linear Subspace of L^2

Linearity is clear due to triangular inequality of L^2 -norm and the fact the sum of $\mathcal{F}_t^X$ -measurable maps is again $\mathcal{F}_t^X$ -measurable. For the closed subset property consider a converging sequence $X_t^n \xrightarrow{L^2} X_t$ as $n \rightarrow \infty$. Then we can extract a almost surely converging subsequence $X_t^{n(k)} \xrightarrow{a.s.} X_t$ as $k \rightarrow \infty$. Then X_t is $\mathcal{F}_t^X$ -measurable as a almost sure limit of $\mathcal{F}_t^X$ -measurable functions (assuming as usual that the underlying probability space is complete). The limit X_t is also in L^2 by completeness of the later.

⁴⁵One can check that the covariance operator defines a compact operator, hence admits a spectral decomposition, hence the square root of the operator is well-defined